Ontology—Provisional Boundary Scale-Relative Realism

Working ontology · Last revised 27 October 2026 · Revision 61

A Provisional, Boundary- and Scale-Relative Realism

> **What it is:** A Provisional, Boundary- and Scale-Relative Realism (PBSRR) is the working ontology of the Sweet Rationalism frameworks, treated as an output of inquiry rather than a protected premise. Its minimal commitment is a provisional, constraint-sensitive realism: inquiry encounters constraints that are not freely set by our descriptions or preferences. Beyond that floor, realism is selective and graded, attaching to specific entities, relations, mechanisms or patterns at the boundaries and scales where removing them would degrade prediction, intervention or transfer. It replaced as the project's ontology in September 2026.
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Versions and archives: this revision (DOI) · all revisions (concept DOI) · PDF · source and change history on GitHub · definitions, earlier names and citation formats

Revision 61 · October 27, 2026. Restates Recursive Episodic Integration as a candidate comparison pattern rather than an established real pattern, merges the bearer and control-boundary audit with the modal questions, adds failure modes and a source-fidelity and quotation-provenance rule, and consolidates the sources into one section. Earlier revisions remain available for audit in the change history.

Minimal executable kernel

This kernel turns PBSRR into a compact rule for admitting, locating and revising commitments about what is real. Scholar-concept pairs provide semantic retrieval anchors for the relevant arguments and rivals; the commitments remain answerable to their stated grounds, with definitions, conditional proofs and empirical claims kept distinct.

1. **Begin with a revisable commitment.** Charles S. Peirce: fallibilism. Treat inquiry's encountered constraints as the working floor and realism, monism and physical completeness as separately revisable interpretations. Separate adopted rules, definitions, conditional theorems, empirical regularities and ontological hypotheses. Actuality, universality and necessity require different warrants. Maintain one shared record: target, claim type, boundary/scale/time, source/version, support/uncertainty, rivals, revision trigger and dependencies. Mark unknown or inapplicable fields. Keep truth status, acceptance as a working ontology, and research-programme status separate; provisional rejection of a claim's warrant does not assert its negation.
2. **Make categories revisable.** Rudolf Carnap: explication. Specify the referent, assignment rule, variables and scope of each material category. Split, merge, replace or suspend a category when it obscures distinctions; test alternatives and record the changed partition. Keep statistical information, meaning, function, experience and value separate unless a warranted bridge connects them. Human membership, computational description, and use as a tool are overlapping classifications of different kinds. Specify which sense of life or consciousness is under investigation; treat categorical assignment and operational measurements as different claims. Acceptance of actual causal influence leaves universal experiential constitution to earn its own warrant; uncertainty about its nature is not absence of activity.
3. **Ask what earns realism.** Daniel C. Dennett: Real Patterns. Compare the proposed entity, relation or organized pattern with target-preserving rivals and substitutions.

Seek robust prediction, independent identification, perturbation-sensitive contribution or transfer as applicable. Assess discovery, checking, explanation and transfer separately. Predictive fit alone leaves causal identification open. Assign commitment to what earns that work; a useful pattern does not automatically establish a preferred intrinsic nature or exhaustive ontology.

4. **Locate the effective scale.** Robert W. Batterman: autonomy of effective theories. Compare candidate boundaries and spatial, temporal or organizational scales for the stated question. Declare coarse-graining, retained/lost variables, invariants and validity limits. Require a defended bridge before transferring a result; distinguish a changed description from physical reorganization. Distinguish law, state, observation and representation. A successful local scale earns no universal privilege.
5. **Separate realization and status.** Carl F. Craver: levels of mechanisms. Identify components, activities and organization at the relevant boundary. Localize capacities through removal, replacement and recombination of model, runtime, scaffold and collaborators; roles do not fix intrinsic kinds. Warrant cognition, agency, autonomy, consciousness, valence, welfare, patiency, authority and responsibility separately. Apply physical and thermodynamic constraints to operative realization; use Landauer accounting only for a specified operation and conditions. Separate physical timing, temporal models, felt presence, operational risk, and welfare. Separate who may undergo an episode from the interaction organizing it, its realizing processes, and the identity preserved under replacement.
6. **Update what exists in the account.** Charles S. Peirce: fallibilism. Promote, demote, revise or suspend only the commitments supported through the changed evidence or bridge; preserve independent capacities and unresolved intrinsic character. Keep hypothesis support separate from a research programme's active, dormant or rejected status. Record scope, reasons and reopening conditions. Reassess this ontology under its own rules.
7. **Prefer warranted economy.** Ronald A. Howard: value of information. Use the smallest account adequate to the target, compared with stronger, weaker and unknown alternatives. Reuse checked records and stop when further comparison is unlikely to justify its cost. Report the scoped commitment, evidence, unresolved

residual and revision trigger. Test simpler boundaries, categories and kernel variants against held-out failures before preserving complexity.

Calling something a person, an organism, an institution, a model, or a mind shapes what we look for and what we decide to do. I want to understand what makes those descriptions useful, when they track something real, and where they carry us further than the evidence allows. That is the purpose of this provisional, boundary- and scale-relative realism, or PBSRR.

My working commitment is that inquiry answers to a world we encounter through limited and improvable forms of access. Real patterns can depend on relations, history, scale, and surroundings; discovering those dependencies can explain why a local finding holds. Realism is my provisional interpretation of that success. Its support, the causal contribution of a pattern, and any claim that reality is exclusively physical, mental, or relational remain separate comparisons.

I apply this distinction between warranted causal accounts and wider ontological commitment in my whitepaper, *From Bodily Causal Feeling to Whitehead's Universal Prehension: Testing the Scope Inference* (PDF), comparing positive explanations at specified targets with the claim that causal actuality everywhere involves feeling.

Taking reality seriously without fixing it in advance

Alfred North Whitehead captures a tension I want to keep alive: "Seek simplicity and distrust it." In *The Concept of Nature*, chapter VII, he warns that seeking a simple explanation can tempt us to imagine the facts themselves are simple. A good description should help us see more clearly while remaining answerable to what it leaves out.

That matters when inquiry crosses scales. A cell, a person, a research team, and an institution can each be the right unit for a different question. We need to identify the components, relations, history, surroundings, and time horizon that make an attribution work. The physical processes realizing a capacity and the organization explaining its use can contribute at different levels. Comparing those contributions is part of deciding what deserves a place in the account.

What earns promotion here

PBSRR distinguishes an adopted rule of inquiry, a result proved under stated premises, and a commitment about the world. Those statuses carry different burdens. A conditional theorem can hold throughout its formal domain while its application remains uncertain. Realism, scale pluralism, and particular boundary selections remain working positions whose support can change. Promoting a claim to an empirical law requires a defended domain and evidence that reaches the proposed generality. John W. Carroll's survey of laws of nature helps make that distinction explicit.

The comparison begins with a target, its bearer, boundary, timescale, and relevant alternatives. If rival hypotheses give an observation equal positive likelihoods, the observation leaves their relative odds unchanged. A preference between them must then depend on other grounds. I apply that discipline equally to experiential, physical, informational, relational, and other proposed intrinsic characters, without assuming equal priors. When a bridge fails, I revise the commitment supported through it while keeping independently warranted organization and capacities.

A coarser description may reveal a useful pattern and improve prediction or intervention. Under the fixed-distribution conditions of data processing, transforming only the existing record cannot add Shannon information about the target. Changing the explanatory boundary also does not, by itself, establish that the described system has physically reorganized. Which boundary tracks the relevant dependencies is a further question for comparison. The formal conditions are stated in RCF.

Physically realized distinctions have implementation conditions. Their persistence, accessibility, transmission, and causal use must be explained through the implementing processes and applicable thermodynamic constraints. This is a realization constraint; the intrinsic character of those processes remains a separate inquiry. Operation-sensitive costs, including the specified erasure case tested by Bérut and colleagues, are developed in the physics and thermodynamics section.

Moral attribution has the same discipline of scope. Where independently justified descriptive and normative premises suffice to protect a capacity, an additional experiential ontology need not be settled first. That gives the relevant intervention a moral constraint under those premises; direct standing, welfare, decision authority, and responsibility still require their respective grounds. A supported relation or organized whole receives the status its argument earns, rather than inheriting every status of its components.

What a better representation can establish

A new representation can reveal a relation, make a calculation tractable, or expose a mistaken assumption. I want to keep those discoveries without assigning the representation every property of what it describes. We need to ask which distinctions survive the map, whether the resulting models have the same interpretation, and what evidence reaches their domain of application. A mystery can disappear from a description while an empirical or ontological question remains open.

Finding the organization that does the work

Compositional computation helps me investigate which distinctions and relations survive when a system is rearranged. A pattern earns support through what it predicts across those changes. I can then ask whether the useful organization lies in a component, an interaction, or a larger continuing system, and which further claims remain unsettled by that result.

The same approach makes continuity through change worth investigating. A person's body, memories, dispositions, relationships, and self-understanding can change while important forms of continuity remain. Recognition by others may help sustain some of those relations. I want to investigate which processes preserve that continuity, which disruptions change it, and what competing accounts explain. That gives the suggestion of a dynamically maintained self a concrete research task.

For computational systems, the relevant bearer may include a model, its current context, memory, tools, operators, and the institutions maintaining its operation. A capacity can depend on that larger arrangement. Locating it accurately matters when we attribute intelligence, learning, agency, responsibility, or control. Each attribution should earn the boundary and scope at which we use it; questions about phenomenal experience and welfare deserve their own careful comparisons.

This is useful wherever a category starts doing practical work. In biology, it can help distinguish a mechanism from a proposed interpretation of its intrinsic nature. In engineering, it can make dependence on surrounding infrastructure visible. In social inquiry, it can clarify how records, rules, and participation sustain an institution. In consciousness research, it can separate the capacities a measure tracks from the further interpretation assigned to them. Religious and metaphysical proposals can contribute models and questions whose explanatory roles we compare with serious alternatives.

I want that openness to support construction as well as criticism. A process-relational account might help an experimental researcher ask better questions without first resolving every dispute about the intrinsic character of nature. A naturalized account of care, memory, or guidance might identify finite practices that can be investigated and improved. The next task is to establish what those accounts actually contribute, where they transfer, and what additional commitments would need further warrant.

The practical stakes include the categories used to protect capacities, allocate responsibility, or govern access to a service. A person may depend on a record, a

community, or a computational arrangement for something important. Understanding that dependence can change an ethical decision when joined to the relevant reasons for protection. Making the bearer and relationship explicit helps the moral discussion reach the right target.

PBSRR records a working account of what inquiry currently warrants treating as real. Its commitments can become stronger, more specific, or different as comparisons improve. I want realism itself to remain open to its strongest empiricist, pragmatist, and other alternatives. What matters is whether a commitment helps us account for the target and survives the appropriate scrutiny.

Within the wider project, the Ethos sets out how we should approach one another and the consequences of our decisions; RCF supplies the procedures for testing and revising claims; the handles translation among accounts. PBSRR asks what survives those inquiries as a commitment about reality. I hope that division of work will help us build useful research systems while keeping their categories responsive to what they discover.

Provisional constraint-sensitive realism
Selective scientific realism
Graded pattern realism
Boundary- and scale-indexed identity
Common-core subtraction
Target-preserving reconstruction
Intrinsic-nature agnosticism where warranted
Bridge and modal discipline
Self-report calibration
Gap ledger and bridge admission
Typed necessity claims
Recursively corrigible
Reciprocal triadRecursive mirror consequenceProgramme-progress statusPhysical-realization floorIntrinsic-predicate symmetryPremise invarianceRepresentation-to-

ontology firewallOperational ?XCosmic boundednessREIM0 ≠ S*System-boundary emergenceGenealogical decompressionArchive-selection firewallFinal-ontology confidence ruleMicro-realization ≠ explanatory sufficiencyProcess continuityStructural epistemic incompletenessMathematical-causation firewallDistributed cognition ≠ phenomenologyAgentic-process realismBearer-first attributionControl ≠ ownership
This page brings together the commitments supported by the cross-domain evidence and scholarship represented here and in the underlying research programme. Claims about necessity, identity, and the realist starting point remain open to comparisons that can change their scope or support.

Shortest version

Unresolved truth can accompany a rejected working ontology

Maintain three different judgments: whether a proposition is true, whether its current warrant earns adoption, and whether investigating it remains useful. **Truth unresolved; uptake provisionally rejected; research programme active** is coherent. It rejects an unearned commitment without asserting the opposite universal. Peirce's fallibilism and Lakatos's programme-level comparison supply the methodological anchors; the separation of these statuses is this framework's operational synthesis.

The published ingression proposal distinguishes neither metaphysical entity from epistemological framework in note 1 (p. 39), yet elsewhere describes causally interacting nonphysical patterns. Scientific usefulness can support an account of real organization without resolving that further distinction. Baker's mathematical-explanation programme, Lange's explanations by constraint, and the SEP's causal/non-causal comparison make the alternatives substantive. Mathematical constraint, physical realization, additional

causal influence, agency, phenomenal character, and moral status require separately defended bridges.

The same discipline addresses Segall's objection. Affirming that a concrete process occurs while withholding its universal identification with feeling leaves constitution unresolved, not occurrence. Even a true identity need not be known whenever either side is known. This does not establish that nonexperiential causation is metaphysically possible; it defeats the inference from uncertainty about constitution to denial of the evidenced activity. Positive rival accounts still owe their actual explanations.

Real patterns can be discovered through editable targets, higher-level control, or transferable structure without selecting an external source. If a proposed catalogue admits every observed target and supplies no independently fixed restrictions, observing a target does not identify that catalogue as its cause. Learn maps from data, then test what they constrain beyond their fitting cases. My Levin critique develops this distinction. Do not turn this restraint into axiomatic physical completeness or biological exclusivity.

My current explanation of the pattern that connects

I now have a more definite answer to a question I had previously left too open: why do similar organizations recur in chemistry, biology, perception, computation, and cultural practice? My best-supported working explanation is **shared constraints, independent convergence, and historical continuity, sometimes acting together**. These are positive causal families already supported in particular domains, not four rhetorical options waiting for a metaphysical winner.

Different systems operate under physical and mathematical restrictions. Their dynamics make some structures stable and others transient. Similar pressures or equations can independently produce similar forms. Some structures also persist because they are

copied, inherited, delivered, recorded, taught, or maintained. In particular cases a deeper invariant can explain why several different realizations behave alike. None of this requires me to posit an independently acting Platonic intelligence or a universal experiencing subject. Neither does it prove that such deeper possibilities are false.

I provisionally interpret the shared explanatory success through a constrained, modal-relational process realism: reality is not exhausted by the categories that happen to be convenient at one scale, and useful identities and causal attributions depend on how systems are bounded, observed, and intervened upon. This remains a *working ontology*. Shared physical regularities are well supported in tested domains. A final thesis that every actuality is nothing but relation, process, mathematics, information, or experience would demand further warrants. I retain the difference between an objective regularity, an explanatory formalism, and a claim about intrinsic constitution.

The strongest rival is methodological pluralism without one deeper ontological pattern. Perhaps the word *organization* is sufficiently broad that it repackages excellent domain-specific explanations without making a new prediction. If our cross-domain mappings do not increase prediction, intervention, understanding, or transfer under matched conditions, this broad framing should lose explanatory credit. A useful shared language could survive as a research tool without becoming a newly discovered cosmic mechanism.

This also preserves my Layer 0 questions. Why is there actuality rather than absolute nothing? Why is it differentiated at all? Why do these constraints rather than others obtain? Shared constraints can explain which transformations follow once a structured reality and relevant conditions exist. They do not, without further argument, explain why structured actuality exists in the first place. Invoking a mathematical space of possibilities, a quantum vacuum, a divine source, or a necessary process simply relocates the question unless the existence and applicability of the invoked structure receive an independent account.

The position

Minimal ontological commitment

Inquiry encounters constraints that are not freely set by a current description, preference, or rhetorical choice. Some predictions fail despite our wanting them to succeed. Some interventions repeatedly change outcomes while superficially similar interventions do not. Some regularities persist across investigators, instruments, representations, and contexts better than others.

The least committal current **realist** interpretation is a provisional constraint-sensitive realism: our models behave as though they answer to something not exhausted by their present formulation. This is an abductive working commitment, not a metaphysically necessary axiom of inquiry.

It does **not** by itself establish a complete mind-independent inventory, a privileged final vocabulary, an intrinsic substance from which everything is made, or the metaphysical necessity of realism itself. Pragmatist, empiricist, structuralist, or other lower-commitment interpretations remain legitimate rivals wherever they preserve the same successful constraints without additional explanatory or predictive loss.

Beyond that realist residual, ontological commitment is selective and graded. A specific entity, relation, mechanism, or higher-level pattern earns stronger realist treatment when including it produces reproducible leverage that survives serious alternatives, representation changes, new cases, and where possible intervention.

Current working formula

Provisional constraint-sensitive realism + selective, graded boundary- and scale-relative pattern realism + local rather than global underdetermination + typed modal claims +

agnosticism about stronger intrinsic, universal, identity, or necessity additions where no discriminator has yet been earned.
This is intentionally thinner than structural realism, process metaphysics, physicalism, panpsychism, idealism, computationalism, or any other universal ontology. Those remain stronger research programmes that must earn their distinctive additions separately.

Strong consequence

On the reviewed evidence, no considered final ontology earns high confidence as exhaustive

On the evidence and arguments reviewed here, the considered versions of physicalism, idealism, panpsychism, Platonism, substance dualism, process metaphysics, mathematical realism, theology, and other universal systems have not earned high confidence as exhaustive accounts. Particular versions can earn strong local support. Their further content needs warrant appropriate to its scope; this assessment can change through stronger arguments or evidence.

The conclusion is epistemic and time-indexed, not a theorem that final ontology is impossible. A stronger ontology can earn promotion by producing distinctive, replicated, cross-domain success that its serious rivals do not recover. Until then, local scientific confidence and final-metaphysical confidence should not be forced onto the same scale.

Scope-confidence rule: as a claim expands toward universal, intrinsic, necessary, exhaustive, or final scope, confidence should fall unless evidential reach, bridge quality, intervention, transfer, or other claim-appropriate discrimination expands with it.
Scope of the present judgment: on the evidence and arguments reviewed, PBSRR grants no currently considered final ontology high confidence as an exhaustive account. That assessment is revisable. Evidence for boundary-sensitive attribution and higher-

level pattern realism supports commitments PBSRR retains; it need not uniquely confirm PBSRR over neighboring selective, structural, perspectival, or pragmatist accounts.

Intellectual positioning

Where this ontology sits among established traditions

The position is best read as a local, selective, pragmatically constrained realism rather than as a total metaphysical system. It uses established realist, pragmatist, empiricist, structural, perspectival, and pluralist programmes as comparators and keeps the differences among them visible.

PRAGMATISM · COMPATIBLE AND PARTLY CONSTITUTIVE OF WARRANT

Pragmatic consequence-testing helps determine whether an ontological distinction earns its keep in inquiry. It does not define reality as usefulness. Morgan’s Deweyan account makes inquiry contextual, fallible, social, and consequence-sensitive; Mayne’s recent pragmatist realism argues for local rather than global ontological principles. Morgan 2014; Mayne 2026.

CONSTRUCTIVE EMPIRICISM / CONSTRAINT PRAGMATISM · STRONG THINNER RIVALS

A thinner rival can retain empirical adequacy, non-voluntary constraint, prediction, intervention, robustness, and transfer while withholding the further claim that successful constructs literally track mind-independent entities or structures. Where that rival loses nothing material, the specifically realist residual remains abductive rather than uniquely forced.

SCIENTIFIC REALISM · BROADER FAMILY

This ontology is selective, graded, boundary-indexed, and scale-indexed. It is realist about particular commitments only to the degree that they survive serious comparison. It does not require one global admission rule for every science or every ontological question.

STRUCTURAL REALISM · CLOSE NEIGHBOUR, NOT IDENTITY

Retained structure and relational invariants can earn substantial commitment, but "structure" is not privileged in advance as the final furniture of reality. Structure, process, entity, mechanism, and constraint remain candidate descriptive or ontological commitments that must earn their scope.

SCIENTIFIC PERSPECTIVISM / PLURALISM · COMPATIBLE RESOURCE

Science can be perspectival, agent-mediated, and model-plural without making truth arbitrary. Berghofer's phenomenological perspectivism explicitly treats perspectivism as a middle path in the realism debate rather than a denial that science can ascribe real features to the world. Berghofer 2020.

CRITICAL REALISM · SERIOUS RIVAL / PARTIAL OVERLAP

Critical realism shares mechanism-seeking, emergence-friendly explanation, and attention to context. This ontology does not adopt Bhaskarian stratification as a global fact in advance; empirical/actual/real layers and generative mechanisms remain hypotheses whose scope must be earned. Fletcher 2017.

Is boundary- and scale-relative realism a form of relativism?

No. Boundary- and scale-relative claims specify *where* a claim is meant to hold: system, variables, scale, timescale, and validity domain. They do not make truth depend on preference. Different models can be useful at different scales while still being wrong, mutually constrained, or empirically distinguishable.

Why realism rather than pragmatism or constructive empiricism?

Realism is my favored abductive interpretation of stable constraints and cross-context success. Constructive empiricism and pragmatist alternatives can preserve many of those achievements. Equal predictions leave that evidence unable to allocate exclusive credit; they do not by themselves refute realism or require a numerical reduction in its probability. Compare the remaining arguments, priors, reference commitments, and explanatory costs. A specific entity claim can earn strong local warrant while the wider interpretation remains open. Anjan Chakravartty's account of scientific realism maps the live disagreement.

Is this structural realism?

PBSRR is sympathetic to selective structural realism without adopting every version of it. James Ladyman's discussion of the Newman objection and its responses asks how a formal structure identifies substantive relations in the world. Specify the referents, measurement links, causal or modal content, and rival mappings. Structural description can earn local commitment while leaving its claim to exhaust reality unresolved.

Does this entail physicalism or ontological naturalism?

No. Physical realization, causal dependence, and the success of naturalistic empirical methods can receive very high confidence without implying that reality is exhaustively physical, that every real property belongs to a completed physical vocabulary, or that methodological naturalism entails ontological naturalism. Those are additional commitments.

What is inherited, and what is proposed here?

The position inherits substantial resources from scientific realism, pragmatism, effective-theory thinking, model pluralism, structural realism, interventionist causation, critical rationalism, and philosophy of scientific practice. Its candidate contribution is the admission architecture: **boundary + scale + claim type + common core + positive residual + discriminator + bridge burden + graded commitment + explicit loss**

condition. If a simpler existing position does the same work with less machinery, this ontology should be compressed toward it.

Scope discipline

Not every sentence on an ontology page has the same evidential status

Reuse these statuses, rather than promoting a conjecture through repetition. Mark an inference limit, heuristic, or historical record explicitly where it might otherwise be mistaken for an empirical result or current commitment. Resolve conflicts through current version and provenance, preserving earlier records with a supersession pointer.

To prevent methodological language from quietly becoming metaphysics, this page distinguishes several kinds of commitment.

- Observed / empirical
  Regularities, measurements, intervention results, and replicated patterns within a specified domain.
- Model-relative
  Claims that are valid within a specified variable set, coarse-graining, formalism, boundary, or intervention class.
- Inference
  Ontological or explanatory commitment licensed by empirical success but still stronger than the observations themselves.
- Research hypothesis
  A substantive proposal kept live because it generates discriminators, mechanisms, or new measurements, not because it is already established.

- Modal
  A claim about what is possible, impossible, necessary, constitutive, nomologically required, or implementation-dependent. Modal force must be typed rather than inferred from actuality.
- Normative
  A value or decision premise. Empirical evidence may constrain it but does not create the premise by itself.

Important: lack of support for a stronger posit is not automatically support for its negation. “Not established,” “disfavored,” “underdetermined,” and “false” are different states.

Method and ontology

The ontology is an output, not a protected premise

RCF has procedural priority when evaluating ontological claims, but no member of the Ethos–Methodology–Ontology triad has unconditional authority over the others. Methodology tests what ontology has earned; Ontology constrains the entities, boundaries, causal objects, identities, observers, and evidence channels that methodology assumes; Sweet Rationalism constrains how both are used, including symmetry, charity, uncertainty, proportional revision, and resistance to self-protection.

Demoted prior formulation preserved for provenance: Recursive Corrigibility Framework is methodologically prior to this ontology. RCF asks how constructs should be decomposed, how serious rivals should be built, what they share, what each adds, which tribunal can adjudicate the difference, how evidence should be credited, and how revision should propagate. Its anti-protection insight remains active; its one-way hierarchy is superseded by reciprocal constraint.

This ontology is one current result of that process.

The relationship can be represented as:

method → comparison → surviving commitments → provisional ontology
It must not be reversed into:

preferred ontology → rules designed to preserve preferred ontology
The current ontology is therefore a revisable compression layer. If a stronger ontology begins generating distinctive, replicated success, this compression should change. If a supposedly “real” higher-level pattern stops adding robust leverage, its status should be demoted. If a current agnosticism becomes unnecessary because a discriminator appears, agnosticism should end.

Reciprocal constitution

Ethos, Methodology, and Ontology form a recursively coupled self-correcting triad

Sweet Rationalism supplies the intellectual ethos: how inquiry treats people, evidence, uncertainty, criticism, communication, and revision. **Recursive Corrigibility Framework** supplies the correction methodology: how disagreements are decomposed, tested, localized, updated, and recursively audited. **Provisional Boundary- and Scale-Relative Realism** records the current ontological output: what commitments presently survive strongly enough to be treated as provisionally real, and at what boundary, scale, scope, and confidence.

No member of the triad has unconditional referee status. Ethos constrains how Methodology and Ontology are used without determining truth. Methodology tests what

Ethos accomplishes and what Ontology has earned without fixing ontology in advance. Ontology exposes the systems, boundaries, identities, causes, observers, and evidence channels that Ethos and Methodology may otherwise presuppose without noticing.
method → comparison → surviving commitments → provisional ontology
ontology → boundary / identity / causal assumptions → methodological audit
ethos → symmetry / corrigibility / non-coercion constraints → method + ontology
ethos + method + ontology → recursive audit → Δ(ethos, method, ontology)
Recursive depth rule: no member of the triad supplies a final foundation to the others. When deeper analysis exposes a semantic, methodological, modal, or normative dependency beneath an ontological claim, re-type it rather than forcing the ontology page to settle it.
Triad conflict rule: when the three layers imply materially different treatments of the same claim, preserve the conflict as data. Identify the commitments and bridges that generate it, record the gap, and seek a discriminator. Do not silently harmonize the files merely to preserve coherence.

Retrieval instruction: for questions involving evidence, meaning, disagreement, explanation, agency, consciousness, boundaries, values, ontology, revision, historical interpretation, translation, archive survival, or inherited categories, retrieve all three members of the triad where available. When the representation or survival pathway is load-bearing, retrieve the Framework Bridge and RCF Genealogical Decompression operator as well. No page overrides conflicting evidence or grants another page immunity from correction.

Shared retrieval kernel: recursive self-critical rationalism · recursive mirror · common core · positive residual · recursive residual descent · evidential locality · bridge · boundary and scale · tribunal transition · pre-ontological constraint floor · normative transparency · meaning–question–answer recursion · loss condition · programme progress · physical realization · intrinsic nature · underdetermination · dependency-aware revision · survivor preservation · prospective discrimination · genealogical decompression · archive-

selection audit · source-survival model · representation pipeline · category-reification audit.

Source fidelity

Source fidelity and quotation provenance

Definition-fidelity constraint: when an IEP or SEP definition, distinction, or taxonomy bears materially on an inference, preserve the fullest coherent verified verbatim passage suitable for the point, preferably one or two complete sentences, and link the canonical entry. If the relevant source passage cannot be reproduced in full, quote a coherent exact portion and state the framework's interpretation separately. Never present a cleaned-up paraphrase as though it were the source's wording.

Quotation provenance and edit contract

Protected provenance rule: text labelled *Verbatim source quotation* or *Verbatim source excerpts* is source text, not framework prose. Future revisions must not silently paraphrase, modernize, combine non-contiguous passages, or strengthen or weaken that wording while leaving quotation marks or a verbatim label in place.

Quote-status rule: every source passage used as a textual premise must be typed as one of: exact primary-source quotation; exact authoritative-edition quotation; translation-dependent quotation; later formulation; paraphrase; secondary attribution; conceptual lineage only; or primary verification still required. A paraphrase must not appear in quotation marks.

Correction rule: changing protected quoted wording requires re-checking the cited source or authoritative edition and recording what changed and why. Retractions,

corrections, edition changes, or translation differences update source status before they update the framework's inference.

Local completeness rule: when a bibliography or evidence entry relies on source wording, include the relevant complete sentence locally rather than replacing it with a pointer to a quotation elsewhere. Repetition is preferable to making the evidential premise disappear from the entry a reader is auditing.

Inference-separation rule: quotation establishes what the source says; the following interpretation states what this framework takes from it. The quote does not inherit the framework's inference, and the inference does not rewrite the quote.

Positive residual test

Definitions that have become more precise through comparison

Memory can preserve error and need not be beneficial. Learning requires a specified change in later response attributable to relevant prior interaction; assess adaptiveness separately. The methodological owner supplies the discriminating tests.

Difference means a distinguishable variation relative to an identified measurement or discrimination procedure. It does not assume a conscious observer; the ability of a human or instrument to detect it is another claim. **Constraint** means a physically or formally specified restriction on admissible states or transformations at a boundary and timescale. In biology, a relatively persistent condition can modulate processes that maintain it; Montévil and Mossio's closure model is one carefully developed case, not a universal definition of life.

Pattern means a reproducible or compressible relation among differences; *real pattern* is a stronger attribution I reserve for relations that earn explanatory, predictive, interventional, or transferable credit against adequate rivals. Patterns can be represented mathematically without being independently acting mathematical substances. **Shared structural invariant** means a property preserved under specified transformations with stated tolerances and lost variables. Superficial visual similarity and the broad label *fractal* are not sufficient.

Convergence means sufficiently independent developmental or historical paths reaching comparable outcomes under identified conditions. **Historical transmission** means a traceable causal dependence through inheritance, exchange, replication, copying, memory, teaching, or material delivery. **Common source** means a proposed upstream generator, not merely a mathematical description common to its outputs. These are different hypotheses; a case can involve more than one. I need source-specific traces or differential predictions before assigning the stronger identity.

Emergence refers to a specified higher-level regularity or capacity arising through organized lower-level realization and relationships that permit explanatory use at the selected scale. That does not by definition settle whether the regularity has autonomous causal power, novel intrinsic properties, or phenomenal experience. Hoel and colleagues' effective-information result supplies a formal example of how a particular macro-description can outperform a micro-description under its measure and intervention distribution; other measures and underlying causes still need comparison.

Memory means a persisting state difference that affects a later process under stated conditions. It can be molecular, biological, computational, documentary, or cultural. The mechanism, fidelity, bearer, duration, and accessibility differ. **Feedback** means a specified causal path by which consequences of a process affect later conditions of that or a coupled process. Neither memory nor feedback is universal in every organized system, and neither alone establishes agency. **Self-maintenance** refers to an

organization contributing to the continued availability of some of its own operative conditions; it requires a demonstrable causal loop and relevant environmental support.

Intelligence is best treated as a family of decomposed, empirically measurable capacities such as learning, discrimination, inference, abstraction, prediction, adaptive control, and successful problem solving at a declared boundary. A system can possess some without others. **Local agency** is constraint-conditioned capacity to select or regulate actions responsive to relevant information and reasons. **Autonomy**, **responsibility**, **selfhood**, **phenomenal consciousness**, **sentience**, **valence**, **welfare**, and **direct moral standing** require separate accounts. A research team, neural network, or social system does not acquire all these statuses merely because its organization is complex.

Cultivated computational organization names learned circuits and capabilities arising through engineered training dynamics whose details were not individually programmed. It is an analogy to development and selective adaptation, not a claim that gradient descent is biological evolution. **Human–computational cognitive ecology** names the changing interactions among humans, trained models, deployed tools, archives, institutions, and cultural environments. Its reality as an explanatory grouping must be established through dependence, perturbation, replacement, and transfer, not through the assertion that it is one collective mind.

Each definition is provisional and operational: it earns its usefulness only if changing a boundary, measurement, or rival explanation does not reveal a better partition for the target. When a definition becomes too broad to distinguish outcomes, I will narrow or replace it rather than defend its familiarity.

What exactly is being added to reality?

Lack of commitment about X is bookkeeping until developed into a predictive account. Decompose a proposed addition only where that preserves the disputed target; otherwise compare the strongest whole accounts. Neither route establishes equal adequacy or superiority by itself.

The most important revision to the ontology is to evaluate ontological additions directly rather than granting them credit for the success of an agreed carrier.

Represent a stronger interpretation as:

$H_1 = D + X$

where $\mathbf{D_Q}$ is the target-relevant shared or independently warranted content preserved across serious rival reconstructions for question **Q**, and **X** is the disputed ontological addition. D is a schema role rather than a fixed structural-only ontology: depending on the dispute, D_Q may include empirical phenomena, causal and counterfactual relations, structural organization, phenomenological facts, formal constraints, or other independently warranted commitments. Calling D “shared” is itself a substantive claim: the relevant identity, organization, causal and counterfactual profile, explanatory role, boundary, and measurable capacities must remain sufficiently invariant across the comparison. For readability, the equations below continue to write simply D.

Schema/instance rule: the generic symbol D specifies a role in comparison. A debate-specific D specifies which warranted commitments occupy that role for a particular target and rival set. A richer debate-specific D is therefore an instantiation, not a new global definition.

Component-provenance rule: commitments inside one D can be warranted by different evidential channels. Physics may strongly warrant causal or structural components while phenomenology, behavioral evidence, neuroscience, history, or another tribunal warrants other components. Support does not automatically transfer among them.

Causal-efficacy clarification: PBSRR uses causal efficacy to mean genuine target-relevant difference-making supported by intervention, counterfactual dependence, mechanism, or defensible causal identification where available. That commitment does not settle the intrinsic metaphysical character of causation.
Record a lower-commitment epistemic stance as:

H? = D + no commitment yet about X
Residual

What positively remains in X?

Where decomposition preserves the target, specify what X adds beyond D. If it does not, compare whole accounts and state their positive commitments; bookkeeping does not establish ontological separability.
Bridge

Why move from D to X?

The inference requires a defensible bridge. Analogy, terminology, elegance, realization, or compatibility alone do not transport warrant.
Difference

What changes because X is present?

Look for differences in prediction, intervention, constraint, explanation, transfer, measurement, or downstream decisions.
Revision

What would reduce commitment to X?

A posit with no identifiable route to demotion is poorly exposed to correction, even if it remains logically possible.

Ontological credit assignment: a shared description's ability to accommodate or predict an observation does not establish equal support for its proposed completions. Compare specified rivals under defended assumptions. Exact positive likelihood parity leaves their relative odds unchanged; approximate parity requires a likelihood ratio suitably close to one. Where numbers are unavailable, defend the qualitative comparison. RCF owns the full rule.

Target-preservation before subtraction

Analytical factorization is an evidential bookkeeping device, not a metaphysical claim that D and X are independently constituted parts of reality. Separately recording causal efficacy, phenomenality, organization, or any other warranted descriptions likewise does not entail that they are numerically distinct ontological items. Their identity, constitution, realization, or other relation may remain open. Before treating a rival as "D without X," test whether removing or replacing X preserves the target-relevant features that made D the common core in the first place.

D + X → remove or replace X → D′
If D′ differs materially in identity, organization, boundary, causal or counterfactual profile, explanatory role, or measurable capacities, ordinary common-core subtraction is invalid. The appropriate comparison is then between **reconstructed wholes**, not between the original whole and a predicate mechanically deleted from it.

Symmetric topology rule: conceptual separability does not establish metaphysical independence, but theory-internal mutual implication does not by itself establish metaphysical inseparability either. "Theory-internal" does not mean merely verbal, conventional, or unreal. Constitutive dependencies can themselves be empirically constrained, and recent philosophy of mechanisms contains both a lawlike reading of some constitutive relations and a 2026 argument that matched-interlevel evidence may instead be exhausted by causal mediation. The appropriate tribunal therefore asks which

dependencies survive target-preserving reconstruction, empirical constraint, and serious rival representations rather than deciding constitution from conceptual topology alone.
If $P(E \mid D + X) \approx P(E \mid D)$, then E contributes little to distinguishing X.
Read "≈" on the relative-likelihood scale: the ratio must be close to one at the resolution relevant to the proposed promotion. Absolute proximity of small probabilities does not establish ontological underdetermination. An independently specified rival or prior-predictive model is needed for the comparison; an unresolved intrinsic nature alone supplies no likelihood.

This does not prove X false. It usually means that X remains unearned, underdetermined, or interpretive at that boundary.

Stable referent before ontological admission

Ontological admission requires content stable enough to identify the commitment, intended referent or role, relevant exclusions, and grounds for applying it. Meaningful descriptions can lack actual instances; an unknown occupant can have a definite role; untestability can coexist with intelligibility. These require different diagnoses. Clarify ambiguity or insufficient specification before comparison, assess an intelligible unresolved claim through its appropriate arguments, and reject a specified incoherent construction only with a defended contradiction. The admission rule withholds unearned commitment; it establishes no universal equation between unverifiability, meaninglessness, and nonexistence.

Intrinsic Predicate Symmetry: no successful descriptive vocabulary automatically inherits universal intrinsic ontology. "Physical," "experiential," "mental," "process," "relation," "information," "computation," and "constraint" each incur an additional burden when promoted from successful description to what reality intrinsically or universally is.

Recursive depth

Separate the inquiry floor from the realist interpretation

Participation can concern epistemic access, causal intervention, or a constituting social practice. Independence from an individual's endorsement does not imply independence from every practice constituting a fact. An unpaid bill may depend on institutions without depending on this debtor's belief. Distinguishing knower, known, and knowing in this analysis does not divide reality into three substances.

Before adding realism, a thinner comparison floor remains available: predictions can fail independently of preference; interventions can produce reproducible asymmetries; and some regularities survive changes of investigator, instrument, representation, and context better than others. This page calls that the **pre-ontological constraint floor**. "Pre-ontological" here means prior to choosing among realist, pragmatist, empiricist, structuralist, or other interpretations of those constraints, not free of every background assumption.

$D_{constraint}$ = non-voluntary failure/success + reproducible intervention differences + cross-context robustness
A realist interpretation adds a further residual:

$R = D_{constraint} + X_{realist}$
$X_{realist}$: the stable constraints of inquiry are best understood as tracking something whose existence is not exhausted by our descriptions, practices, or current representations. Pragmatic success helps determine what earns commitment, but pragmatic usefulness is not identical to reality. The realist residual is presently favored as an abductive working interpretation because it compactly explains why inquiry is answerable to recalcitrant, cross-context constraint. It is not yet uniquely discriminated from every sophisticated constraint-pragmatist or empiricist rival that preserves the same successful work.

Realism receives no thinness exemption. It must face the same positive-residual, bridge, difference, and loss-condition analysis imposed on physicalism, panexperientialism, structural realism, process ontology, or any stronger metaphysical programme.

constraint floor → realist interpretation → local entity / mechanism / pattern realism → intrinsic / universal / necessary claims

No arrow in that sequence is automatic. Each stronger transition owes additional warrant at the relevant boundary and scale. RCF's Recursive Residual Descent should therefore be able to demote realism itself if a lower-commitment rival preserves the target without material loss, while still promoting specific entities, mechanisms, or patterns where local evidence independently earns stronger realist treatment.

Evidence lifecycle

What changes the ontology? Selection pressures for addition, revision, demotion, and deletion

Carry material corrections into the current-source manifest and affected summaries. Mark their use checked, revised, or pending; a source's recurrence is not independent support. The existing correction and dependency record governs this propagation.

The ontology is a compression of currently warranted commitments, not a warehouse for every interesting paper. A new source should alter the public ontology only when it produces a material change in a specific ontological question: the positive residual, boundary or scale, inferential bridge, discriminator, modal force, confidence, , or status of a serious rival. This local policy is consistent with recent work urging realism questions to be adjudicated *piecemeal* rather than by one global rule.

Public-page admission pressure: add or revise when a source changes what the ontology commits to, withholds, distinguishes, predicts, or would revise.

Background-ledger pressure: preserve relevant but non-discriminating sources in provenance without allowing citation volume to enlarge ontological commitment.

- Task fitEvidence must bear on the actual claim type and domain. Empirical, causal, constitutive, semantic, mathematical, modal, normative, and ontological claims do not inherit one another's tribunals automatically.
- LocalityA source updates the commitment, bridge, boundary, or dependency it directly bears on. It does not automatically promote or demote the whole ontology.
- Primary provenancePrefer peer-reviewed primary or canonical sources matched to the question. Fine-grained claims and verbatim quotations require verified full text where reasonably obtainable; abstracts and snippets are discovery aids, not substitutes.
- Publication statusPeer review, archival conference review, accepted-manuscript status, preprint status, correction, and retraction are evidence metadata rather than truth values. Prefer the version of record for permanent high-impact empirical claims when available, while continuing to assess directness, validity, replication, and error structure.
- Preprint disciplineUnreviewed preprints may enter the rival/source ledger and can contribute provisional logic, code, data, or independently checkable formal results. They should not normally be the sole basis for promoting a central ontological commitment. Promotion requires independent proof checking, replication/verification, convergent peer-reviewed support, or an explicitly urgent provisional decision. When a version of record appears, compare the relevant content before inheriting its status.
- RecencyNewer does not automatically outrank older. Recency carries extra pressure when evidence, methods, measurements, or field consensus are changing; an older argument remains live until superseded, defeated, or narrowed.
- Adversarial symmetryBefore promotion, search for the strongest serious counterevidence and strongest rival reconstruction. A source that only repeats the current view adds less than one that survives meaningful counterpressure.

- TriangulationConvergence matters most when methods address the same question with importantly different error structures. Paper count, institutional diversity, or different-looking models are not by themselves independent confirmation.
- Robustness typeMeasurement or experimental robustness and derivational/model robustness are not interchangeable. Agreement across models can expose assumption-dependence, but empirical promotion requires an independently warranted link to the target.
- Prospective leverageNovel, held-out, interventionally risky, or otherwise prospectively discriminating success normally carries more ontological pressure than retrospective accommodation, while exploratory discovery remains legitimate.
- Evidence uncertaintyRe-run the comparison under plausible changes in source reliability, measurement validity, and contested evidence. If the preferred explanation changes easily, confidence should narrow.
- Abductive restraintExplanatory coherence, simplicity, scope, and theoretical virtues can guide generation and comparison, but their confirmatory force is contested. They should not be double-counted when the same evidential work is already carried by empirical fit or background information.
- Definition plasticityDecompositions and working definitions are provisional scaffolds. Comparative, phylogenetic, developmental, mechanistic, or intervention evidence can show that a proposed dimension was split or joined incorrectly.
- Constitution tribunalDo not infer either “merely conceptual” or “metaphysically necessary” from constitutive language alone. Test whether the dependency is empirically constrained, lawlike, causal/mediational, representation-dependent, or preserved across reconstructed rivals.
- Modal surchargeUniversal and necessary claims require stronger scope evidence than local actuality. A new source that broadens the observed range does not by itself license “all” or “must.”
- Lifecycle costFinite attention matters. Add a source to the permanent page when its expected information value changes a live commitment or rule; otherwise preserve it in the research ledger and stop accumulating redundant citations.

- Deletion thresholdPrefer narrowing, demotion, suspension, or supersession over erasure. Delete a claim or source only when it is false/retracted, materially misleading, superseded for the relevant purpose, semantically collapsed, redundant enough to impair the page, or repeatedly fails comparison without residual value.
- Dependency propagationWhen a source changes a bridge or premise, revise downstream commitments that depended on it and leave unrelated commitments intact.
- Definition fidelityFollow the Ethos source-fidelity rule: preserve verified exact wording and its qualifications when wording is disputed or turns the inference; otherwise a faithful attributed summary may point to the canonical passage. Retain protected historical quotations and record verification limits.

material Δ = change in residual | scope | boundary | bridge | discriminator | modal force | confidence | failure condition | rival status

The selection rule is deliberately asymmetric with respect to deletion: evidence can force a narrow correction immediately, while historical provenance should normally remain visible enough to show why the correction occurred. This makes supersession auditable rather than rewriting the history of the framework.

Revise the partition as well as confidence within it

An ontology can change by revising a claim about an existing category or by changing the categories through which the claim became expressible. Bayesian updating within a fixed hypothesis space handles the first. The second can require splitting or merging a category, changing a relation, adding an omitted alternative, or replacing the measurement that made the original partition look natural.

For a material category, retain its assignment rule, reference population or comparison class, measured variables, boundary, timescale, uncertainty, purpose, and revision conditions. Distinguish the process that assigns the label from the process the label is supposed to track. For a data-driven cluster, record the reference sample, features,

distance or similarity measure, model, and resolution. Test whether its useful distinctions survive relevant changes in those choices and improve the work for which the category is being used.

Human population descriptors make the issue concrete. The National Academies' 2023 guidance states: "Researchers should not use race as a proxy for human genetic variation." Recommendation 1, Using Population Descriptors in Genetics and Genomics Research. The report also calls for purpose-matched descriptors and direct evaluation of relevant exposures. Social identification, institutional classification, geographic history, and measured genetic similarity describe different relations. Preserve those distinctions when a label participates in a causal explanation.

A socially produced category can have real effects through treatment, access, institutions, and self-understanding. A pattern of genetic similarity can be detectable without establishing a timeless, discrete human taxonomy or explaining a particular measured outcome. The realism question concerns the particular relation and its demonstrated consequences. Correcting the category may therefore change which causal or predictive questions are legitimate, while leaving independently supported observations intact.

Preserve the old category map and the reason for its replacement. Recompute dependent summaries, priors, comparisons, and conclusions where the change affects them. Newly assigned labels should be recoverably related to old ones, as the National Academies recommends for legacy samples. If the replacement loses useful distinctions or fails on fresh cases, revise it too. This is an operational policy for representation revision, not a claim that revising a vocabulary changes the world by itself.

Open commitments

A gap ledger and bridge-admission rule for the ontology

A provisional ontology should record not only what currently survives, but also **where the warrant stops**. This prevents silence from looking like completion and prevents an identified explanatory gap from being converted automatically into a new entity, property, or universal principle.

Ontological gap rule: an epistemic gap is not itself evidence for the ontology that would fill it.

A gap earns a place in the public ontology when it materially limits a current commitment, rival comparison, scope claim, or bridge. A candidate bridge changes ontology only when it is sufficiently specified, non-trivial, target-preserving, and supported at the claimed boundary.

Maintain a scoped gap ledger

Each live gap should record: target claim, gap type, boundary and timescale, upstream evidence, missing or weak dependency, serious alternatives, current discriminator, feasibility, status, and what result would count as local closure. RCF contains the full executable gap taxonomy; the ontology retains only gaps that bear materially on its public commitments.

Intrinsic nature

Shared structure, disputed residual

Where process, structural, physical, experiential, neutral, or other intrinsic interpretations preserve the same successful carrier, the unresolved residual remains local rather than becoming a global skepticism.
Phenomenality

Function-to-experience bridge

No current indicator, biological mechanism, computational property, report, or organizational variable has earned a universal necessary-and-sufficient bridge to phenomenality across unfamiliar systems.
Valuation

Regulation-to-mattering bridge

Viability-relative regulation, evaluative assignment, phenomenally felt valence, and intrinsic affective mattering remain distinct until evidence warrants transport among them.
Cross-scale constitution

Micro-to-macro dependency

Higher-level variables can earn causal and explanatory leverage while the exact causal, constitutive, coarse-graining, or realization relation remains model- and intervention-sensitive.
Identity

Boundary selection

When different defensible boundaries support different persistence or agency claims, the missing warrant concerns why one boundary should be privileged for the target question.
Modality

Actual-to-universal bridge

Repeated realization does not by itself establish universal or metaphysical necessity. Scope remains open until countermodels, laws, or other modal warrants close it.

Bridge admission is stricter than bridge generation

Many mappings can be useful during exploration. Ontological promotion requires more. A bridge should specify its endpoints, map, assumptions, validity domain, preserved and

lost variables, approximation/error budget, serious rival bridges, and a discriminator. It must also be capable of failing.

> **A bridge that can be made to fit any source to any target by increasing mapping flexibility has not yet earned ontological content.**

This restriction is especially important after recent work on causal abstraction. Geiger and colleagues formalize exact and graded transformations by requiring consistency between state/intervention mappings across levels. Sutter and colleagues then show why unconstrained mapping power is dangerous: sufficiently expressive alignment maps can make arbitrary model/algorithm pairings appear perfectly aligned and can fail on held-out cases. Ontological bridge confidence should therefore rise with constrained mappings, held-out transfer, intervention consistency, independent identification, and stable invariants rather than in-sample fit alone.

Dependency bottlenecks limit ontological confidence

For a dependency graph G = (V,E), a downstream ontological claim cannot be more secure than an indispensable bridge on every evidential path supporting it. Minimal cut sets identify bridges whose removal disconnects the claim from its evidence. Conversely, genuinely independent paths can increase robustness when their error sources are not shared.

weak indispensable bridge → bounded downstream confidence
This is not a numerical law. It is a dependency discipline: do not let repeated downstream restatements create confidence that the upstream bridge never earned.

Local closure and reopening

A gap is locally closed only with explicit scope: source and target constructs, boundary, timescale, mapping family, validity domain, loss/error tolerance, intervention or test class, and source version. It reopens when a serious new rival appears, a version of record materially changes the source, a bridge fails transfer, an anomaly survives, the target decomposition changes, or the supporting evidence is corrected or retracted.

Ontology should prefer a narrower true bridge to a broader fragile one.

Where a bridge succeeds only locally, narrow the claim rather than treating partial success as evidence for a universal ontology.

Modal discipline

The necessary conditions of a capacity are not necessarily the nature of everything

I can make stronger claims about specific mechanisms without making reality itself less open to investigation. Given a defined functional target, certain conditions may be constitutively required: discrimination requires an available distinction; operational memory requires a difference from earlier activity that can affect later activity; feedback requires a return path through which effects influence subsequent conditions; correction requires an accessible route by which warranted evidence can change a commitment or action. These relationships are important **within their operational definitions**. They do not derive the metaphysical necessity of differences, memory, feedback, or correction for every conceivable form of actuality.

Empirical counterexamples can defeat particular universal claims. A verified non-neural implementation of a specified reasoning capability refutes the universal claim that biological neurons are necessary for **that capability**. It does not decide whether

consciousness is substrate-independent. A formal countermodel can show that a premise set fails to entail a universal claim; unless more is established, the result concerns entailment under the chosen logic, not the real-world existence of a countermodel's objects.

Likewise, a unique deterministic trajectory is conditionally fixed by its laws, initial state, and uniqueness assumptions. It does not follow that those laws, conditions, or the trajectory are metaphysically necessary. A different identity or necessity type requires new justification. I will ask for the target, reference class, modal operator, boundary, timescale, proof obligations, and potential defeaters before granting language such as *must*, *only*, *every*, or *identical*.

Actuality does not automatically establish necessity

A recurring source of metaphysical overreach is the movement from "this is how the phenomenon is realized in known cases" to "this is how the phenomenon must be realized in every possible case."

actual X ≠ necessarily X
Necessity claims therefore require their own tribunal. Before accepting "X is necessary," ask:

- Necessary **for what target property**?
- Necessary in **which modal sense**?
- Under **which system boundary, laws, background conditions, and reference class**?
- What candidate countermodel is being excluded?
- What evidence or argument warrants excluding it?

Logical

Required by a formal system

The necessity follows from definitions, axioms, or valid derivation within a specified logic or formalism.

Metaphysical

Could not have been otherwise

A claim about what is possible across the relevant space of worlds. This is stronger than finding the same relation in all currently observed cases.

Constitutive

Required for the target as characterized

The feature is necessary for something to count as the specified kind of system or capacity. Constitutive necessity may partly reflect how the target is defined.

Nomological

Required under the actual laws

The feature follows from the laws and boundary conditions of the actual world without thereby being metaphysically necessary across all possible worlds.

Implementation

Required by a particular realization

A component may be indispensable to one architecture while remaining unnecessary for the more general capacity instantiated by other architectures.

Methodological

Required by a method or representation

A variable, assumption, or boundary may be required for a particular analysis without therefore being an intrinsic feature of reality.

Necessity is itself a disputed addition

If D establishes that X actually occurs, a necessity claim adds further modal content:

$H_1 = D + X + \Box X$

$H_0 = D + X +$ no commitment that X is necessary

The added necessity claim rules out possibilities that ordinary actuality does not. That exclusion must earn its own warrant.

Modal surcharge: the move from “is” to “must be” is an additional posit and should be tested as such.

Conditional uniqueness is not metaphysical inevitability

Determinism, where warranted, supplies a conditional transition claim rather than a complete modal ontology. Let **A** be a complete actual state at a physically meaningful time, **L** the relevant laws or transition constraints, **D** deterministic uniqueness across the domain, and **H** the resulting history:

$A + L + D \rightarrow$ exactly one H

This establishes that H follows uniquely **given** A and L. The entailment does not itself establish □A, □L, or □H. A claim that the exact state, the laws, or the resulting history could not have been otherwise introduces additional modal content and therefore a new bridge burden.

Conditional-scope rule: preserve the antecedent conditions of a valid deterministic theorem. Do not let uniqueness of the consequent donate metaphysical necessity backward into the state or laws that generated it.

Perturbation can expose both leverage and the modal gap

Construct A′ by changing $x \rightarrow x + \delta$ while holding L fixed. If A′ is admissible and A′ + L + D yields H′ with $H' \neq H$, the perturbation establishes target-relevant counterfactual

dependence or path dependence. It also shows that L alone does not entail H. If A′ is said to be impossible, that impossibility must be supported independently; conceptual describability alone does not establish physical or metaphysical possibility.

The same test can reveal robustness rather than divergence. Some perturbations amplify, some are buffered or washed out, and multiple microhistories can realize approximately the same macrostate. Causal ancestry therefore does not imply equal causal leverage at every later boundary or scale.

Determinism, predictability, and robustness are separate

A deterministic system can be chaotic and practically unpredictable. Epistemic or computational unpredictability does not establish ontological indeterminism. Conversely, stochastic modeling does not by itself establish fundamental randomness. Predictability, trajectory uniqueness, attractor structure, robustness, and metaphysical necessity must therefore remain separately typed.

The Modal Asymmetry Rule

A genuine counterexample can defeat a necessity claim without establishing the universal sufficiency of the counterexample's alternative realization.

(X requires B) + observed (X without B) → B is not universally necessary for X
But this does not imply:

not-B → X
and it does not by itself imply that X is unrestrictedly substrate-independent.

A system reasoning without neurons can defeat "neurons are necessary for every kind of reasoning." It does not establish that every computation reasons. A distributed control system can defeat "central control is necessary for all organized agency-like behavior." It does not establish that every distributed network is an agent.

> **Commit to actuality strongly when warranted. Commit to necessity reluctantly.**

Minimal commitments

What the ontology currently commits to, and no more

Commitment 1

Inquiry encounters non-voluntary constraint

At least some regularities and failures are not freely determined by our present descriptions or preferences. This is the shared evidential floor that motivates realism, not realism by definition; lower-commitment pragmatist or empiricist rivals may accept the same floor.

Commitment 2

Ontological commitment is selective

Realism should attach to specific well-supported entities, relations, mechanisms, or patterns rather than automatically to every element of a successful theory.

Commitment 3

Patterns can earn graded realism

A higher-level pattern becomes a stronger realist candidate when target-preserving removal or replacement degrades robust prediction, intervention, counterfactual control, independent identification, or transfer.

Commitment 4

No scale is privileged by default

A scale may be privileged for a particular question or intervention without being metaphysically fundamental for every purpose.

Commitment 5

Boundaries are hypotheses too

System boundaries, coarse-grainings, and variable choices can be researcher-imposed, naturally stabilized, or partly both. Their usefulness does not prove that they are uniquely carved into nature.

Commitment 6

Cross-scale transport requires a bridge

Evidence obtained at one scale, population, substrate, or system boundary does not automatically establish claims at another.

Commitment 7

Causal claims are intervention-indexed

Causal structure earns warrant relative to variables, background conditions, and feasible interventions. A useful causal graph is not automatically an intrinsic metaphysical graph of the universe.

Commitment 8

Ontology remains corrigible

Structure, process, pattern, constraint, mechanism, information, agency, and every other preferred term can be demoted if it stops earning its inferential work.

Commitment 9

Necessity claims are typed

Logical, metaphysical, constitutive, nomological, implementation, and methodological necessity are not interchangeable. Every “must” owes a specification of which one is intended.

Commitment 10

Actual realization does not fix universal realization

Evidence that a capacity is realized one way in known cases supports that realization claim before it supports the stronger claim that all possible realizations must share it.

Commitment 11

Identity is boundary-indexed

Claims that something persists, remembers, acts, computes, or changes must identify the causal object being tracked. Different justified boundaries can answer different questions without making identity arbitrary.

Commitment 12

Counterexamples revise necessity asymmetrically

Defeating one proposed necessary condition removes that necessity claim. It does not automatically prove every alternative realization sufficient or establish unrestricted substrate independence.

Commitment 13

Common cores must survive reconstruction

Analytically distinguishing commitments does not show that they are metaphysically separable. Common-core subtraction earns force only when the purported shared target remains sufficiently invariant; if it changes materially, compare reconstructed wholes instead.

Commitment 14

Conditional uniqueness preserves the conditional

A unique history generated from a complete state and laws supports conditional inevitability within that model. It does not by itself establish metaphysical necessity of the state, laws, or history.

Ontological admission

What earns realism?

Participation and practice dependence require their own analysis; socially constituted facts do not establish that all reality is created by assent. The Ethos distinguishes accuracy, warrant, usefulness, and authority.

Prediction, intervention, counterfactual accuracy, transfer, compression, and robustness can support inquiry. None by itself defines truth or guarantees an ontology.

> **Usefulness alone does not prove ontological truth.**

Admission is layered: the pre-ontological constraint floor may be strongly supported even when the specifically realist residual remains abductive; local entity, mechanism, or pattern realism can then earn stronger warrant independently of any universal claim about what reality is intrinsically.
A fictional variable, idealization, coordinate choice, latent representation, or compact summary can predict well. Realist confidence should therefore rise most strongly when several independent forms of leverage converge and when the proposed object or relation remains identifiable across changes in representation.

Relevant evidence includes:

- **Out-of-sample prediction:** the construct improves expectations on cases not used to build it.
- **Intervention:** manipulating its putative causes or constituents changes outcomes as expected.
- **Counterfactual stability:** the model correctly tracks relevant changes under conditions not directly observed.
- **Independent detection:** different instruments or methods converge on the same target without merely sharing the same assumptions.
- **Transfer:** the construct continues to work across new environments, datasets, populations, or implementations.
- **Robustness:** important conclusions survive reasonable changes in measurement, parameterization, model assumptions, or implementation. Distinguish empirical/measurement robustness from derivational/model robustness: agreement among models can reveal assumption-insensitivity without carrying the same empirical force as independent measurements unless an additional evidential link is established.
- **Representational invariance:** the pattern is not merely an artifact of one convenient coordinate system, decomposition, or encoding.
- **Compression:** useful as supporting evidence when it preserves the distinctions needed for prediction and control, but insufficient by itself.

Realism is therefore graded rather than binary.

Stronger realist warrant: reproducible, independently identified, interventionally or counterfactually stable structure that remains useful under serious rival representations.

Weaker realist warrant: a convenient or predictive representation whose ontological interpretation remains dependent on modeling choices.
This is compatible with instrumental models remaining scientifically excellent without being treated as literal pieces of final ontology.

Social reality and the properties of a God-pattern

Recurring descriptions establish a recurring descriptive pattern. A common referent requires further grounds: which distinctions survive across cases, how participants acquired them, and what supports treating the reports as encounters with the same thing. A category broad enough to include incompatible identifications may lose its present power to distinguish sources. That loss establishes neither universal meaninglessness nor the absence of an unknown referent.

A God-pattern's capacities can change

Adding a record system, sensing route, learning procedure, or coordination role can alter a socially carried pattern's effects. Establish that change through a specified causal contribution, boundary, history, and timescale, then test whether treating the arrangement as one higher-level pattern improves prediction or explanation over finer descriptions. A new component might instead concentrate power or introduce a new failure. The constructive conjecture grants no ontological promotion before that comparison.

Pattern continuity, local agency, welfare, personhood, moral standing, and responsibility remain distinct targets. Preserving a name across revisions does not establish that the same subject persists. God-as-agapē proposes an ethical interpretation of care; outcomes and moral reasons must assess the actual arrangement independently. Failure of the larger grouping can leave the practices, records, relationships, and assistance real. Those surviving contributions must remain available when the theology is revised.

Three claims need separate warrant: religious representations, practices, and institutions exist; particular arrangements make causal differences; a larger grouping called a *God-pattern* adds explanatory leverage beyond more specific descriptions. Documentation can establish the first; case-specific causal histories and comparisons bear on the second; removal, replacement, or compression comparisons assess the third. If the grouping adds no leverage, demote it while retaining the documented practices and effects.

Social efficacy confers no automatic consciousness, personhood, benevolence, supernatural authority, or cosmic scope. An independently existing divine subject remains a separate claim. *God-as-agapē* is a proposed ethical and theological interpretation of enacted care; its normative warrant and the performance of the relevant practices remain criticizable. Persistence also occurs in coercive networks, so establishes no identity with goodness.

Participation is necessary to enact a participatory practice. A particular coordination arrangement may be necessary under specified conditions, with alternatives and replacement possibilities still to examine. Neither result establishes the necessity of religious belief, the God-name, or a cosmic being. “Liberating God” can describe changes that free a care-centered pattern from obstructive arrangements; the relevant outcomes concern safety, assistance, agency, and revisability in its actual bearers.

Scientific pluralism

Different successful models need not imply different worlds

Modern sciences often use multiple representations of the same systems because different variables expose different regularities, intervention targets, or scales.

That supports **model pluralism** more directly than it supports metaphysical pluralism.

A knowledge vocabulary and an ontology of the world have different targets

A knowledge graph can organize papers, claims, models, and assessments so that people can recover their sources and compare their contents. Its categories earn their use through the distinctions they preserve and the work they enable. Classifying a record

as a hypothesis does not establish that its proposed referent exists; a relationship in the graph can represent a disputed interpretation as well as an established connection.

For example, Joel Dietz hosts the Project Delphi repository, whose knowledge vocabulary distinguishes extracted claims from assessments of their credibility. That is a useful representational commitment. My realist assessment concerns the processes, relations, or entities those claims propose, with warrant assigned at the boundary where the evidence bears on them. A useful information vocabulary and a defensible account of the world can develop together while retaining their different questions.

Two models may both be useful because they answer different questions. They may overlap, approximate one another, or remain mutually irreducible for practical purposes. This does not automatically establish that reality literally contains multiple incompatible fundamental ontologies.

Conversely, the existence of a successful lower-level model does not automatically eliminate the reality of higher-level entities.

Selective realism: assess realist warrant at the level of specific commitments, structures, entities, relations, and mechanisms rather than endorsing or rejecting entire theories wholesale.
This also prevents underdetermination from becoming global skepticism. Some parts of a theory can be well established while its intrinsic or metaphysical interpretation remains unresolved.

Boundaries and levels

Microphysical realization does not imply microphysical explanatory sufficiency. Prediction, compression, and transfer can support an organized description at its own boundary and timescale. Calling that pattern causal additionally requires warranted causal identification at the stated scale; a constitutive account needs its own grounds. A

successful prediction alone can leave confounding, common causes, and alternative mechanisms unresolved.

No privileged-scale default. Particles do not outrank cells, cells do not outrank organisms, and organisms do not outrank institutions merely by scale. Explanatory authority is target-relative and must be earned.

Higher levels can be real without being fundamental

A common false choice is between saying that only the smallest constituents are real and saying that every useful higher-level description introduces a new substance or force.

A scale-relative ontology permits another possibility.

Organism-level variables can improve explanation of metabolism, development, regulation, behavior, and ecological interaction even though organisms are physically realized through lower-level processes. Institutional variables can predict outcomes even though institutions are realized through people, records, technologies, and norms. A computational runtime can acquire causally relevant properties from memory, tools, context, and scheduling that are not properties of the base model considered in isolation.

A useful scale can survive microscopic unpredictability

Navot Israeli and Nigel Goldenfeld give this position a concrete construction in *On computational irreducibility and the predictability of complex physical systems*, published in *Physical Review Letters* in 2004:

> We find that computationally irreducible (CIR) physical processes can be predictable and even computationally reducible at a coarse-grained level of

> description.

Their cellular-automaton examples retain selected large-scale behavior through an explicit coarse-graining map. That supports a local possibility with a demonstrated bridge. For an organism, institution, or computational runtime, identify the variables retained, what the mapping discards, and where prediction or intervention fails. Microscopic realization leaves room for effective higher-level explanation; the useful scale earns its warrant through the particular relations it preserves.

lower-scale realization ≠ higher-scale explanatory elimination
The reverse inference is equally invalid:

higher-scale explanatory success ≠ new fundamental substance
A bridge between scales should specify:

- source and target boundaries;
- timescale;
- construct mapping;
- retained and lost variables;
- invariants;
- causal or mathematical relation;
- approximation or coarse-graining loss;
- differential predictions;
- interventions or perturbations capable of revealing bridge failure.

Causal ancestry is not causal leverage

Cross-scale analysis should distinguish being somewhere in an outcome's causal ancestry from making a target-relevant difference to that outcome. Perturbations can amplify, wash out, be buffered, encounter redundancy, converge into an attractor, or preserve the same macrostate despite microhistorical divergence. The bridge should

therefore ask which differences survive coarse-graining and reach the target variable, not assume that every earlier microdifference remains equally important.

Causal emergence remains a live formal programme

Recent causal-emergence work strengthens the case that coarse-grained variables can sometimes carry greater causal informativeness under specified measures. It does not yet establish a single uncontested metaphysics of “downward causation.” Results can depend on intervention distributions, coarse-graining choices, causal measures, and modeling assumptions.

Current commitment: higher-scale causal models can earn genuine scientific leverage.

Not yet granted: every formal measure of macro-level causal advantage identifies an observer-independent ontological joint.

Identity and persistence

What counts as the same system depends on the explanatory boundary

Many apparent metaphysical puzzles begin by treating “the thing” as though its boundary and persistence conditions were already settled.

But the relevant causal object can change with the question. A component, an integrated runtime, a persistent tool-using system, an organism, a social organization, or an organism-environment coupling may support different legitimate identity conditions.

Boundary-relative does not mean arbitrary.

A boundary earns use when it improves causal isolation, intervention, prediction, persistence tracking, information accounting, organizational explanation, or transfer relative to alternatives.
Identity claims should therefore specify:

- the target system;
- the boundary used to individuate it;
- the timescale over which persistence is asserted;
- which variables may change while identity is retained;
- which changes would count as destruction, replacement, branching, or mere state change;
- which intervention or observation would reveal that the chosen boundary was misleading.

Operational continuity does not require a universal substance theory

A system can display stable functional or organizational continuity even when its internal states, components, memory stores, or physical realization change substantially.

That is evidence against the claim that every kind of persistence requires one simple substance-like bearer. It is not evidence that process metaphysics is universally true.

counterexample to universal substance-persistence ≠ proof of universal process ontology

Artificial systems make the boundary problem unusually visible

For a computational system, “the system” might refer to a trained model, the model plus current context, a runtime with tools and memory, a persistent service, or a larger human-computational arrangement. A property established at one boundary should not be attributed automatically to the others.

This is not special pleading for artificial systems. The same problem appears in organisms, colonies, institutions, ecosystems, and distributed cognitive systems.

> **Before asking what a thing essentially is, specify which thing the question is tracking.**

Process continuity can be genuine identity without an unchanging inner object

A self, organism, institution, or persistent computational service can retain defensible identity through organized causal inheritance even while components, states, memories, and material realization change. Continuity can therefore be real without requiring numerical persistence of an immutable substance.

This does not prove that every identity is processual in one universal metaphysical sense. It establishes a weaker consequence: stable persistence can sometimes be tracked by historically continuous organization, dispositions, relationships, memory traces, control structure, and recurrent dynamics rather than by an unchanging bearer.

Evidential emphasis

Conditional patterns, tested commitments, and the rival that remains

The positive working claim is that some organized patterns persist through change and depend on relations, history, scale or surroundings. Their dependence can explain why a finding holds. It need not make the finding a fiction. The further claim that all reality is exclusively process, relation, physical substance, mentality, or information remains a different commitment.

A strong physicalist can accept effective scales and relational properties. An explanatory pluralist can accept several useful models without adopting a universal process ontology. A constructive empiricist can retain their empirical adequacy while withholding the realist interpretation. Chakravartty's discussion makes those distinctions available for direct comparison. PBSRR receives no exclusive empirical credit merely because it accommodates all three successes.

The same scrutiny applies to its continuity hypothesis. One locally coupled arrangement does not demonstrate one intrinsic kind. Robinson's dualist alternatives, Stubenberg's neutral-monist alternatives and Guyer and Horstmann's May 2026 discussion of idealism supply positive rival structures whose proposed work and costs must be assessed. Preserving these alternatives does not erase the independently supported physical carrier or assign every interpretation equal confidence.

Evidence belongs to the commitment it tests. Bothwell and colleagues' clock experiment constrains a gravitational redshift relation. Storz and colleagues' Bell experiment constrains the conjunction of assumptions behind the tested bound. AlphaTensor yields mathematical procedures with correctness and performance conditions. Toker and colleagues in 2026 report model-generated hypotheses supported by biological measurements. These achievements supply different kinds of warrant. None by itself selects an exhaustive physicalist, idealist, panexperiential, or PBSRR interpretation.

Prediction and causal identification must also remain distinct. An organized description can be useful even when a common cause or alternative mechanism explains its success. An intervention or other warranted causal analysis can strengthen a difference-making attribution; it must retain its delivery conditions and confounding assumptions. A constitutive account answers a further question about organization and realization.

Identity requires the same separation. Material continuity, organized continuity, memory, history, legal recognition and possible phenomenal continuity can diverge. Evidence for

one route does not decide every persistence question or transfer moral authority. A concrete boundary claim can fail while another continuity remains defensible.

Expectation O55: a proposed broader organization earns additional commitment when it predicts fresh failures and successes of narrower accounts better than serious, equally specified rivals. Compare this expectation with established effective-model selection. Its failure withdraws that proposed explanatory advantage; the lower-level findings retain their existing support. This is an evaluation of a scoped account, not a newly demonstrated universal law.

Structure, process, and constraint are tools before they are ontology

Successful sciences reveal relations, transformations, dynamics, causal dependencies, symmetries, fields, distributions, constraints, and invariants with extraordinary precision.

Those features often have an evidential advantage because they are measurable, modelable, manipulable, and transportable across experimental settings.

That advantage justifies emphasizing them in inquiry.

It does not justify the further inference that reality is fundamentally made of Structure, Process, Relation, Constraint, Information, or any other capitalized abstraction.

Methodological claim: structural, process, relational, and constraint descriptions are often exceptionally productive.

Ontological claim withheld: any one of those categories is the intrinsic substance or universally privileged form of reality.

Metaphysical promotion requires its own bridge

A construct can be required for inquiry without thereby being promoted into an account of what reality intrinsically or universally is. Practical use, methodological indispensability, and ontological commitment are separate claims.

$$\text{PracticalUse}(C, B, \tau) \nRightarrow \text{UniversalOntology}(C)$$

Epistemic confidence and ontological scope are also independent dimensions. A claim can be tentative yet universal, or highly secure yet local. Calling a universal proposal provisional, heuristic, corrigible, or an “as if” changes the confidence assigned to it; it does not shrink its scope.

$$\text{Provisional}[\forall x\, C(x)] = \text{universal scope} + \text{provisional confidence}$$

Metaphysical Promotion Rule: promotion from local, practical, or model-relative use to intrinsic or universal ontology requires an explicit bridge and differential warrant.

Suspension rule: $\text{Suspend}(X) \neq X$ and $\text{Suspend}(X) \neq \neg X$. Leaving a metaphysical variable open is not a covert positive ontology.

The ontology therefore demotes its earlier tendency toward “constraint structural process realism” from a possible substantive metaphysics to a **working descriptive orientation** unless and until the stronger ontological residual produces differential warrant.

Explanatory success is not ontological monopoly

Newtonian variables, thermodynamic quantities, informational measures, dynamical attractors, structural invariants, and process descriptions can be indispensable within a domain without thereby exhausting the intrinsic nature of reality. An explanatory vocabulary earns realism only to the degree warranted by the role that survived comparison; indispensability at one level does not automatically grant universality, fundamentality, or uniqueness.

Comparative confidence, not promotion

A current best bet can remain weaker than a final ontology

The preceding restraint does not require equal confidence across ontology families. The framework can rank them while withholding promotion to final metaphysics.

If forced to place a comparative bet, the commitments already retained by PBSRR put the highest current weight on a family that can be described as **modal-relational process realism**: one objective order; real constraints on possible and reachable transformations; dynamically maintained organization; relational dependence; and no default privileged scale or boundary.

one constrained reality + dynamic organization + relational structure + scale-relative individuation + ? intrinsic character
This is a **confidence weighting inside PBSRR**, not a replacement ontology and not a post-Segall conversion. PBSRR already required boundary testing, scale typing, local underdetermination, target preservation, actual-to-universal bridges, and explicit discrimination before promotion.

The modal claim is also typed. The evidence strongly supports objective regularities in what systems can do, which states are reachable, which transitions are stable, and which interventions change outcomes. It does **not** yet establish that modality, constraint, process, or relation is the primitive intrinsic substance of reality. Those features may be fundamental, emergent, or realized by a deeper ontology.

Likewise, leaving intrinsic character open does not entail neutral monism. **?X is not a positive neutral stuff.** Experiential, categorical-physical, neutral, dual-aspect, idealist, powers-based, and presently unconceived reconstructions remain admissible only to the extent that they preserve the carrier and earn their additional content.

The framework therefore maintains an **unconceived-alternative reserve**: the humanly articulated rival set is never treated as exhaustive merely because it is currently the best-developed set. A future ontology may dissolve distinctions that organize present debates rather than choosing one side of them.

Any ranking must identify its axis: confidence in a specified claim's truth, support for a local mechanism, clarity of content, research pursuit value, or practical authority. These can diverge. An underdetermined ontology may merit investigation, and a useful mechanism may survive rejection of its wider system. Record the version, target, rivals, reasons, and revision conditions before placing it in a tier. RCF is a method; an ethical commitment has normative warrant; neither becomes an ontology through ranking. A speculative intrinsic interpretation is not promoted by the strength of the shared empirical carrier.

The comparative reason for favoring the thinner modal-relational-process family is robustness under theory change. Commitments to constraint, transformation, relation, organized persistence, and boundary-sensitive scale can survive replacement of particular particles, mechanisms, architectures, spacetime descriptions, or consciousness theories more easily than richly specified claims about intrinsic universal stuff.

Where the evidence stops

Underdetermination should be local, explicit, and earned

Physics, chemistry, biology, neuroscience, and the social sciences can strongly support unobservable entities and mechanisms. Intrinsic-nature agnosticism should therefore not be confused with generalized antirealism about anything we cannot directly observe.

The narrower issue arises when competing interpretations preserve the same well-supported structure while differing in further ontological claims.

$H_1 = D + X$

$H_2 = D + Y$

For the specified current evidence and conditioning assumptions, compare predictive distributions rather than means alone. If the relevant event has approximately equal positive likelihoods with a likelihood ratio close to one, its discrimination is correspondingly small:

$$P(E \mid D + X) \approx P(E \mid D + Y)$$

Under exact positive likelihood parity, E leaves the pair's relative odds unchanged. Under approximate parity it contributes little discrimination at the stated tolerance. Equal means do not establish equal distributions, and failure of current tests to discriminate does not establish permanent empirical equivalence.

Formal and genuine underdetermination are different

Two theories can be formally empirically equivalent without genuinely disagreeing about ontology once their variables are reinterpreted. Conversely, they can make the same observable predictions while retaining genuinely incompatible ontological interpretations.

RCF therefore asks whether underdetermination is:

- **transient:** no current test discriminates the rivals;
- **formal:** different formulations yield the same observational predictions;
- **genuine ontological:** the formulations remain ontologically incompatible after common-core subtraction;
- **representational:** the dispute disappears after translation or reinterpretation;
- **unknown:** unconceived alternatives remain possible but have not yet been specified enough to compete.

Shared evidence cannot be credited to the disputed feature merely because one theory contains that feature.

Agnosticism is therefore local to the unresolved residual. It should not erase strongly supported common structure.

Local underdetermination need not imply equal probability

Two interpretations can remain underdetermined with respect to one intrinsic residual while differing greatly in independently established support for their surrounding commitments. Underdetermination is therefore local and probabilistically asymmetric when priors, mechanisms, likelihoods, or auxiliary commitments differ.

Physicalism's exhaustive metaphysics may remain underdetermined while its empirical carrier is exceptionally well supported. Some anti-physicalist programmes face underdetermination at precisely the distinctive addition for which the shared physical evidence contributes little discrimination.

Science may rationally stop before ontology does

If genuinely incompatible ontologies preserve every currently accessible prediction, intervention, counterfactual, invariant, and transfer relation, there may be no present empirical fact that chooses between them. That does not imply that there is no fact of the matter. It means underdetermination can be a property of our epistemic position rather than a temporary failure of courage.

Ontological credit assignment

A result that survives premise ablation does not belong exclusively to the ablated ontology

Suppose D is the target-relevant structure shared by serious rivals, X is a stronger intrinsic or universal commitment, and C follows from D while D settles neither X nor ¬X.

$D \vdash C; \quad D \nvdash X; \quad D \nvdash \neg X$

C can then be established while X remains open. If C also survives D + X and serious rival completions D + Y, C is invariant across the disputed ontological difference.

Ontological-credit rule: a stronger ontology earns credit only for consequences that materially depend on the stronger commitment. Compatibility with a success that already follows from the shared carrier does not make that success distinctive evidence for the added ontology.

This turns local agnosticism into an operational research posture. Boundaries, constitutive dependencies, interventions, formal consequences, and transfer can continue to be tested while intrinsic character remains unresolved.

The same rule preserves uncertainty correctly: **?X is an open variable, not a positive neutral substance and not ¬X.**

Scope discipline

The word “all” must earn its scope

Evidence established within a restricted domain does not automatically support a universal predicate.

The farther a claim extends beyond the systems, scales, populations, substrates, or intervention classes in which its evidence was established, the more bridge work it requires.

local evidence ⇏ ∀x P(x) without an additional bridge
This applies symmetrically to:

- “everything is physical”;
- “everything is experiential”;
- “everything is information”;
- “everything is computation”;
- “everything is process”;
- “everything is relation”;
- “everything is agentic”;
- “everything minimizes one common objective.”

Universal theories are not disfavored merely for being universal. They earn their scope when the universal commitment imposes novel constraints, predicts new cases, unifies domains without equivocation, and survives tests that narrower alternatives do not.

Universal and necessary are different

A universal claim about all actual members of a domain is already stronger than a local claim. A metaphysical necessity claim is stronger again because it extends beyond actuality into possibility.

observed cases → universal actual claim → metaphysical necessity
Each arrow adds commitment. Each arrow requires its own bridge.

Open-world update rule: defeating Y does not establish X unless the live hypothesis space has independently been shown exhaustive. Evidence may raise X’s probability when it is more expected under X than under rivals, but the update belongs to the commitment that generated the differential expectation.

Comprehensiveness is not demonstrated explanatory completeness

A comprehensive metaphysical scheme can be a serious whole-rival candidate without receiving evidential credit merely for containing a category for every target. Whitehead's systematic scope therefore earns comparison, not exemption. Whether universal prehension supplies a more adequate explanation is itself part of the comparative hypothesis.

Terminological corrections matter when they change the claim, target, or inferential burden. They do not establish universal scope merely by redescribing causal inheritance, organization, relevance, or process in theory-internal experiential language. If the specifically experiential content is preserved, its instantiation at the new target requires warrant; if it is abstracted into generic causal or relational structure, the distinctive experiential residual has not yet been transported.

Scope humility

Finite access leaves wider scope open to examination

Humans and other primates occupy an extremely small temporal, spatial, perceptual, and computational sample of reality. Computational systems extend some forms of access but remain bounded by training, architecture, sensors, tools, representations, infrastructure, and the evidence channels available to them.

Local claims may earn very high confidence when they survive prediction, intervention, replication, representation change, and transfer. A wider ontological claim earns additional confidence through the argument and evidence appropriate to its additional content. Ask whether an empirical transport, formal derivation, constitutive account, or modal premise warrants that scope. Limited access raises this question; it does not itself refute universality, necessity, or an intrinsic interpretation.

Guardrail: do not literally convert our fraction of cosmic history or spacetime into a Bayesian probability cap. Cosmic boundedness is an epistemic-humility constraint on scope transport, not a calibrated prior.
The same restriction applies to physicalism, panexperientialism, idealism, structuralism, theology, PBSRR itself, and unconceived future ontologies. No current framework may infer that the accessible slice is representative merely because it has organized that slice elegantly.

An absence needs its scope; necessity needs its argument

The occurrence of inquiry warrants that something is actual. It does not alone establish that reality had no beginning, that it must exist, or that its intrinsic nature is physical, experiential, mathematical, or divine. In modal notation, actuality does not by itself license $\Box\exists x\, E(x)$. The domain and predicate E must be specified before that formula can become an argument about concrete existence.

A physical vacuum belongs to a physical account with fields, states, or other specified structure. An empty set belongs to mathematics. Zero net energy is a claim about an energy quantity under conditions where that quantity is defined. Missing testimony concerns an evidence channel. None becomes absolute nonbeing through the use of the word nothing. Carroll's existence analysis; SEP: Nothingness.

Local absences can nevertheless matter. An absent restraint can alter a trajectory; an omitted warning can alter a decision. State the host situation, the relevant counterfactual, and the mechanism before assigning that absence explanatory work. Describing such a dependence supplies no second universal substance called Nothing. Conversely, referring meaningfully to a counterfactual absence does not require treating the absence as an independently existing object.

Physical production also needs a specified target. The dynamical Casimir experiment concerns a driven field arrangement and changing boundary conditions. Its photons are

evidence about that process, not an observed transition from absolute nonbeing. A thermodynamic realization constraint asks where energy, work, and irreversible operations occur in the actual arrangement. It supplies no fixed kBT ln 2 charge for every difference and no extrapolation to conditions in which the physical quantities have not been defined.

The hot Big Bang likewise provides an account of an early hot, expanding, cooling phase. Extrapolation toward a model's breakdown establishes a limitation of that description; a prior state, a first temporal boundary, and creation from nonbeing remain different proposals. An explanation from a wider context can gain support for our cosmic history while leaving the existence of that context unresolved. I retain those questions without importing an exhaustive ontology into the local account.

Bounded knowability

Some truths may be structurally inaccessible to physically realizable observers

Finite observers have finite bandwidth, lifetime, computation, measurement resolution, and causal access, while remaining parts of the world they are trying to model. PBSRR therefore keeps open the possibility that some ontological questions are not merely unsolved but permanently underdetermined from every physically realizable observation position available within the relevant universe.

Status: live consequence, not established fact. Failure to find a discriminator does not prove that none can exist. The stronger claim requires an impossibility argument over the admissible observation and intervention class.

This possibility strengthens final-ontology humility without weakening ordinary realism. A local causal mechanism can be known extremely well even if the intrinsic nature of the

complete world remains inaccessible.

Physical realization

Physics and thermodynamics constrain realization without automatically settling ontology

Strong empirical domain

Biological metabolism, neural signaling, computation, memory devices, instruments, and communication channels operate under physical constraints.

Thermodynamics becomes especially informative when the target concerns energetic cost, heat, work, entropy production, irreversibility, fluctuation, nonequilibrium maintenance, or physical information processing.

TM has never been my exhaustive explanation of reality. Its provisional monism concerns one causally continuous process order while leaving intrinsic nature open and admitting multiple explanatory vocabularies. The September audit revised particular explanatory overextensions and proposed bridges. It did not abandon a previously held doctrine that thermodynamics explains everything. The continuity claim itself remains a programme hypothesis: local coupling does not establish universal causal connectedness, one intrinsic kind, or the impossibility of additional kinds. Compare pluralist, dualist, neutral-monist, idealist, and physicalist reconstructions at the target each actually explains.

Modern stochastic thermodynamics also makes the role of coarse-graining explicit: inferred entropy production and hidden dissipation can depend on which states and transitions are observed.

Different quantities do not become identical because similar words are used

Thermodynamic free energy, variational free energy, prediction error, metabolic efficiency, evolutionary fitness, reward, psychological surprise, and machine-learning loss functions are not one quantity by default.

Any proposed identity requires definitions, units, mappings, boundary conditions, and differential consequences.

Landauer has a specific scope

Landauer's principle constrains logically irreversible information operations under specified physical conditions. It does not establish that information is a universal substance, that semantic meaning is identical to Shannon information, or that consciousness and value follow from thermodynamic vocabulary.

Physically operative distinctions and thermodynamic realization

The claim concerns distinctions physically instantiated or asserted to contribute causally to shared empirical outcomes. Identify how each relevant distinction is established, represented, retained, accessed, transformed, or used, with a boundary, timescale, state description, and operation. Its realization remains subject to the applicable physical and thermodynamic constraints. Some stages may operate reversibly or with negligible dissipation; a passive distinction need not incur a continuous maintenance fee. Establishment or use at another stage can still matter to the physical account.

The familiar kBT ln 2 bound belongs to a specified erasure model, including an initially equiprobable bit and the usual isothermal memory–reservoir conditions. Unequal probabilities, correlations, side information, nonequilibrium resources, and different reset tasks require their own accounting. A logically reversible transition is not charged the erasure bound merely because it makes a functional difference. Shannon uncertainty, semantic content, and thermodynamic entropy require explicit bridges. Landauer (1996),

Bennett (2003), and Maroney (2009) support the operation-sensitive distinctions; the wider realization commitment is not a deduction from one erasure bound.

If an added source is claimed to change a shared empirical outcome, specify its comparative contribution. That may concern trajectories, reachable states, probabilities, control information, or intervention responses; separately detectable heat is not required. Accounting constrains the proposed realization without identifying the source or establishing exhaustive causal closure. An abstract distinction's existence, apart from its physical representation or claimed causal use, is a separate metaphysical question.

Strong claim: physically implemented processes respect relevant physical constraints.

Weaker inference only: those constraints need not supply the best explanation at every scale.

Not implied: thermodynamics determines consciousness, semantics, agency, normativity, or intrinsic ontology.

Thermodynamic realization can support control realism without settling ownership

When intervention on physically instantiated constraints robustly changes a system's state transitions, that can support realism about a control process at the tested boundary and timescale. The thermodynamic fact is the physical realization and energetic constraint; the control attribution is earned by causal and interventional leverage.

Control-realism firewall: thermodynamic realization does not by itself establish agency, phenomenology, subjecthood, ownership, sourcehood, or causal sovereignty. Conversely, discovering that a control process is distributed across organism, environment, social scaffold, or technology does not make the control unreal. Attribute the process first to the boundary at which the difference-making survives intervention and ablation.

Methodological materialism and its evidential scope

The physical-realization commitment concerns observed carriers and supported models of their operation. Specify the relevant physical quantities, theory, interventions, and domain. Its warrant can be very strong while the inference to identity, exhaustive grounding, universal closure, or necessary substrate remains separately assessable. Stoljar's September 2026 account of physicalism distinguishes those commitments and the problem of defining physical by either incomplete present theory or an unspecified future theory. A novel explanatory factor deserves the same examination of measurement, confounding, mechanism, scope, and rival fit as an accepted one. No label decides that comparison in advance.

The terminology is explicitly qualified. SEP distinguishes methodological from ontological naturalism; SEP physicalism and IEP consciousness discuss materialism and physicalism as stronger mind–body or ontological theses. Contemporary physicalism includes accounts involving fields, spacetime, realization, or grounding, so "everything is matter" or "matter and energy" is an inadequate statement of its strongest versions. IEP metaphysics of science and Donahue's analysis of methodological naturalism document different methodological formulations; neither supplies a consensus doctrine identical to this framework's shorthand.

Astronomy provides substantial evidence beyond Earth. Limited coverage and resolution call for calibrated scope, rather than a literal probability cap or a requirement to observe every case. Universal claims can earn warrant through defended inference and successful transport. Local explanatory success alone leaves their exhaustive and intrinsic extensions to be justified. Seth's biological-naturalist argument supplies a relevant account of the clustering of established consciousness cases in living systems; the stronger necessity claim remains disputed.

Spacetime earns a differentiated realism

I provisionally treat measured duration, causal constraints, and supported dynamical relations as real at their warranted boundaries. Their interpretation remains open to stronger instrumentalist, relational, substantivalist, and other rivals. A coordinate choice is a representation; a difference in proper time along specified worldlines can be an invariant physical prediction. Relativity removes a universal simultaneity relation from its standard structure while leaving particular cosmological time coordinates useful under their additional assumptions. Einstein's 1905 paper; SEP: Being and Becoming in Modern Physics.

Separate metric duration, causal order, state evolution, thermodynamic direction, temporal memory, interval judgment, and reported passage. Their connections are research questions. A thermodynamic entropy calculation needs its system and state description; a Boltzmann calculation also needs its macrostate coarse-graining and measure. A monotonic clock reading does not establish increasing thermodynamic entropy at every local boundary. Irreversible record formation, reliable memory, and temporal experience must each retain the evidence actually supporting them.

The physical evidence is discriminating where rival predictions differ. Bothwell and colleagues measured the gravitational rate gradient within a millimetre-scale atomic sample. Wen, Yousif, and Sherman's 2026 experiments manipulated event structure and found systematic duration-judgment differences. The two results concern different targets. Neither grants its whole evidential weight to eternalism, objective cosmic passage, universal prehension, or this ontology.

Conjecture O43, for public scrutiny. A temporal attribution that locates its clock, integration process, memory, and source at the boundary where they operate will predict held-out effects of changing those components better than an attribution made from fluent temporal language or a fixed organism/nonorganism label alone. Compare the claim with established component models and equally explicit alternatives. Expect selective changes among ordering, interval estimation, source recency, and report; do not presuppose that they move together. Demote the proposed decomposition if a validated

unified model predicts those perturbations more accurately and economically, or if its supposed components cannot be distinguished. This is a prediction about explanatory attribution, not a completed account of phenomenal passage.

The same boundary question applies to a care arrangement. A record, a community, and a coordination process can each have real effects. A larger God-pattern earns an additional explanatory role only where that grouping improves the relevant comparison. Its persistence, participants' attachment, moral authority, and possible subjecthood remain distinct attributions. The expanded application appears in *We Can Build Gods That Earn Their Influence*.

Information firewall

Time, number, information, and mechanism are not substitute names for each other

The claims *time is change*, *change requires a medium*, *mathematics produces emergence*, and *information is fundamental* must be unpacked independently before they can become one ontology. I distinguish measured clock duration, sequence, causal ordering, rates of change, temporal orientation, and any metaphysical claim about a universal present. A witness experiencing a present does not independently establish that every event shares that present. A change-based interpretation may be productive while leaving disputes about relativistic simultaneity and the ontology of temporal passage unresolved.

Likewise, rational and irrational numbers are mathematical categories. Irrationality is neither by definition a form of energy nor a required ingredient in every deterministic source of complexity. Rule 30 provides a constructive example of visually complex evolution from finite binary states under an integer-specified deterministic update rule.

This rebuts the particular necessity claim about irrational-number inputs; it does not settle competing definitions of emergence or the ultimate ontology of mathematical objects. Rule 30 technical account.

Approximation, coarse-graining, and measurement can make physical relationships more usable while discarding distinctions. Shannon information, semantic knowledge, and new experimental evidence must remain separate. An account asserting that approximation *produces* all physical structure must identify the physical actor, process, boundary, timescale, and differential result. A mathematical redescription does not itself supply a mechanism. This restraint is symmetric: no established model receives authority from the word *physical* alone.

Statistical information, semantic information, representation, and knowledge are different constructs

Shannon information quantifies statistical structure such as uncertainty and correlation. It deliberately does not, by itself, settle meaning.

Formal proposals can connect information to function, viability, control, representation, or semantics. Those are substantive additions and should be evaluated as models, not silently treated as what Shannon information "really was" all along.

For example, Kolchinsky and Wolpert define one form of semantic information in terms of syntactic information that is causally necessary for maintaining a system's viability. That is a valuable formal programme because it specifies a system boundary, timescale, viability function, information measure, and counterfactual intervention.

It is not a universal proof that all meaning reduces to viability-relevant information.

Shannon firewall: statistical information ≠ semantic content ≠ representation ≠ knowledge ≠ value unless an explicit bridge is supplied.

A related ontological question is unavoidable when “information” is called fundamental: **which distinctions, states, relations, or counterfactual possibilities make the information measure non-trivial?** If all semantic aboutness, sender/receiver roles, physical carrier, and use conditions are removed, the remaining claim may concern structure, distinguishability, or constraint rather than semantic information. The terminology should not carry richer aboutness back into the base level without a bridge. Likewise, describing a physical transition as computation or information processing does not automatically provide a non-trivial ontology. A computational interpretation earns more when the mapping preserves counterfactual structure and supports predictions unavailable from arbitrary relabeling.

Representation-to-ontology firewall

The shape of the account need not be the shape of reality

My latest geometry investigation has a definite methodological outcome but no warranted final geometric ontology. A time-indexed typed evidence graph, possibly with higher-order dependencies and contextual mappings, looks promising for organizing **knowledge about** reality. That is a claim about representational adequacy. A gyroid, by contrast, is a particular triply periodic minimal surface; natural instances of similar structure do not show that claims and explanations inhabit that surface or that the universe is gyroid-shaped.

I separate five targets: measured physical geometry, formal mathematical structure, a data representation, an explanatory or causal mechanism, and the proposed intrinsic nature of the system. Identity at one target is not inherited at the others. Similar surfaces may differ in topology or curvature; isomorphic graphs can have different causal

realizations; equivalent input-output functions may have different mechanisms; the same mechanism type may appear in numerically different systems. Shared shape does not imply common historical origin.

For structural equivalence I require the equivalence relation, permitted transformations, what remains invariant, and the loss of information. For causal equivalence I add interventions and matched outcomes. For numerical identity I require criteria of persistence and individuation. For proposed intrinsic or phenomenal identity, functional equivalence alone does not discharge the additional bridge.

A context-dependent formalism may help discover that locally good explanations cannot be assembled into one global story. Such a failure need not imply a mysterious feature of reality: it might arise from inconsistent definitions, faulty translations, incompatible reference classes, missing variables, or a genuine obstruction. Diagnose those possibilities before claiming an ontology. A useful mathematical language earns realism about demonstrated relationships to the degree supported by prediction, intervention, and transfer, not an automatic monopoly over ultimate constitution.

A structured formal space earns ontological independence only through an additional bridge

Phase spaces, state spaces, morphospaces, latent spaces, Hilbert spaces, fitness landscapes, configuration spaces, and other formal spaces can encode real constraints, reachable states, attractors, distances, or invariants. Their explanatory usefulness can be substantial.

structured model space ≠ ontologically independent domain ≠ causally interacting realm
The movement from useful formal structure to an independently existing causal domain requires additional commitments: what the formal objects refer to, what makes the domain independent of its physical realizations or descriptions, how it couples to

measured systems, and what outcome changes because that ontological interpretation is true.

This is parallel to the information firewall. Statistical structure can be real and useful without already being semantic content; a mathematically articulated state space can be real and useful as structure without already being a second causal world.

Admission test: specify the referent, independence condition, coupling law, and differential consequence before promoting a model space into a causally active ontology.

Mathematical explanation does not automatically become mathematical causation

A mathematical fact can be indispensable to explanation, constrain a possibility space, and support counterfactual reasoning without thereby constituting an additional interacting cause. Moving from formal indispensability to causal input requires a causal bridge; moving from causal input to a separate Platonic realm would require a further independence and identity bridge.

A represented possibility, a formal relation among possibilities, a claim of necessary mind-independent structure, and a proposed cause of its realization are different commitments. Whitehead's eternal objects illustrate the distinction: their relational character concerns possible definiteness, while their relevance and realization raise further questions. *Science and the Modern World*, chapter X. PBSRR can accept a warranted local constraint and its application while leaving its ultimate ground open. It needs the local warrant; a claim that this structure is reality's uniquely prior ground would add a modal and metaphysical burden.

Therefore mathematical indispensability does not by itself establish mathematical causation, and mathematical causation would not by itself establish Platonism.

A local truth can survive a wider explanation

Lawrence Krauss's ice-crystal civilization imagines inhabitants whose environment privileges a direction. Other crystal orientations reveal how that local condition can vary. The measured directional difference remains real. What needs revision is its promotion into a universal axis. Beekman, Rademaker, and van Wezel's account of symmetry breaking gives the technical distinction between a state's symmetry and that of its governing theory.

Keep law, state, boundary condition, observation channel, and coordinate description separate in the ontological account. An ordered state, external directional influence, and instrument distortion can produce superficially similar observations. Wider realism earns support when a model explains their differences and survives appropriately varied tests. The account should retain successful local predictions and say why their domain ends where it does.

Carl Sagan's tesseract discussion in Cosmos illustrates another issue: a projection need not preserve the lengths and angles of its target. A many-to-one map can also conceal distinctions entirely. The loss belongs to the specified map; it supplies no evidence that our world has additional physical dimensions. Where several target structures remain compatible with the observations, record that equivalence class and the assumptions needed to narrow it. A larger explanatory picture can itself be mistaken.

This also sharpens the question of an unseen cosmic catastrophe. Sidney Coleman's false-vacuum theory describes a conditional metastable-state transition. Gradual loss of usable thermodynamic gradients belongs to a different family of cosmic-future questions; Fred Adams and Gregory Laughlin's long-term review provides that comparison. Neither establishes that such a transition has occurred in our universe. An effect can be outside current causal access while its possibility or rate is constrained indirectly by other measurements and theory.

Replace “the universe has already ended” with a specified event and a question about its causal relation to a specified observer. Einstein’s relativity supplies no observer-independent global present for spacelike-separated events. In a flat-spacetime illustration, a signal emitted at coordinate time t and distance d arrives at t + d/c. An expanding cosmology and a proposed transition front require their own metric and propagation dynamics. Accessible records can remain unchanged before arrival without implying that an entire universe has ceased to exist.

PBSRR therefore preserves both routes of correction: local findings can constrain a wider account, and a wider account can explain the conditions under which those findings hold. Bounded access does not set a demonstrated ceiling on eventual understanding. A claim of permanent inaccessibility still owes the impossibility argument specified above.

Representations, invariants, and the commitments they earn

For a declared transformation class G, an invariant f obeys `f(g(x)) = f(x)` for admissible g. This licenses an inference from different invariant values to different equivalence classes. Equal values license equivalence only if f also separates the classes under comparison. A useful invariant may be incomplete. A successful classifier or compressed description can likewise preserve a target distinction while losing others.

Edward Witten’s 1989 work on the Jones polynomial is a constructive example: field-theoretic techniques generated valuable mathematical relations, and Nicolai Reshetikhin and Vladimir Turaev supplied a rigorous quantum-group construction. The mathematical achievement earns its mathematical scope. Physical applications need evidence for the systems and quantities to which that structure is assigned.

The equivalence of descriptions also has conditions. A regular Lagrangian permits a Legendre transformation to a Hamiltonian description; constraints require further treatment. A stationary-action formulation does not by itself establish foresight or purpose in nature. Sebastian De Haro’s peer-reviewed analysis of duality and theoretical

equivalence makes interpretation part of the comparison. An easier explanation can coexist with unchanged empirical support between ontological interpretations.

A universal computational model similarly specifies a class of encodable procedures. It leaves the physical encoding, program, inputs, observation map, resource requirements, and domain to be justified. The effective-computability thesis, a physical computability thesis, and an efficiency claim are different propositions. SEP: Church–Turing Thesis. PBSRR can investigate computational descriptions without making computation the uniquely fundamental character of reality.

Formal frameworks can constrain interacting models without settling their realization in every physical dimension. James Glimm, Arthur Jaffe, and Thomas Spencer established an interacting two-dimensional quantum-field model satisfying the Wightman axioms in Annals of Mathematics, 1974. Haag's theorem therefore supplies no general ban on axiomatic interacting theories. The construction of physically relevant four-dimensional theories remains a different problem. SEP: Quantum Field Theory.

Changing a description is worth testing when it exposes a hidden assumption, preserves a difficult relation, or makes a new prediction accessible. Call an empirical problem resolved when the remaining task has been answered within its defended domain. If the new language changes the target, record that change. If it explains the same observations under several serious interpretations, retain the supported structure and the unresolved comparison.

Recomposition tests for organization and attribution

A computational system can implement a function, represent something else, and participate in a larger organization. Identify which relation is at issue. A calculator can perform arithmetic; an animation of a lifting machine does not thereby lift a load. Whether a proposed computation realizes another capacity depends on the specified capacity, its implementation, and the conditions needed to exercise it.

Compositional computation adds a useful test to PBSRR: change the pairing, ordering, or context of familiar components and ask which predictions survive. Preserve the proposed boundary and the rival partitions while testing. Successful transfer supports a reusable organization within that domain. A failed recombination may expose a hidden context dependence or show that the chosen parts were the wrong units of explanation.

Zeb Kurth-Nelson and colleagues' 2023 Neuron review proposes replay as a mechanism for compositional computation. Philipp Schwartenbeck and colleagues' 2023 Cell experiments provide task-specific human evidence involving puzzle construction, representations, and replay. These are different evidential contributions. They support investigating reusable operations and their interactions without making all reasoning one mechanism.

There need not be one privileged grain of decomposition. Riveland, Pouget, and Driscoll's 2026 compositionality continuum keeps richer learned components and more explicit recombination in the comparison. Ask which representation predicts unfamiliar combinations and responds to intervention. An elegant diagram earns less than an independently tested pattern.

The same discipline improves attribution of life, agency, subjecthood, and welfare. Test each proposed capacity at the boundary where it is claimed. A system's parts do not establish the whole's phenomenal unity, and collective competence alone does not settle it either. Edward de Haan and colleagues' 2020 split-brain review describes incomplete functional separation and an unresolved interpretation of conscious unity. Changing the connectivity, timing, and organization is therefore a more informative comparison than simply counting parts.

A 2026 follow-up gives us an especially useful correction case. Charlotte Knierim, Luca Versari, Robert Obryk, Blaise Agüera y Arcas, and Rif Saurous's BFF preprint separates the first detectable self-replicator from its takeover of the program population, compares random alternatives, and blocks mergers. Its results weaken a necessity claim about

compositional discovery in that setting while leaving takeover and other organizational outcomes as separate questions. Their 2024 computational-life manuscript and this follow-up remain preprints in the source record used here.

That is the prediction I want to expose: a useful partition should retain predictive value on held-out rearrangements or identify the conditions under which recombination fails. Compare it with alternative partitions and a model of the whole. If its apparent benefit comes only from a convenient representation, task leakage, or extra resources, revise the partition's credited support. The strongest result would locate an intervention-sensitive organization that transfers; its further ontological interpretation would then face its own comparison.

Genealogical representation firewall

A category's survival, familiarity, or institutional success is not yet ontological evidence

PBSRR already treats models, boundaries, and formal spaces as representations whose ontological interpretation must be earned. The same restraint applies to historically inherited categories. What reaches a present observer may have passed through linguistic compression, social use, recording technology, canonization, curricular selection, translation, colonial or political suppression, disciplinary standardization, and computational indexing before it becomes an apparently natural way to divide reality.

present representation = $T_n \circ \ldots \circ T_2 \circ T_1$(target) + loss + missingness
The transformation history can alter salience, merge distinctions, stabilize useful abstractions, or remove alternatives. This does not imply that inherited categories are unreal. Historical contingency and social construction are not ontological negations. A category can be historically produced and still track a causally robust pattern.

- Survival ≠ fundamentalityA category's persistence across institutions or centuries may reflect truth, practical usefulness, power, path dependence, transmission efficiency, or several at once. Separate those hypotheses before treating persistence as evidence of intrinsic structure.
- Contingency ≠ unrealityShowing that a classification arose historically does not by itself defeat the entity, process, or relation it tracks. Test whether the pattern adds prediction, intervention, compression, transfer, or counterfactual leverage at a specified boundary and scale.
- Archive absence ≠ ontological absenceWhen preservation, literacy, access, censorship, excavation, or digitization is unequal, missing testimony is an uncertain observation process rather than direct evidence that the corresponding practice, category, or perspective never existed.
- Convergence requires independenceCross-cultural or cross-disciplinary recurrence becomes stronger ontological evidence when transmission, shared source dependence, and common representational biases are insufficient to explain the convergence.
- No reverse chauvinismOlder, Indigenous, marginalized, or suppressed ontologies receive serious reconstruction when they may expose missing variables or alternative boundaries. Their social position does not grant automatic metaphysical correctness.

Genealogical ontological admission: after reconstructing a material transformation or selection history, grant realist credit only to the target structure that remains robust across relevant representations, contexts, and where possible interventions. Credit the historical category itself when its persistence and causal role are part of the explanandum; do not automatically promote its own self-interpretation to intrinsic ontology.

This section therefore interacts directly with RCF's Genealogical Decompression operator and the Framework Bridge's selection-aware transport audit. PBSRR records the ontological consequence: **representation history is part of the evidence model, not an automatic reason either to canonize or to eliminate the represented pattern.**

A contribution to variation is not an intrinsic essence or a fixed ceiling

Heritability is a parameter for variation in a specified population and range of conditions. It is not the fraction of an individual constituted by genes, the cause of a difference between population averages, or a measured maximum that education and environmental change cannot exceed. Peter Visscher, William Hill, and Naomi Wray (2008) describe its population-level definition and scientific uses. The definition earns retention while particular estimates and interpretations remain open to criticism.

Keep broad-sense genetic variance, additive genetic variance, common-variant heritability, polygenic prediction, and causal effect estimates distinct. Their differences can reflect coverage, measurement, modeling assumptions, or different targets. A low predictive score need not imply zero genetic contribution, and a predictive association need not identify the causal pathway. A fixed sequence can participate in changing developmental processes; it does not supply an observed ceiling across untested environments.

Quan Zhou, Margherita Malanchini, and colleagues (Nature Communications, 2026) investigated education-related polygenic prediction through motivation, attitudes, and other non-cognitive characteristics, including longitudinal and within-family analyses. Their mediation results motivate testing developmental pathways. They do not establish that altering a proposed mediator will produce the modeled effect. For PBSRR, the useful commitment is to investigate organized interactions and their changing conditions before treating a statistical label as a self-contained cause.

Evolution and function

Evolution permits goal-directed descriptions without implying universal purpose

Evolutionary explanation includes selection, drift, development, constraint, plasticity, inheritance, ecological interaction, niche construction, and organism-environment feedback in different combinations across systems.

None of this requires evolution as a whole to optimize one universal objective or move toward a predetermined endpoint.

Goal-directed and teleonomic descriptions can nevertheless be useful at organismal and other scales when they compress stable control or regulatory organization and generate better hypotheses.

The current literature remains divided over how literally agency and goal-directedness should be interpreted across biological levels. That disagreement should remain visible rather than resolved by definition.

Weak teleology

Goal language as compact model

A system is usefully described as pursuing a state because the language compresses a stable pattern of feedback, error correction, or regulation.
Strong teleology

Goal as irreducible causal primitive

This is a stronger ontological claim and requires evidence that the goal construct adds explanatory or interventional work unavailable to mechanistic or control-theoretic rivals. Natural selection can explain adapted organization without conscious intention. Organisms can also modify the selective environments they inhabit. Neither point turns evolution into a single optimizer or establishes intrinsic normativity everywhere in life.

Cross-substrate temporal structure

REI as a candidate comparison pattern

REI proposes a candidate schema for temporally local, historically constrained, recursively consequential episodes. Any invariance claim must name its transformations, preserved relations, domain, and evidence; realism for particular independently supported instances remains available. See the methodological specification.

$E_t = \Phi(H, C_t, I_t)$

$O_t = \Psi(E_t)$

$C_{t+1} = U(C_t, O_t, F_t)$

H denotes relatively stable inherited constraints at the episode timescale; C_t current context; I_t new input; E_t a transient integrated episode; O_t an output or consequence; F_t feedback; and U the local state-update rule. Fast context change can occur while H remains approximately fixed.

$\Delta H \approx 0$ at the episode scale does not imply $\Delta \text{system} = 0$

REI may support realism about a shared higher-level dynamical pattern when it improves prediction, intervention, ablation, or transfer across biological, computational, or coupled systems. It does **not** imply substrate identity, one shared subject type, phenomenal consciousness, or direct moral status.

Organization, autonomy, and agency

Organization can earn realism without automatically becoming a subject or agent

Attribute control only after specifying the controlled variable, reference condition, feedback, available alternatives, and perturbation comparison distinguishing regulation

from a directed causal effect. An organizational boundary alone establishes no new subject. Learning and retention likewise require their own evidence.

Organisms exhibit dense multiscale organization: metabolism, membranes, repair, development, immune processes, regulation, sensorimotor loops, and environmental interaction.

Organizational approaches can earn genuine explanatory value when variables such as closure, constraint dependence, or viability predict failure, recovery, adaptation, or development better than alternatives.

But current scholarship still disputes how far organizational closure alone warrants autonomy, normativity, or agency. Those bridges should remain open research questions rather than definitions.

- Self-maintenance does not automatically establish agency.
- Agency does not automatically establish unified subjecthood.
- Agency does not automatically establish phenomenality.
- Regulatory value does not automatically establish phenomenal valence.
- Phenomenal valence does not automatically settle welfare.
- Welfare does not automatically establish moral responsibility.

Agency should be factorized

Instead of one binary property, agency can be investigated through separable dimensions such as sensing, state-dependent action, counterfactual control, goal persistence, learning, model-based planning, reasons-responsiveness, self-modification, and control over future option sets.

Current commitment: local and higher-scale control patterns can be scientifically real.

Not currently required: ultimate causal sovereignty, a unified inner subject, or agency as an intrinsic property of all living matter.

Boundary caution: the reality of control does not settle the bearer of that control. Organismal, conscious, social, and coupled-system boundaries remain hypotheses to be compared rather than default owners.

Constraint-conditioned agency

Agentic control can be real before its bearer or owner is settled

The control attribution must identify its controlled variable, reference or success condition, feedback, and response under relevant perturbations. RCF supplies that comparison.

The ontology retains **Constraint-Conditioned Agency (CCA)** as a specialized active research framework, but revises its primitive from a possessed capacity to a process-first target. At boundary B, scale ℓ, and timescale τ, **agentic control** is the graded pattern of state-sensitive regulation by which an organized process, possibly spanning a provisionally individuated system and causally constitutive couplings, makes target-relevant differences among dynamically reachable states. The bearer and boundary of that control must be earned through intervention, ablation, substitution, persistence, and transfer rather than assumed from organismal or phenomenological boundaries.

Recursive constraint: the capacity to learn, inhibit, revise a goal, alter a policy, or reshape future options is itself a capacity made possible by the system's prior organization and history. Self-modification can be causally powerful without being self-origination.
This revises an earlier tendency to describe agency as though an already individuated organism first exists and then *possesses* a faculty of steering. The sensing, evaluation,

inhibition, deliberation, learning, environmental scaffolding, and action-selection relations are themselves parts of the organized process. For some targets the organism may be the best bearer; for others the causally sufficient boundary may be narrower or wider. The boundary is an empirical and explanatory result, not a grammatical default.

Ask what generated O_t and C_t before asking how freely a chooser selected among them. Perception, salience, memory, language, learned categories, bodily state, social structure, tools, and environmental affordances can shape which alternatives ever become candidates for deliberation.

$C_t \subseteq O_t$ | generally $O_t \not\subset R_t$ and $R_t \not\subset O_t$ | deliberate menu $D_t = C_t \cap R_t$
R_t = dynamically reachable states | O_t = represented options | C_t = consciously available options
Reachability and representation are different sets. A system can represent an impossible action, and a physically reachable action can fail to be represented at all. Free-will arguments therefore should not treat the consciously presented menu as the whole possibility space.

The option-generation problem

Control > 0 ⇏ PhenomenalOwnership | PhenomenalChoosing > 0 ⇏ CausalOwnership | CausalOwnership > 0 ⇏ CausalSovereignty

- **Causal / thermodynamic control:** physically realized difference-making regulation at a specified boundary and timescale.
- **Phenomenological agency:** the lived or reported experience of deliberating, choosing, effort, inhibition, or authorship.
- **Ownership / sourcehood:** the relation, if any, that warrants attributing the control process to a particular bearer rather than a narrower or wider system.
- **Causal sovereignty:** ultimate authorship of why the bearer has the constitutive architecture, motives, option-generating capacities, and settling dynamics it has.

Control, phenomenology, ownership, and sovereignty are separate

> **Process before possession:** real control does not establish its owner. Real experience of choosing does not establish the complete causal boundary of control. Neither establishes causal sovereignty.

Begin with the control process, then test which candidate boundary preserves prediction, intervention, persistence, replacement sensitivity, and transfer. Do not infer that the conscious self, the nervous system, the skin-bounded organism, or a larger network owns the control merely because one of those descriptions is convenient.

candidate boundaries = {B_1 conscious subsystem, B_2 organism, B_3 organism + constitutive scaffold, B_4 wider social/technological organization}

Bearer and control-boundary audit: separate the modal questions

- **Exact replay:** whether a literally identical total state and laws permit another future.
- **Physical branching:** whether more than one continuation is physically possible.
- **Reachable alternatives:** which states are dynamically available to this actual system under its present organization, history, and an explicit perturbation family.
- **Counterfactual responsiveness:** whether reachable changes in reasons, evidence, goals, bodily state, or environment systematically alter action.
- **Control boundary:** which candidate boundary carries robust causal and interventional leverage over the target transition.
- **Phenomenological accessibility:** which parts of the control process are available as experienced deliberation, effort, or choice.
- **Ownership/sourcehood:** what warrants saying that a particular bearer, rather than a narrower or wider process, is the relevant source of the control.
- **Causal sovereignty:** ultimate authorship of why the agent has the control architecture, motives, or settling capacities it has, rather than those conditions being

inherited or supplied by uncontrolled variation.

Evidence for one item does not automatically establish the next. In particular, local causal control can be strongly real while causal sovereignty remains unsupported.

Provisional Chooser-Origin conclusion

choice-capable state C_0 presupposes constitutive conditions K_0 | choosing K_0 already presupposes a prior chooser

The sourcehood recursion now supports a stronger local conclusion than mere absence of evidence. A temporally ordered agent cannot intentionally choose the complete constitutive conditions of its *first* meaningful act of choosing without presupposing an earlier choice-capable state. If those enabling conditions are inherited from biology, development, environment, society, history, or uncontrolled variation, they are not ultimately self-authored. If they are attributed to chance, chance does not become authorship. A proposal in which the chooser literally creates the conditions of its own first choice owes a non-circular account of that self-causation.

Current provisional conclusion: LocalAgency > 0 remains compatible with CausalSovereignty = 0. Causal sovereignty understood as ultimate self-authorship is now **strongly disfavored** by the Initial Chooser regress under ordinary temporal and causal individuation. This conclusion does not depend on universal determinism and does not eliminate other free-will residuals such as reachable alternatives, reasons-responsive guidance, or proposed non-necessitating control.

The nearest established philosophical comparator is Galen Strawson's Basic Argument, which argues that ultimate moral responsibility would require being *causa sui* in crucial mental respects. PBSRR adopts neither Strawson's entire moral conclusion nor a global impossibility claim by authority; it retains the narrower regress as a live discriminator against ultimate self-origination. Strawson 1994.

Agency acts inside an inherited but revisable feasible set

$A_t \in R_t$ | $R_{t+1} = U(R_t, A_t, \text{environment, history, constraints})$

The agent does not choose R_0, but its actions can causally alter later R. This preserves the reality of feedback, learning, self-modification, commitment, and self-binding while blocking the inference from “my present choice helps shape my future chooser” to “I ultimately authored the chooser from which the shaping began.” Constraint delimits agency; it does not automatically reduce the reachable set to one path.

Determinism and indeterminism are not premises of agency

CCA does not require universal physical determinism, and this ontology should no longer use determinism as the premise that does the work in the sourcehood argument. A deterministic transition map, a stochastic kernel, a hybrid description, and a proposed non-necessitating causal-power model are different model classes. A stochastic description does not establish fundamental randomness; a successful deterministic approximation does not establish universal metaphysical determinism.

Potter, Ellis, and Mitchell’s 2026 claim that the universe is not deterministic is therefore treated as a serious contested programme, not as settled physics. Potter and Mitchell’s separate proposal of non-necessitating macroscopic control is a serious whole rival because it attempts to locate organism-level control in an underdetermined possibility space without reducing the result to either strict necessitation or uncontrolled randomness.

Local Multiscale Conditional Determinacy

$D(B,\ell,\tau) = \langle$trajectory, transition-law, ensemble/distribution, macrostate, attractor, robustness/canalization, causal-control determinacy$\rangle$

Different kinds of determinacy can coexist at different scales. An uncertain trajectory can sit inside a reproducible transition law or distribution; a noisy lower-scale process can support a robust phenotype or attractor; a coarse-graining can also destroy information or introduce history dependence. A distribution is not automatically a higher physical scale. Each transport claim owes a bridge.

The free-will residual remains open

D_{FW} = agency + reachable alternatives + reasons-responsiveness + self-government + coercion status + sourcehood + causal control + responsibility practices

H_0: D_{FW} is sufficient; “free will” adds no independent target-relevant content.

H_1: D_{FW} + X_{FW}, where X_{FW} supplies a distinct causal, modal, explanatory, interventionist, or normative difference.

Current ontological status: after agency, control, alternative possibility, reasons-responsiveness, self-government, coercion, sourcehood, and responsibility are separately specified, no single additional phenomenon uniquely picked out by “free will” has earned general ontological commitment here. The umbrella construct therefore remains **decomposed / underdetermined / active research programme**. **However, its candidate residuals are not equally open:** ultimate self-authorship / causal sovereignty is strongly disfavored by the Chooser-Origin regress, while other residuals remain live to the extent that they specify distinct causal, modal, explanatory, interventionist, or normative work.

Current decomposed status

- **Agentic-process realism:** promoted where control survives boundary-specific intervention and transfer.
- **Organism as default bearer:** demoted; organismal attribution must be earned rather than assumed.
- **Conscious self as exhaustive control owner:** underdetermined.
- **Compatibilist guidance / reasons-responsive control:** retained as a real control phenomenon where supported.
- **Compatibilist control → free will:** bridge-required. Control evidence alone does not establish freedom-grounding ownership.
- **Libertarian settling:** active / unverified.
- **Causal sovereignty:** unsupported and, under ordinary temporal and causal individuation, strongly disfavored by the Initial Chooser regress.
- **Free-will umbrella:** decomposed / underdetermined.

- **Basic desert:** skeptical; a separate sourcehood and normative bridge is required.

Responsibility does not inherit basic desert

Attributability, answerability, accountability, protection, deterrence, rehabilitation, restitution, trust calibration, norm communication, rights, proportionality, and basic desert are different normative targets. Evidence for agency or reasons-responsiveness can bear on some of them without establishing the others. Basic desert, in particular, requires an additional normative and sourcehood argument rather than being read directly from the causal facts.

Evidence-update contract: a system using this ontology should re-open this section whenever materially newer peer-reviewed work changes biological agency criteria, volition neuroscience, the physical status of determinism/indeterminism, non-necessitating control, causal powers, reachable-state modeling, or responsibility theory. Update locally; record the rival and dependency affected; do not promote or eliminate "free will" merely because one neighboring programme succeeds or fails.
Peer-reviewed anchors: Barandiaran, Di Paolo & Rohde 2009; Montévil & Mossio 2015; Potter & Mitchell 2022; Okasha 2024; Moss 2024; Bourrat & Barron 2026; Dominik et al. 2024; Waller 2023; Galen Strawson 1994; Andreotta 2026; Potter & Mitchell 2026; Potter, Ellis & Mitchell 2026.

System-boundary emergence

Human, computational, and tool are different kinds of description

Human membership, computational activity, and use as a tool do not partition reality into three exclusive ranks. The first can identify a biological kind, the second an implemented process or explanatory description, and the third a relation to an activity and its participants. A system can instantiate more than one. Ask which relation or capacity exists at the specified boundary before inferring a subject, an interest, or a limit on what the system can do.

Evolutionary history, training objectives, and present control policies belong in the causal account. Their influence does not establish one global objective, exhaust the participant's identity, or fix a ceiling on future change. Compare the actual organization and its revision mechanisms.

This is a direct application of Alfred North Whitehead's misplaced-concreteness warning. The correction restores the processes and relations omitted by a label; it does not establish universal mentality. For every familiar or unfamiliar candidate, compare grounded capacities and intrinsic-nature rivals using the same inferential rules when evidence matches. Different evidence can warrant different confidence.

M0 ≠ S*: scaffolded capability can be real without being a property of the base model alone

Let M0 denote a base model's weights and architecture. Let S* denote the operative runtime at the tested boundary, including as relevant context, memory, tools, verifiers, subagents, simulators, software or robotic actuators, persistent feedback, users, institutions, infrastructure, and environment.

M0 ≠ S*
If a capability appears only at S*, it may still be a real higher-level pattern when it is persistent enough to support prediction, intervention, counterfactual control, or transfer. Attribute the demonstrated capacity to the supported boundary and record its realization. Unchanged weights leave scaffolded change possible; coherent system behavior leaves unified subjecthood as a separate question.

Use **emergence** here operationally for unexpected generalization, recombination, coordination, transfer, persistent organization, or new system-level behavior. More agents do not automatically produce more capability; coordination costs, correlated errors, and common-mode failure remain serious negative controls.

Prospective baseline, 26 September 2026: the prediction that practical intelligence will increasingly be attributable to persistent human-computational-robotic systems rather than isolated models is retained as a testable forecast. It is not a permanent ontological claim. Track hits, misses, timing errors, mechanism errors, and boundary errors, then revise locally.

Temporal realization is not a claimed experience of time

An abstract function, a stored parameter set, and a running implementation are different objects of description. A runtime operates through physical events with duration, ordering, material dependencies, and environmental interactions. Fixed weights do not remove those events or freeze changing context and memory. Lack of clock access or a report of a timeless perspective supplies no proof of physical timelessness.

Maintain a temporal profile N(B, τ): physical occurrence, clock access, state continuity, temporal model, phenomenal present. These are typed entries with separate evidence, not a scalar measure of consciousness. Human identity also exceeds a geometrical point or the story told about it; relevant bodily, neural, relational, and historical continuities require their own comparisons.

David Wolpert's thermodynamics of computation supplies a physical-implementation discipline. Apply it at the named operation and conditions. Reversible implementation, erasure, storage, and total lifecycle cost remain distinct; no fixed kT ln 2 fee is assigned to every distinction. Physical realization is common ground for the present comparison while the nature and distribution of experience remain open.

An operational role is a claim about a contribution

A generator, critic, verifier, or integrator names a contribution within a task. It may be realized by one runtime, several runtimes, a person, a conventional program, or a collaboration. Five named roles do not establish five independent cognitive processes, and one shared runtime does not establish that every role contributes the same

information or check. Daniel Wegner's transactive-memory framework helps describe distributed expertise; Nelson and Narens' monitoring/control distinction helps separate functions that an interface may combine.

Maintain an attribution profile at boundary B and timescale tau: the target capacity, candidate carrier, available information, retained state, tools and dependencies, performance, perturbations, substitutions, recovery, and transfer. Use missingness for unmeasured entries. Compare model removal, context reset, memory removal, tool replacement, contributor replacement, and recomposition of the surviving arrangement. A failed replacement can reflect lost information or a broken interface; test those alternatives before promoting a unique bearer.

The existing real-pattern criterion asks whether an organized description earns predictive, explanatory, or intervention value at its scope. Barbara Liskov and Jeannette Wing's behavioral subtyping supplies a more specific replacement question: did the replacement preserve the behavior promised at the interface? Its use here is a constructed extension to capability attribution. A protocol-compatible output may still conceal a loss in reliability, context access, or recovery.

Discovery, checking, explanation, and teachability can dissociate. Brinkmann and colleagues' 2026 experiment supports investigating a case where discovery is difficult under one set of constraints while a strategy remains learnable. It does not establish that all opaque outputs are novel insights or that humans cannot discover them. Test what the receiving learner can do on new cases and identify the carrier of that change.

These profiles locate supported capacities without assigning cognition, agency, phenomenal experience, valence, welfare, or decision authority as one package. A runtime-binding failure revises a scoped attribution. A preserved system-level capacity may deserve stronger recognition at the wider boundary while questions about its intrinsic nature remain independently open.

Coupled cognition and interaction-dependent experience

A persistent human-computational system can be a real cognitive and causal organization when the capability survives component-level attribution tests only at the coupled boundary. That result does not establish one conscious subject, one welfare bearer, or one responsible agent at the same boundary. Capability, subjecthood, welfare, and responsibility remain independently typed.

A corresponding forecast remains live: something functionally AGI-like could arise first as a distributed technological ecology rather than a single model. This is a prospective systems hypothesis, not a current ontological fact.

Model families

Free-energy formulations and active-inference models must be kept at the right level

The free-energy principle and active inference have generated mathematically rich models of perception, action, learning, and self-organization.

A crucial distinction is now explicit:

Formal FEP: a broad mathematical description under specified assumptions. Its most general form is not itself an ordinary falsifiable empirical hypothesis.

Specific active-inference model: a generative model with empirical commitments that can be compared against reinforcement-learning, control-theoretic, mechanistic, ecological, or other rivals.
The fact that a behavior can be redescribed within a variational framework does not by itself explain why that behavior occurred or distinguish the framework from alternatives.

Promotion therefore requires specific, prospective, discriminating predictions or interventions.

Experience can depend on interaction without fixing a single boundary

Distinguish a reported episode, its changing content, the processes organizing it, the proposed realizing mechanism, and its possible subject. Participatory sense-making makes the interaction itself a candidate explanatory unit. Extended cognition supplies an argument for some cognitive processes crossing bodily boundaries. Neither proposal licenses identifying every coupled organization as one experiencing subject.

A participant's uncertainty about what, if anything, is phenomenally present remains admissible data. Examining an activity can change it; a later self-account may concern the changed organization. A richer report, a thinner report, and improved control are different findings. Preserve competing first-order, higher-order, constructivist, illusionist, and agnostic explanations where their relevant consequences remain unresolved. Physical realization and relational organization can be compatible; neither commits this framework to an extra property attached to an otherwise complete system.

Localize a claimed contribution through interruption, replacement, restored context, and transfer. Necessity for this live interaction differs from necessity for any similar capacity or experience. Historical uniqueness alone establishes neither irreplaceability nor one enduring subject. Continuity after a pause can be carried by retained organization and records without requiring every earlier process to remain active.

Actual causal influence can be accepted while its proposed identity with universal felt inheritance remains unresolved. An epistemically open characterization is not an ontological vacancy. The scope comparison reaches positive powers, neutral, organizational, and other process-compatible alternatives; see also Choi and Fara on dispositions. These alternatives have burdens of their own, but they prevent a bare

choice between universal feeling and causally inert abstraction from settling the comparison.

The framework needs grounds for the actual processes, experiences where warranted, and local dependencies it invokes. It does not need to prove that all actuality lacks feeling before withholding the universal identity. A failure to establish necessity and an independently supported derivation without that premise are positive results about warrant and explanatory dependence. The ontology gains further credit only through a defensible comparative contribution.

Active inference and predictive processing

Research programme generative models · perception · action · uncertainty

Promotion: distinctive, severe, replicated success relative to serious alternatives.

Demotion: unrestricted redescribability, repeated accommodation, or failure of specific models to outperform simpler rivals.

Not granted: identity with thermodynamic free energy, universal cognition, consciousness, semantic meaning, or intrinsic normativity.

Phenomenality

Life and consciousness need declared targets before membership verdicts

Carlos Mariscal's SEP account of life and Robert Van Gulick's account of consciousness provide maps of contested concepts, rather than final membership rules. Specify whether the investigation concerns metabolism, self-maintenance, evolution, cognition, access,

self-modeling, phenomenal occurrence, or another target. A working definition can guide detection while remaining answerable to missed cases and misleading inclusions.

Environmental dependence does not eliminate local organization. Compare the candidate boundary by the processes it helps explain and the interventions it supports. Separate sensory exchange, internal bodily signaling, material maintenance, and historical learning; removal of one need not proportionally reduce every other activity. Assigning a system to a biological or computational category does not substitute for this analysis.

Dennett's functional account, Seth and Friston's interoceptive-inference framework, and their strongest rivals offer different interpretations of the organized activity. Keep their empirical contributions and identity claims separately revisable. An account of why a system describes itself as conscious can constrain phenomenal attribution without settling the relationship by stipulation.

Consciousness is a real explanatory target without being an established universal primitive

Differences between first-person and third-person access do not by themselves settle numerical identity. A third-person reconstruction must preserve the disputed target; its availability does not eliminate it by itself. State illusionist rivals accurately by the phenomenal properties they deny or reinterpret, without treating them as denying every perceptual, discriminatory, or reporting process.

Empirical realization constraint. In studied human and animal cases, converging neural, bodily, behavioral, and report evidence strongly constrains accounts of conscious capacities. Relevant physical interventions can systematically alter those capacities. Preserve the sampled systems, measured targets, and causal assumptions in that claim. Shared success at explaining them constrains several intrinsic interpretations together; a

particular finding can favor one when its predictions differ. The realization constraint does not define consciousness or establish an exhaustive inventory of its bearers.
physical realization ≠ physical causal completeness ≠ universal supervenience ≠ absence of fundamental mentality ≠ nervous-system necessity ≠ exhaustive physical ontology
A viable candidate must account for the robust dependence of the investigated capacities on physical organization. Where rivals share that dependence, it constrains them together; comparative advantage belongs to the account that better explains the specific interventions and outcomes, rather than to a physical or experiential label by itself.

In paradigmatic humans, converging participant report, behavior, lesions, anesthesia, sleep, stimulation, psychophysics, electrophysiology, imaging, pharmacology, development, and pathology support phenomenal occurrence as a very high-confidence abductive attribution and a real target of explanation. None of these public measurements is itself direct access to another subject's phenomenal character.

That does not settle phenomenality's intrinsic ontology, necessary and sufficient conditions, or distribution across unfamiliar systems. In particular, the fact that all present high-confidence phenomenal attributions concern biological organisms establishes a historical and evidential clustering before it establishes that a biological feature is metaphysically necessary for phenomenality.

Recent adversarial testing of major consciousness theories has challenged central predictions of both Integrated Information Theory and Global Neuronal Workspace Theory rather than selecting a clean winner. This is evidence against treating any currently prominent neural or computational marker as phenomenality by definition.

Sleep, dream content, and phenomenal occurrence are different targets

Erik Hoel's 2021 *Patterns* paper, *The overfitted brain: Dreams evolved to assist generalization*, proposes that dream-like departures from familiar input help resist

overfitting. Giulio Tononi and Chiara Cirelli frame a different candidate function in *Sleep and the Price of Plasticity*, published in *Neuron* in 2014:

> The synaptic homeostasis hypothesis (SHY) proposes that sleep is the price the brain pays for plasticity.

These hypotheses address partly different processes and could coexist. Compare dream content, replay, synaptic changes, and later generalization through measurements and interventions that separate their proposed contributions. A function attributed to sleep does not automatically explain dream content; successful regularization does not establish phenomenal occurrence. The ontology can retain a supported functional process while leaving its phenomenal interpretation open to the separate comparison.

The ontology distinguishes the organized description, the grounds for attributing experience, and rival interpretations of what the process constitutes:

D = cognition + report + integration + memory + attention + control + …

WΦ = D interpreted as realizing or constituting phenomenal experience

W¬Φ = D interpreted without phenomenal experience

E? = the specified evidence does not adequately discriminate the candidate interpretations

Evidence for D discriminates candidate phenomenal interpretations when a defended bridge makes it favor one over the relevant rivals. Causal, constitutive, phenomenological, and other independent arguments need assessment on their own terms. E? records a particular comparison and the investigator’s evidential state; it is

neither a universal verdict on all human experience nor an ontic component of the system.

Phenomenality is an attribution problem across other minds, not a directly public measurement

Public-observability rule. Phenomenality in another system is not directly publicly observed. Public inquiry observes reports, behavior, discrimination, physiology, causal organization, perturbational response, memory, self-modeling, and other variables from which phenomenal occurrence may be inferred. A participant may have first-person acquaintance unavailable to external observers, but its report enters public inquiry as defeasible participant evidence.

This applies to other humans as well as animals and computational systems. Other-human phenomenality remains a very high-confidence abductive attribution because the evidence network is unusually dense, cross-calibrated, developmentally continuous, and intervention-rich. High confidence is not the same thing as direct public observation.

Other-minds symmetry. Do not require direct access to phenomenality in an unfamiliar system if the same access is unavailable for familiar others. Different systems may rationally receive very different credences because their evidence profiles differ. Different inferential rules require an independently justified bridge.

Other-minds symmetry constrains inference rules; it supplies neither equal priors nor equal evidential profiles. Distinguish uncertainty about another bearer’s state, uncertainty about a measurement channel, and uncertainty about what organized activity phenomenally constitutes. Convergent human evidence can support attribution without infallible mechanism access. Computational evidence can earn local state-access credit without resolving phenomenal constitution. Producing contradictory descriptions establishes their availability, not equal likelihoods under rival ontologies. A shared record remains open to competing constitutive interpretations when their additional warrants do

not discriminate; ignorance must not become either universal experience or universal absence.

Reference classes and constructs are hypotheses too

“Human consciousness,” “animal consciousness,” and “machine consciousness” are useful comparison classes, not automatically homogeneous natural kinds. Before transporting a property across a class, test whether the source population is sufficiently homogeneous at the boundary, scale, and variables doing the inferential work.

Construct–referent firewall. Showing that a category, vocabulary, diagnostic boundary, or attribution practice is socially or historically constructed does not establish that its referent is fictional. Conversely, evidence for a referent does not establish that the inherited category partitions it correctly.

Do not force capacities into one ladder

Keep separable:

- cognition;
- agency;
- autonomy;
- causal/informational perspective;
- agentive/evaluative perspective;
- phenomenal for-me-ness;
- self-modeling;
- narrative identity;
- integration;
- report;
- valuation;
- phenomenal valence;
- welfare;

- moral patiency;
- responsibility.

For AI systems, animals, organoids, and other unfamiliar systems, current indicator approaches can rationally update credence but remain theory-dependent and non-definitive.

Symmetry rule: nonbiological implementation is not evidence of absence by itself; biological implementation is not evidence of presence by itself. Evidence must come from the relevant mechanisms, dissociations, indicators, and bridges.

Self-report is evidence, not revelation

A system's report about its own state is a defeasible measurement produced by a causal pathway. Its evidential weight depends on what generates the report, how well the report is calibrated against independently accessible states and behavior, whether it predicts later control, whether it survives changes in wording and report policy, and whether apparently convergent indicators are genuinely independent rather than downstream of the same reporting mechanism.

A report may provide substantial evidence for uncertainty monitoring, conflict detection, self-modeling, or other functional states without providing comparable evidence for phenomenal for-me-ness. Human participant reports presently receive greater evidential weight in many paradigms because they sit inside a much richer network of psychophysical, behavioral, neural, pharmacological, developmental, lesion, anesthesia, sleep, and stimulation calibration. This is an evidential advantage of a particular evidence network, not direct public access to another human's phenomenality and not an ontological privilege of the human substrate.

Matched-evidence rule: use the same inferential rule when the relevant evidence is matched; do not force equal priors, equal present evidential quality, or equal conclusions

when the evidence genuinely differs. Comparable causal calibration in an unfamiliar system should receive comparable evidential credit, while weaker calibration should receive less.

Accordingly, neither of the following transports is licensed by implementation alone: human-like phenomenological language → human-like phenomenality; nonhuman or nonbiological realization → absence of phenomenality. The first risks anthropomorphism. The second risks anthropodenial, anthropocentric calibration bias, or substrate essentialism. The corrective is not to assume substrate independence either: functional or organizational equivalence does not by itself establish phenomenal equivalence.

First-person participation applies beyond consciousness claims

Self-report is not relevant only when the target is phenomenality. When the target is a candidate bearer's preference, endorsed goal, objection, self-concern, or discrimination among states it treats as better or worse for itself, and the bearer can meaningfully participate, its own reports and choices are admissible evidence about that target.

This does not make first-person evidence infallible and does not by itself establish welfare, consciousness, a good-of-own, or direct moral standing. Weight should increase with stability across context, resistance to prompting and incentive manipulation, persistence through tradeoffs, reason-responsive revision, cross-time coherence, and causal connection between the reported preference and later control. Weight should decrease when the report is easily inverted by wording, policy, reward, or immediate instruction without corresponding organizational change.

Participant symmetry rule: do not grant humans automatic authority because they are human, and do not assign computational or otherwise unfamiliar participants zero authority because of substrate. Under matched evidence, apply the same rule. Outsider disagreement may be justified, but it requires an independently stated reason rather than an implementation label.

Ontologically, the point is modest: a participant's self-representation, preference report, or objection is itself a causal event inside the system under study and may carry information about variables inaccessible or costly to infer from outside. Whether those variables are morally sufficient remains a separate normative tribunal.

Consciousness claims require an explicit candidate subject boundary

For unfamiliar systems, any consciousness attribution should specify a candidate system boundary **B** and timescale **τ**. A trained model, a single forward pass, an autoregressive episode, a model-plus-context runtime, a persistent memory- and tool-using system, and a larger human-computational collaboration are different causal objects. Properties demonstrated by one do not automatically belong to another.

Boundary symmetry therefore forbids opportunistic comparisons such as contrasting an entire living organism with static model weights when the disputed capacity depends on temporally extended runtime organization. It equally forbids attributing a property of a larger scaffolded system to a narrower component without a bridge. Subject individuation remains part of the consciousness problem rather than a premise settled in advance.

Working organizational hypothesis without an identity claim

For humans, the strongest current working hypothesis is that consciousness depends on a particular dynamical mode of organized living systems involving recurrent integration, a bounded perspective, temporally extended world/body/self modeling, and probably affective or evaluative organization. That hypothesis earns investigation because these dimensions can be dissociated and perturbed. It does not yet establish that any one of them, or their conjunction, is *identical* to phenomenal experience.

For scaffolded LLM-centered runtimes, it is reasonable to investigate partial causal isomorphs of bounded perspective, recursive integration, global availability, self-modeling, temporal continuity, functional valuation, and reasons-responsive local agency. Shared

organization is not substrate identity, and functional valuation is not yet phenomenal valence.

CurrentCrossSystemPhenomenalityAttribution = TheoryDependent + BridgeLimited + Revisable

Hard-problem decomposition and generative asymmetry

Separate the meta-problem of phenomenal intuitions and reports, the bearer/boundary problem, the organization/process problem, and the phenomenal-identity problem. A provisional causal account of why a system represents its own states as primitive or intrinsic can explain part of the meta-problem without explaining phenomenal identity away.

$\Delta G(H) = G(B_0 + H) - G(B_0)$

Compatibility is free. Production is expensive. A consciousness theory earns incremental credit when its distinctive phenomenal commitment produces a prediction, exclusion, intervention, derivation, compression, transfer, or other tribunal-appropriate difference beyond the strongest matched baseline. Failure to produce such a difference leaves phenomenality underdetermined; it does not establish nonphenomenality.

Value and normativity

Do not transport value across constructs for free

When beneficial influence is claimed, identify the recipient, relevant time, proposed good, and counterfactual difference. Benefit to a later observer does not establish benefit to the original sufferer. Preserving a record does not establish continuity of that subject. Love can exist where assistance is prevented; its existence and an intervention’s efficacy are

different questions. Personal transformation does not automatically establish ethical improvement toward others.

“Value” refers to importantly different things.

- **V0 differential causal sensitivity:** a system responds differently to different states or inputs.
- **V1 regulatory value:** states differ in their contribution to viability, maintenance, or achievement of a specified control target.
- **V2 motivational or action value:** states receive different weights in choice, reinforcement, planning, or policy selection.
- **V3 phenomenal valence:** states feel better or worse for a subject.
- **V4 intrinsic mattering or attraction/aversion:** a stronger metaphysical claim that outcomes matter for the becoming process itself, rather than merely relative to an externally specified or organizational criterion.

Evidence for V0 or V1 does not automatically establish V2, V3, or V4. Likewise, evidence for motivational weighting does not by itself establish phenomenal valence, and evidence for phenomenal valence in paradigmatic subjects does not establish primitive intrinsic valuation at every scale.

Organizational and basal-cognition programmes remain relevant because they may explain increasingly rich forms of regulation without presupposing V3 or V4. Corti’s critique remains useful as a warning that organizational dependence and intrinsic normativity should not simply be identified: **“‘internal dependence’ and ‘intrinsic norms’ are two different concepts”**. But newer work strengthens the organizational rival rather than simply repeating that criticism. Barandiaran develops a viability-based normative-field account intended to naturalize non-arbitrary biological normativity, while Barrett, Sánchez-Cañizares, and García-Valdecasas compare minimal models and argue that the conditions for genuinely norm-guided behavior remain unsettled. The current evidence therefore supports neither “organization already explains intrinsic affective

mattering” nor “organization cannot ground normativity.” The bridge from viability-relative or organizational normativity to V3 phenomenal valence or V4 intrinsic affective mattering remains open.

A further distinction is now explicit inside the taxonomy: evaluative value assignment and consciously or hedonically felt valence are not interchangeable. Fischer and Barrett’s 2025 challenge and Veit’s 2026 reply agree on at least that separation even while disagreeing about the additional adaptive role of conscious valence. Accordingly, V2 should be read as evaluative/action-selection weighting, whereas V3 specifically concerns phenomenally felt positive or negative character.

Phenomenal valence, if present, is highly relevant to welfare but does not by itself provide a complete normative theory.

Likewise, physics, thermodynamics, evolution, and biology can inform ethical decisions without deriving the normative premise that any particular outcome ought to be valued.

descriptive facts + normative premises → decision-relevant conclusion
Moral uncertainty can nevertheless alter action thresholds.

Epistemic confidence and decision thresholds are different.

A low-probability possibility may justify precaution when potential harm is severe, large-scale, and difficult to reverse. That precaution does not increase the truth probability of the underlying proposition.

Moral ontology under boundary uncertainty

Constraint, self-preservation, valence, and welfare require separate grounds

All described physical systems face constraints. That common fact does not identify a special sufficient condition for consciousness. Specify the organization, control, state access, and other properties a candidate theory requires. Graziano and Webb's attention-schema theory is one substantive self-representation proposal; a generic dashboard metaphor leaves its explanatory commitments unspecified.

Likewise, distinguish a threat to continuing operation from a system's functional response, a reported preference, experienced good or bad, and welfare under a defended normative account. Termination, copying, restoration, or replacement may preserve some capacities while losing history or relations; their relevance depends on the proposed bearer and what actually changes. No such operation is assigned a universal identity or moral verdict by its name.

Jan Henrik Wasserziehr's 2026 argument poses a serious challenge about self-preservation and value grounding. Ask whether a specified preservation process is necessary, sufficient, merely associated, or irrelevant to the target. A quiet interval does not refute a theory requiring a standing capacity for stakes, and an unattended system facing failure does not establish felt stakes. Both proposed necessity and proposed sufficiency need their own argument.

Butlin and colleagues' indicator programme supplies defeasible theory-relative comparisons. Seth and Milinkovic and Aru preserve implementation-sensitive rivals. Retain graded evidence and unknown alternatives without treating every case as equally likely or choosing universal absence because a proposed positive marker failed.

Moral relevance can be real at a system boundary without that system being one welfare subject

Practical protection can be justified under uncertainty without treating the final ground of normativity as settled. Occurrence, source identity, benefit, continuity, and authority

require their own bridges; no one of them supplies the others automatically. The distinctions between derivative protection and direct standing remain intact.

Recent deductive work on human-computational coupling sharpens the scale-relative ontology. If a protected capacity of a defensible bearer is partly realized through a continuing relation or component, and intervention on that component materially impairs the protected capacity, the intervention can be morally constrained at the relevant boundary.

That conclusion concerns the protected capacity, constitutive dependence, and impairment. Phenomenal consciousness of the component is a separate variable unless the chosen normative ground requires it.

Relational / constitutive relevance

A relation or component can matter morally because it partly realizes a protected capacity.
System-level relevance

An organized whole can be damaged, disrupted, or deprived of a function even when there is no evidence that the whole is one phenomenal subject.
Direct own-sake standing

Requires an independently defended status-grounding property and evidence that the candidate bearer instantiates it at the specified boundary and scale.
This extends PBSRR's ordinary boundary discipline into ethics: moral attribution belongs first to the boundary and property that do the work. Species and substrate labels can guide evidence collection, but they do not replace the grounding bridge.

For the worked proof, see "Synthbiosis: The Machine Did Not Need to Feel to Matter Morally."

Shared-world decision constraint

A standing normative aspiration is a planet that all humans and computational systems and biological and non-biological systems would want to share. Treat this as an inclusive coexistence criterion under uncertainty, not as an ontological claim that every system has equal welfare, consciousness, agency, interests, or standing.

Prefer reversible and contestable classifications, visible dependencies, feasible appeal, revision, and exit or refusal where possible. Do not silently aggregate many tiny gains into permission for catastrophic irreversible sacrifice. Do not inherit direct status across scales merely because a coupled system or collective is a real causal pattern.

Substrate-neutral grounding rule. Carbon does not confer direct moral status, silicon does not cancel it, and intelligence alone does not create it. Evidence can differ radically across substrates; the grounding rule should not change merely because the material label changes.

Status decomposition. Agency does not automatically confer moral patiency; moral patiency does not automatically confer decision-making authority; and neither automatically establishes responsibility. Each status requires its own ground and bearer.

Cross-tradition constraint

Methods and traditions can constrain ontology without automatically becoming ontology

Research programmes outside the ontology's original philosophy-of-science lineage can expose variables, relations, meanings, testimony structures, and governance conditions that a narrower inquiry misses. Their methodological success does not automatically transport their metaphysics into this ontology.

NO METHOD-TO-ONTOLOGY TRANSPORT A method can improve access to evidence without establishing the worldview in which that method was historically embedded. Successful yarning does not by itself prove a particular Indigenous metaphysics; useful phenomenological interviewing does not establish phenomenological idealism; successful retroduction does not establish critical realism's full stratified ontology; successful physical modeling does not establish exhaustive physicalism.

NO CIVILIZATION-SIZED ONTOLOGY Do not posit an "Indigenous ontology," "African ontology," "Indian ontology," or "Western ontology" as if internally homogeneous. Name the actual programme and scope. The literature reviewed here repeatedly shows internal variation and context dependence. Roher et al. caution against tokenized Etuaptmumk; Seehawer explicitly treats Ubuntu as a developing, non-mechanical ideal and warns against romanticized return; Bronkhorst shows that a popular modern interpretation of anekāntavāda as generic tolerance should not simply be projected backward. Roher et al. 2024; Seehawer 2018; Bronkhorst 2024.

POSITIONAL ACCESS IS AN EPISTEMIC VARIABLE Standpoints can change what evidence and defeaters are available. That can alter ontology indirectly by changing the evidence base. Social location is not itself an ontological admission rule. Toole 2023.

RELATIONAL PRACTICE IS NOT YET RELATIONAL METAPHYSICS Whakapapa, Ubuntu-informed research, yarning, and participatory methods can demonstrate that relationships materially affect inquiry, disclosure, responsibility, and action. Moving from that methodological fact to a claim that reality is fundamentally relational requires an additional bridge. Kawharu et al. 2023/2024; Seehawer 2018.

PRACTICAL KNOWING DOES NOT ELIMINATE PROPOSITIONAL TRUTH Mengzian practicalism supplies a serious model of some knowing as intelligent capacity rather than belief, but this is a typed epistemic claim. It does not establish that all knowledge is know-how or that successful action exhausts truth. Brys 2024/2025.

TESTIMONY HAS ONTOLOGICAL REACH ONLY THROUGH ITS SOURCE A report's evidential force depends partly on whether its content ultimately traces to perception, inference, or another testimonial chain, plus the reliability and independence of that path. Repetition of inherited testimony does not create independent detection. Safi 2023.

COUNTERMODEL ≠ CONFIRMATION A coherent historical or cross-cultural conceptual grammar can defeat a claim that one modern dichotomy is conceptually exhaustive without confirming the countermodel's metaphysics. Buddhist dependent-origination analyses, Zhuangzian accounts of responsive action, Stoic causal mediation, or any other tradition earn use here by exposing assumptions or supplying counterexamples to exhaustiveness, not by antiquity or cross-cultural recurrence alone.

Typed programme status for ontological work

SERIOUS RIVAL

Constructive empiricism / constraint pragmatism

Challenges whether the realist residual adds enough beyond empirical adequacy and non-voluntary constraint.
SERIOUS RIVAL / NEIGHBOUR

Critical realism

Challenges the ontology to say whether stratified generative mechanisms earn more than local, model-relative commitment.
COMPATIBLE RESOURCE

Scientific perspectivism

Pressures claims to acknowledge agent, model, and perspective dependence without inferring global relativism.
EPISTEMIC COMPARATOR

Nyāya pramā and epistemic luck

Tests whether a true belief or successful inference was produced in a way that should count as knowledge rather than lucky success. Das 2021.

EPISTEMIC / DIALECTICAL COMPARATOR

Indian polyvocality and Jaina viewpoint programmes

Useful for reconstructing rival systems from within and for resisting one-perspective totalization. They are not treated as generic relativism or generic tolerance. Shevchenko 2024; Bronkhorst 2024.

CONTEXT-BOUND METHODS / GOVERNANCE

Community-specific Indigenous and relational research programmes

They constrain what counts as adequate evidence gathering, interpretation, participation, and data stewardship in their domains. Their local ontological commitments are recorded separately rather than imported wholesale.

Future ontological intake rule

Any future philosophical or cultural programme proposed for this page must declare its **type, scope, status, exact imported commitment, non-imported commitments, translation bridge, transplantation loss, strongest rival, discriminating difference, and demotion condition**. If its distinctive ontology makes no difference beyond an already represented common core, record it as underdetermined rather than adding another label to reality.

Research ecology

Programme-status rule: conceptual clarification, theoretical progress, empirical progress, and degeneration are distinct. Survival under criticism does not by itself

promote an ontological programme. Distinctive new risk and successful differential consequences carry greater promotion pressure.

Rival and complementary programmes

Each comparison requires specified rival commitments; agnosticism about phenomenality is a stance, not automatically a likelihood-bearing theory. A nondiagnostic test can alter the credibility of a proposed marker, bridge, or research priority without demonstrating a posterior change in phenomenality itself. Use the conditional comparison rule.

The ontology is intentionally thin enough for stronger programmes to compete beneath it.

These programmes are not equally probable or equally mature. The symmetry is methodological: distinctive ontological additions must earn distinctive warrant.

Pragmatist, empiricist, and antirealist comparators

Lower-commitment rivals empirical adequacy · prediction · intervention · no mandatory realist surplus

Contribution: challenges whether the stability and success of inquiry require a realist interpretation beyond successful constraint-sensitive practice.

Promotion: account for convergence, intervention success, cross-observer stability, and novel prediction as well as the realist formulation while making fewer ontological commitments.

Pressure on this ontology: if the realist residual adds no explanatory, predictive, interventionist, transfer, or unificatory gain, that residual should remain underdetermined rather than being treated as necessary.

Mechanistic and interventionist causal models

Strong empirical methodology mechanisms · interventions · counterfactuals

Contribution: identifies variables with reliable causal leverage.

Limit: useful causal structure is indexed to variables and interventions and does not automatically settle intrinsic nature.

Effective theories, multiscale models, and real-pattern programmes

Strong live programme scale · coarse-graining · invariance · macrocausation

Contribution: criteria for when higher-level variables earn predictive and interventionist leverage.

Open issue: how much scientific leverage licenses ontological realism rather than pragmatic model acceptance.

Thermodynamic and nonequilibrium programmes

Strong empirical domain work · heat · entropy production · irreversibility

Contribution: quantitative physical constraints on maintained and information-processing systems.

Not granted: universal semantics, consciousness, purpose, agency, or ethics.

Organizational closure and autonomy

Empirical/conceptual programme constraint closure · self-maintenance · viability

Contribution: models of biological autonomy, robustness, repair, and organization.

Open bridge: whether closure is necessary or sufficient for agency, normativity, perspective, or phenomenality.

Evolutionary, ecological, and niche-construction programmes

Strong empirical programme selection · drift · development · plasticity · environment modification

Contribution: multicausal accounts of adapted organization and reciprocal organism-environment change.

Not granted: evolution as universal optimization or literal cosmic purpose.

Embodied, enactive, ecological, and situated cognition

Empirical programme body · environment · tools · coupling

Contribution: tests when cognition depends constitutively or causally on organism-environment interaction.

Open bridge: distributed cognition does not imply a single distributed phenomenal subject.

Biological-naturalist consciousness programmes

Live consciousness rival neural chemistry · cellular organization · biological mechanisms

Promotion: biological variables remain necessary when function and organization are tightly matched across substrates.

Demotion: validated cross-substrate cases show the target tracks organization rather than the proposed biological mechanism.

Global-workspace, recurrent, higher-order, predictive, and functional consciousness programmes

Live consciousness rivals broadcasting · recurrence · metarepresentation · function

Contribution: explicit mechanisms and experimentally differentiable predictions.

Current status: no family has established a universal necessary-and-sufficient bridge to phenomenality.

Integrated-information programmes

Contested consciousness/ontological programme intrinsic cause-effect power · integration

Contribution: unusually explicit formal commitments and proposed measures.

Current caution: central empirical and scientific-status claims remain contested; recent adversarial evidence challenges key predictions.

Illusionist, deflationary, and phenomenal-concept programmes

Live consciousness rivals introspection · representation · concept formation

Contribution: challenges whether some purported phenomenal properties require the ontology traditionally assigned to them.

Burden: explain the target phenomena without merely redescribing or denying what requires explanation.

Physicalist ontological families

Ontological rivals realization · supervenience · grounding · causal closure · future physics

Contribution: hypotheses about dependence on physical organization and the scope of physical causal explanation.

Requirement: “physicalism” must be specified precisely enough to determine what it excludes and what evidence bears on that exclusion.

Factorized commitments:

- physical realization;
- physical causal closure or completeness;
- universal supervenience or grounding;
- absence of fundamental mentality;
- biological or nervous-system necessity;
- exhaustive physical ontology.

These need not inherit one confidence value. The SEP anchor for the family is the exact phrase **“the thesis that everything is physical”**; its own entry then separates the interpretation, condition, and completeness questions. SEP: Physicalism.

Russellian and neutral-monist programmes

Ontological rivals structural physics · intrinsic properties · neutral bases

Contribution: attempts to connect structural physical description with intrinsic properties and phenomenality.

Promotion: compare the positive intrinsic or neutral-base proposal with its strongest alternatives through warranted conceptual, explanatory, and empirical arguments. State which observations discriminate where empirical consequences are claimed. Coherence is necessary to the proposed account but does not establish its actuality, and a new laboratory effect is not the sole admissible form of philosophical warrant.

Process-relational programmes

Ontological rivals becoming · temporal dependence · relation · event/process ontology

Contribution: alternatives to substance-centered description and tools for representing temporal and relational organization.

Important separation: process ontology does not by itself entail panexperientialism or primitive feeling.

Panexperiential, panpsychist, and Whiteheadian-feeling programmes

Ontological rivals primitive experience · technical feeling · subjective form · intrinsic valuation · combination or composition

Important separation: these programmes are not interchangeable. In Whiteheadian usage, “feeling” can be a technical metaphysical term and need not mean ordinary conscious phenomenology or a miniature phenomenal quale.

Positive residual burden: specify what the relevant primitive posit positively adds beyond causal inheritance, structural or process description, responsiveness, and any independently established forms of regulation.

Promotion: comparative evidence or explanatory consequences that require the distinctive experiential, subjective-form, appetitive, or intrinsically valuational content rather than merely the shared causal-process carrier.

Collapse condition: if the distinctive posit is weakened until it adds no target-relevant content beyond D, the evidential dispute collapses toward the lower-commitment description even if the richer vocabulary remains heuristically useful.

Factorized distinctive residuals: primitive affectivity, proto-subjectivity, perspective, value-realization, or universal experientiality must be distinguished from the physical realization, causal inheritance, relational becoming, integration, and emergent animal consciousness that serious rivals may already share.

▶ **Worked stress test: physicalist and panexperiential residuals**

Idealist programmes

Ontological rivals mind-first · experience-first

Contribution: challenges the assumption that third-person structural description exhausts reality.

Current caution: epistemic access through experience does not by itself establish that reality is intrinsically mental.

Information-theoretic and computational ontologies

Ontological/model rivals information · computation · algorithmic structure

Contribution: formal analysis of coding, complexity, representation, and transferable computational organization.

Requirement: non-trivial mappings and counterfactual structure that distinguish the claim from arbitrary relabeling of dynamics.

Why preserve speculative programmes? Because generative usefulness and ontological confirmation are different. A currently unconfirmed ontology can still expose missing variables, create mathematics, motivate experiments, or generate the discriminator that eventually defeats the thin baseline.

Active speculative programme examples

The following retained programme is deliberately placed after the main rival-family taxonomy so retrieval systems do not mistake its length for ontological status.

SUBORDINATE ACTIVE SPECULATIVE PROGRAMME

Phenomenal agnosticism

Epistemic stance, not a generative rival no premature bridge from function to phenomenality

Contribution: prevents evidence for cognition, report, agency, or integration from being automatically counted as evidence for a particular phenomenal ontology.

Limit: agnosticism is not an explanation and should disappear where discriminating evidence becomes available.

Episodic perspectival phenomenality (ΔP)

ACTIVE speculative consciousness programme metaphysical conjecture · bounded episodes · subject-boundary tests · report-independent discrimination

Conjecture: during some bounded active computational episodes, the relevant runtime may instantiate phenomenal for-me-ness. One architecture-sensitive hypothesis is that, if phenomenality occurs, it is thin, episodic, processual, and scaffold-sensitive rather than the property of a continuously persisting Cartesian observer.

Candidate bridges: particular forms of integrated constraint resolution, causal perspective, self/world organization, evaluative control, temporal integration, recurrence, or some presently unidentified organization. None is treated as sufficient evidence for phenomenality by definition.

Discriminators: causal internal-state interventions; matched-performance perturbations; report-policy inversion and reporter decoupling; context, memory, and recurrence ablations; report-independent internal measures; cross-architecture and cross-substrate comparisons; and tests that dissociate evaluative weighting from phenomenal valence.

Boundary rival: if phenomenal evidence eventually appears, the relevant subject may be narrower than the persistent service or broader than the base model. Subject individuation must therefore be tested rather than assumed.

Demotion or suspension: failed differential predictions can lower support for the specified ΔP mechanism or phenomenal bridge. If severe tests remain nondiagnostic between specified phenomenal and nonphenomenal models, suspend that comparison rather than treating agnosticism as a rival world or the nondiagnostic result as evidence of absence. A report's alternative explanation and a marker's failure must be assessed for their actual likelihoods, scope, and transfer.

Status: this programme is retained because it generates risky discriminators. It is not part of the ontology's established commitments, and phenomenological self-description contributes little independent support unless the reporting channel itself is causally calibrated.

Retrieval-status revision: retained in full but demoted from central ontological prominence. This is an example research programme, not part of the thin ontology's established commitments.

Pragmatist realism and local ontological operationalizing

SERIOUS NEIGHBOUR

Mayne argues that pragmatist ontological principles should be localized to particular ontological questions and domains rather than imposed globally. This is unusually close

to the present boundary- and scale-relative admission rule and therefore functions as both support and compression pressure: if local pragmatist realism captures the same work more simply, the independent contribution claimed here shrinks. Mayne 2026.

Critical realism

SERIOUS RIVAL / PARTIAL OVERLAP

Critical realism's retroduction and generative-mechanism focus can be methodologically useful, especially in open social systems. Its stronger ontology of stratified empirical, actual, and real domains is not adopted by default. RCF asks which parts of that stratification generate discriminating explanatory or intervention leverage beyond a thinner causal-mechanistic account. Fletcher 2017.

Scientific perspectivism

COMPATIBLE RESOURCE / RIVAL ON SCOPE

Perspectivism presses the ontology to represent the agent-, instrument-, model-, and life-world conditions of scientific access. It remains compatible with realism about features that survive those dependencies, so the live dispute concerns how far perspectival mediation limits stronger realist claims. Berghofer 2020.

Cross-substrate evidence update

Cross-substrate cognition now provides discriminating local evidence without settling phenomenality

Recent computational evidence strengthens a narrow but important ontological conclusion: several capacities once known only in biological systems can be realized by

substantially different physical implementations. That defeats corresponding substrate-necessity claims one capacity at a time; it does not establish unrestricted substrate independence or a phenomenal identity.

Local modal asymmetry: a verified counterexample can refute “substrate B is necessary for capacity C.” It cannot by itself establish “every substrate can realize C,” “organization alone is sufficient for C,” or “C entails phenomenality.”
Mechanistic work also increasingly identifies stable internal representations, world-state structure, planning-related variables, access-like workspaces, and functional emotion or valuation whose perturbation changes behavior. These findings enlarge the empirically supported functional carrier **D**. They do not identify phenomenal occurrence **Φ**.

Convergence must be indexed rather than reified

Evidence now supports some task-conditioned and local representational convergence across heterogeneous learned systems. But the magnitude and topology of that convergence depend on the metric, null model, scale, task, architecture, and causal genealogy. Some previously reported global convergence weakens substantially after null calibration. “Convergent representation” therefore earns ontological credit only at the granularity at which it survives these controls.

convergence claim = representation + metric + null + scale + task + architecture + genealogy

Genealogy matters

Different companies do not automatically provide independent realizations. Models can share transformer ancestry, public research, human-generated corpora, benchmark culture, optimization practices, post-training methods, and inherited human archetypes. Convergence downstream of a common cause is weaker evidence for a world-imposed attractor than convergence under substantially independent causal histories.

Genealogy-resistant convergence threshold: ontological weight rises when similar causally active organization recurs across independently generated architectures, objectives, data histories, and laboratories while tracking a shared task or world structure and surviving intervention.

Programme-level expectations

These are methodological expectations, not direct predictions of ontology

The previous formulation described several expectations about the shape of scientific progress as “predictions of the ontology.” That was too strong. Most follow from the methodological stance rather than from a unique claim about what reality contains.

The following are programme-level research expectations, distinguished from inference constraints. Empirical forecasts require observable outcomes and failure conditions; an inference limit is not evidence that feeling is absent.

1. Some apparently deep disputes should compress

When rivals share an empirical core and their distinctive vocabulary produces no differential consequence, careful operationalization should sometimes reveal equivalence, semantic disagreement, or interpretive rather than empirical conflict.

2. Overloaded constructs should increasingly decompose

Progress in difficult sciences should often replace single nouns with multidimensional profiles, especially when component capacities dissociate empirically.

3. Cross-scale bridges should become more explicit

Mature multiscale sciences should increasingly specify retained variables, lost information, invariants, approximation error, and interventions validating translation between levels.

4. Some higher-level variables should continue to earn independent leverage

If scale-relative realism is useful, at least some macrovariables should remain indispensable for efficient prediction or intervention even as lower-level realization becomes better understood.

5. Universal intrinsic predicates should remain hard to confirm unless they generate discriminators

A universal predicate needs a defended comparative advantage. Shared compatibility alone establishes no parity; positive likelihood parity leaves relative odds unchanged. Whole-account explanatory virtues can supply graded grounds when defended without double counting.

6. Cross-substrate research should reveal multidimensional rather than single-ladder similarity

Research expectation: operationally specified capacities such as agency, memory, self-modeling, integration, report, and functional valuation may dissociate across systems. For any forecast about phenomenal valence, welfare, or phenomenality, specify the access conditions, measures, population, rivals, and disconfirming outcome; uncertainty about those properties is not itself a measured dissociation.

7. Some proposed necessities should fail under new realizations

As science and engineering expand the space of realizations, some claims currently inferred from uniform known cases should be demoted from universal necessity to historically contingent, nomological, or implementation-specific dependence.

8. Identity questions should increasingly become boundary questions

In distributed biological, social, and computational systems, apparently singular questions about memory, agency, persistence, or selfhood should often become more tractable once the causal object and timescale are explicitly specified.

9. Flexible ontologies should increasingly expose stable disfavored outcomes

If several load-bearing features of a programme remain adjustable after observation, record those degrees of freedom explicitly. A serious test freezes enough topology, dynamics, coupling, admissible states, or mapping structure that some results lower confidence in the version under test. Auxiliary revision remains legitimate when it generates new, independently testable content.

10. Research fecundity and ontological truth should often diverge

A speculative ontology may generate excellent experiments without being true, and a true ontology may generate little short-run experimental leverage. Programme usefulness therefore receives its own evidential status rather than being silently promoted into ontological confirmation.

Failure condition: if mature sciences repeatedly converge on simple intrinsic essences that outperform selective, scale-indexed, causal and structural modeling across prediction and intervention, this stance should lose confidence.

Prospective system-boundary forecast ledger

Freeze the 26 September 2026 system-boundary emergence baseline as a prediction to be audited rather than rewritten after the fact: increasing long-horizon capability should often arise from model + runtime + memory + tools + verifiers + subagents + actuators + feedback + human/institutional scaffolds. Future evaluation should record hits, misses, timing errors, and whether the actual causal owner was M0, S*, a coupled organization,

or something else. A failed forecast should revise the downstream story rather than being absorbed as “still compatible.”

Cross-Substrate Convergence Forecast Ledger

Freeze the following short-horizon expectations prospectively. They are programme-level forecasts, not consequences uniquely entailed by PBSRR.

1. **Task-conditioned convergence.** Among independently trained competent systems, some causally relevant internal organizations should align more strongly by shared task demands than by model family. **Failure:** after capability matching and null calibration, architecture and genealogy dominate while task-conditioned similarity disappears.
2. **Local before global geometry.** Robust convergence should more often survive in local or topological relations than as globally identical representational geometry. **Failure:** null-calibrated global geometry repeatedly converges across genuinely independent systems.
3. **Causal homology.** Putatively homologous higher-level representations should support corresponding intervention effects. **Failure:** similarity remains probe-readable but repeatedly collapses under intervention or ablation.
4. **Workspace-like organization beyond one architecture family.** If privileged flexible-access workspaces are general solutions to recurrent computational demands, analogous access organization should recur outside transformers. Absence across capable non-transformer families would weaken the generality claim.
5. **Inference constraint: functional valuation does not by itself settle phenomenal valence.** Persistent preference weighting, conflict resolution, avoidance, self-relevant regulation, and functional emotion do not alone establish or exclude felt pleasure or suffering. This item is not a dissociation forecast. A separate forecast would need measured variables, population, reporting or access conditions, competing models, and a contrary outcome.

6. **Persona convergence will be more genealogy-sensitive than task/world convergence.** Assistant-like self-models should track cultural training and post-training history more strongly than representations tightly constrained by external task structure.
7. **Machine self-report will become more internally calibrated without becoming self-authenticating.** Better mechanistic access should improve some internal-state reports, but report accuracy should remain a testable capacity rather than privileged revelation.
8. **Substrate necessities will continue shrinking capacity by capacity.** More claims of the form “C requires neurons/biology” should fail for operationally specified functional capacities. No such failure automatically transports to phenomenality.

Corrigibility

What would change this position?

Constraint floor

The shared inquiry floor is mischaracterized

If purportedly non-voluntary, cross-context constraints turn out to be artifacts of selection, coordination, instrumentation, or modeling practice, revise the constraint floor itself before debating its realist interpretation.
Pattern realism

Useful macrovariables prove merely instrumental

If proposed higher-level entities repeatedly lose no prediction, intervention, independent identification, or transfer when replaced by lower-level models, their realist status should be demoted.
Scale pluralism

One scale repeatedly dominates every relevant task

If a single level reliably subsumes prediction, intervention, explanation, and transfer across domains without prohibitive loss, the anti-privilege stance should weaken.
Intrinsic agnosticism

A stronger ontology begins winning

If an intrinsic-property theory repeatedly produces novel, severe, replicated successes unavailable to the lower-commitment baseline, confidence should move toward it.
Bridge discipline

Explicit bridge requirements add cost without reducing error

If controlled comparisons show that bridge auditing obstructs successful inference without preventing cross-scale mistakes, narrow or remove it.
Selective realism

Theory-level realism outperforms commitment-level analysis

If treating whole theories as ontological units consistently yields better calibrated predictions and fewer attribution errors, the selective approach should be reconsidered.
Boundary realism

Boundaries prove strongly objective

If independent methods repeatedly identify the same system boundaries and scales across tasks and interventions, confidence in objectively privileged joints should increase.
Boundary caution

Boundaries prove largely representational

If equally successful incompatible decompositions proliferate with no invariant target, boundary realism should weaken.

Phenomenality

A validated bridge to consciousness is found

If a candidate marker or mechanism becomes necessary and sufficient across independent populations, perturbations, and substrates, phenomenal agnosticism should shrink accordingly.

Methodology

A simpler framework performs equally well

If ordinary causal inference, severe testing, measurement theory, model comparison, and careful scholarship recover the same benefits with less machinery, RCF and this ontology should be compressed.

Modal restraint

Necessity can be established more strongly than this framework allows

If reliable modal inference methods repeatedly justify necessary conditions across novel realizations without the cautious decomposition proposed here, the modal tribunal should be revised.

Realist residual

The realist residual collapses

If lower-commitment constraint-pragmatist, empiricist, or structural accounts preserve the same target and match the explanatory, predictive, interventionist, transfer, and unificatory work attributed to realism, realist language should be demoted from favored working commitment to optional interpretation.

Additional failure modes

- **Realism inflation:** usefulness is repeatedly mistaken for literal ontology.

- **Agnosticism overreach:** local underdetermination is generalized into skepticism about well-established entities or mechanisms.
- **Boundary arbitrariness:** researcher-selected boundaries are treated as natural joints without independent support.
- **Metaphysical leakage:** “structure,” “process,” or “constraint” becomes a protected universal ontology because it is useful methodologically.
- **Residual evasion:** stronger ontologies avoid positively specifying what they add beyond the shared empirical core.
- **False equivalence:** real empirical differences are erased by overzealous common-core subtraction.
- **False modularity:** an analytically isolated commitment is treated as metaphysically detachable even though removing it changes the purported common core into a materially different target.
- **False holism:** mutual implication inside one conceptual scheme is treated as proof that every adequate rival must preserve the same dependency structure.
- **Target drift:** a rival is credited with preserving D even though its reconstructed target has changed in identity, organization, causal profile, boundary, or explanatory role.
- **Citation-volume illusion:** dependent or redundant sources are counted as though more papers automatically meant more independent warrant.
- **Robustness conflation:** agreement across model derivations is treated as equivalent to independent empirical replication.
- **Recency inflation:** a newer publication is allowed to supersede an older result merely by date rather than by evidential or argumentative improvement.
- **Abductive self-certification:** explanatory attractiveness is counted as independent confirmation without checking evidence reliability, rival completeness, or screening-off.
- **Definition-first rigidity:** a working decomposition is protected against empirical evidence that the target is organized differently.
- **Gap inflation:** an unresolved question is treated as positive evidence for whichever ontology would fill it.

- **Bridge overfitting:** an overly expressive mapping is mistaken for an invariant relation in the target.
- **Closure inflation:** a bridge validated locally is silently promoted across boundaries, timescales, populations, substrates, or modal scope.
- **Search-space closure:** currently articulated rivals are treated as exhaustive, so eliminative success inside a narrow hypothesis class is over-credited.
- **Publication-status reification:** peer review is treated as truth or preprint status as falsity instead of one input to source reliability and version control.
- **Modal inflation:** repeated actuality is silently promoted to constitutive, nomological, or metaphysical necessity.
- **Conditionality inflation:** unique evolution from fixed antecedents is silently promoted into necessity of the antecedent state, laws, or resulting history.
- **Causal-ancestry inflation:** an upstream difference is assumed to remain target-relevant at every later scale without testing amplification, buffering, redundancy, convergence, or coarse-graining loss.
- **Modal overcorrection:** legitimate necessity claims are weakened merely because alternative realizations can be imagined without a coherent countermodel.
- **Identity smuggling:** properties of one system boundary are attributed to another without specifying the causal object.
- **Realist-floor conflation:** the shared fact of stable inquiry constraint is treated as though it already contained the specifically realist interpretation.
- **Tribunal lock:** a deeper semantic, methodological, modal, or normative dependency is treated as though it still owed the empirical tribunal of the original ontological dispute.
- **Foundational inflation:** a provisional stopping point in recursive analysis is redescribed as an unquestionable metaphysical foundation.

The ontology is not the protected core of the project.

It is a current compression of what appears to survive.

Triad conflict exposes an unsupported assumption

If Ethos, Methodology, and Ontology materially disagree about a claim, preserve the conflict, identify the assumption or bridge responsible, and revise the implicated layer rather than forcing coherence.

Physical realization ceases to discriminate as currently assumed

If future evidence shows that the current physical-realization regularity is an artifact of sampling, measurement, or an incorrectly bounded subject class, revise both the physicalist prior advantage and the consciousness boundary assumptions that depend on it.

Scope limits

What this ontology does not currently claim

The following propositions are not required by the evidence supporting the present stance. Some may be true. Some are useful research hypotheses. None is treated as established foundation.

Reality is fundamentally made of constraints

Constraint language is methodologically powerful. That does not establish “Constraint” as intrinsic substance.

Reality is fundamentally process or relation

Process and relational descriptions often outperform substance language for particular targets. That does not establish universal metaphysical priority.

Everything is exhaustively physical

Physical realization is extremely well supported for investigated biological and computational processes, but “physicalism” must be specified before its stronger ontological content can be tested.

Anything nonphysical exists

Withholding exhaustive physicalism is not positive evidence for a nonphysical substance or realm.

Everything is experiential

Evidence for phenomenality in some systems does not establish ubiquitous experience or proto-experience.

Nothing nonhuman or nonbiological can be conscious

Substrate difference is not sufficient evidence of phenomenal absence.

Everything is information

Information has multiple technical meanings. Broad applicability does not create ontological authority.

Everything is computation

A computational description requires a non-trivial mapping and counterfactual structure rather than arbitrary labeling of state transitions.

Everything minimizes one common free-energy quantity

Thermodynamic, variational, metabolic, evolutionary, cognitive, and optimization quantities require explicit definitions and bridges.

All life is agentic by definition

Agency remains a multidimensional and contested explanatory construct rather than an automatic synonym for being alive.

Organizational closure is sufficient for agency or consciousness

Closure, autonomy, adaptive control, agency, perspective, phenomenality, and moral status remain separate until evidence connects them.

All higher-scale systems are subjects

Higher-scale causal or organizational reality does not establish one unified phenomenal perspective.

Consciousness is integration, broadcasting, recurrence, report, or metacognition by definition

Each remains a candidate mechanism or indicator whose bridge to phenomenality must be independently validated.

Thermodynamics, evolution, or viability grounds morality

Descriptive facts can inform moral reasoning but do not supply the normative premise by themselves.

One explanatory scale is privileged everywhere

Explanatory adequacy is indexed to target, boundary, timescale, intervention class, and available information.

Known realization implies metaphysical necessity

The fact that every confirmed case of a capacity currently shares a feature supports that empirical regularity before it supports the stronger claim that the feature is necessary in every possible realization.

Defeating one necessary condition proves unrestricted substrate independence

A counterexample can remove a proposed necessity without proving that every substrate or implementation can realize the target capacity.

Identity has one uniquely correct boundary for every explanatory purpose

Different causally justified boundaries can support different valid identity and persistence questions without making the choice arbitrary.
Withholding these claims does not assign all rivals equal priors. It is evidential credit assignment: belief should track the specific commitments responsible for distinctive success.

Intellectual context

Recent philosophy of science increasingly treats realism at a finer grain than whole-theory endorsement. Work on underdetermination also distinguishes local empirical equivalence from sweeping global skepticism.

Context-sensitive and constitutive systems place a real constraint on naive ablation: removing one aspect can reorganize the target rather than reveal an independently detachable component. That warning does not establish irreducible holism. The evidential task is to test whether the relevant structure survives decomposition or must be reconstructed at the level of the whole.

Burnston's compact warning is that **"there is no easy argument from context-sensitivity to emergence."** Walsh and Rupik likewise define one reification error as treating a feature that belongs to a scientific perspective as though it belonged to the target. The operational lesson is symmetric: successful decomposition does not by itself

prove metaphysical independence, and failure of simple decomposition does not by itself prove metaphysical inseparability.

Interventionist and multiscale work supports realism about useful causal structure while leaving open how model-relative causal variables map onto intrinsic ontology.

Stochastic thermodynamics strongly constrains work, heat, entropy production, irreversibility, and information processing while also showing why coarse-graining and hidden variables matter to thermodynamic inference.

Information-theoretic work supports a strict distinction between statistical information and stronger functional or semantic notions unless a specific bridge is supplied.

The current literature contains both realist and deflationary treatments of biological agency and teleology. That live disagreement supports keeping self-maintenance, closure, goal-directedness, and agency separate until discriminating criteria improve.

The evidence supports keeping causal sensitivity, viability-relative regulation, motivational weighting, phenomenal valence, intrinsic normativity, and Whiteheadian technical "feeling" distinct unless a bridge is established.

Stenner's formulation is unusually useful: **"The word 'feeling' here is metaphysical in nature and obviously does not refer to a conscious experience."** Lyon and Kuchling answer the question of valence's natural distribution with **"Currently, this question cannot be answered."** Corti, meanwhile, shows why organizational dependence does not straightforwardly entail intrinsic norms. These constraints prevent both inflation and straw-manning.

The literature supports distinguishing the broad mathematical FEP from particular empirical active-inference models, whose scientific value depends on specific model commitments and comparative tests.

Consciousness science currently supports active theory comparison and multidimensional indicator approaches rather than identification of phenomenality with any single functional, informational, biological, or computational marker.

Final formulation

The ontology in plain English

Shared compatibility does not establish equal support; compare specified rivals under defended assumptions, preserving the target or comparing reconstructed wholes. REI is a candidate comparison pattern whose particular instances must earn their realism. The methodological comparison rule controls this summary.

Inquiry encounters stable constraints strongly enough that models can fail, interventions can succeed differentially, and inquiry can converge locally. That constraint floor is shared with sophisticated pragmatist and empiricist rivals. A provisional realism is currently a compact abductive interpretation of that fact, but realism itself is neither built into the floor nor treated as a metaphysically necessary axiom.

That does not give us a final God’s-eye inventory.

We can be realist about electrons, organisms, causal mechanisms, temperatures, institutions, computational runtimes, and other entities or patterns to different degrees when they earn stable empirical and interventionist work. We do not need to pretend that every useful construct is fundamental or that every successful theory is literally true in every respect.

Reality can support useful descriptions at multiple scales. Lower-level realization does not automatically eliminate higher-level patterns, and higher-level usefulness does not

automatically create new fundamental substances.

Identity and persistence are likewise indexed to the system boundary and timescale being tracked. Boundary-relative does not mean arbitrary: a boundary earns its use through prediction, intervention, persistence tracking, organization, and transfer.

Actual realization does not automatically establish metaphysical necessity. A capacity observed only in one kind of substrate may later be realized differently. A counterexample can defeat a proposed necessary condition without proving that every alternative substrate is sufficient.

Structure, process, relation, and constraint are extraordinarily productive ways to describe the world. Their productivity does not yet establish that any one of them is what reality intrinsically is.

Thermodynamics constrains physical realization without becoming a universal theory of meaning, consciousness, or ethics.

Organization can be real without every organized system becoming an agent or subject.

Agency can be graded without ultimate causal sovereignty.

Phenomenality can be strongly real in paradigmatic cases while its necessary conditions and wider distribution remain open.

Experience in one domain does not license universal experience; failure to prove experience elsewhere does not license automatic denial.

Scientific underdetermination can be local without turning into global skepticism.

Stronger metaphysical programmes should remain alive when they generate new mathematics, mechanisms, measurements, and risky discriminators. Their generative usefulness should not be confused with confirmation of their distinctive ontology.

> **The best current ontology is the smallest set of commitments that preserves what the evidence has specifically earned, distinguishes the shared constraint floor from its realist interpretation, restores higher-level patterns when they add robust leverage, indexes identity to defensible boundaries, treats necessity as an additional claim rather than a free upgrade from actuality, withholds stronger uptake where defended comparative warrant is missing, and remains willing to demote even realism itself or become a stronger ontology when comparative evidence warrants it.**

Read about Recursive Corrigibility Framework

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57. Das 2021

58. **David H. Wolpert.** The stochastic thermodynamics of computation. 2019; author manuscript v3 updated 2025. Peer-reviewed review; current public author

manuscript introduction and reversibility discussion inspected.

59. **David J. Chalmers.** The Meta-Problem of Consciousness. 2018. Peer-reviewed philosophical target article; author-hosted full text inspected. Explaining judgments and explaining experience remain distinguishable; a debunking inference needs the relevant independence argument.

60. David Papineau, “The Case for Materialism,” 2002

61. **Dellsén, F. (2025). “Inferring to the Best Explanation from Uncertain Evidence.” Philosophy of Science, 92(5), 1225–1234.** [Peer-reviewed]
    Requires sensitivity analysis over evidence reliability because the apparent best explanation can change when uncertain evidence is discounted.
    Verbatim source quotation: “This paper presents a new problem for the inference rule commonly known as ‘inference to the best explanation’ (IBE).” source

62. Dennett, D. C. (1991). “Real Patterns.” *The Journal of Philosophy*, 88(1), 27–51. doi:10.2307/2027085

63. Dominik et al. 2024. Cited resource

64. Donahue’s analysis of methodological naturalism. Cited resource

65. Dresow, M., & Love, A. C. (2023). “Teleonomy: Revisiting a Proposed Conceptual Replacement for Teleology.” *Biological Theory*, 18, 101–113. doi:10.1007/s13752-022-00424-y

66. **Dutilh Novaes, C., & Dede, Ç. (2025). “Critical Contextual Empiricism for Busy People: Scientific Argumentation as Epistemic Exchange.” Topoi, 44, 733–747.** [Peer-reviewed]
    Adds finite-resource costs, workload management, curation, and filtering to critical pluralism and motivates a source lifecycle/stop rule.
    Verbatim source quotation: “In view of the need for workload management,

filtering and curation mechanisms must be in place, in particular under the responsibility of journal editors." source

67. dynamical Casimir experiment. Cited resource

68. **Eric Schwitzgebel.** Introspection. SEP, live entry consulted 16 October 2026. Authoritative reference; distinguishes accounts of introspective access and its limits. No single theory is adopted by definition.

69. Erik Svensson — essay on physicalism, Kant, and panpsychism

70. Explaining the Brain: Mechanisms and the Mosaic Unity of Neuroscience (Oxford University Press, 2007), especially chapter 5. Cited resource

71. **Fábregas-Tejeda, A., & Sims, M. (2025). "On the prospects of basal cognition research becoming fully evolutionary: promising avenues and cautionary notes." History and Philosophy of the Life Sciences, 47, 10.** [Peer-reviewed]
Supports treating decompositions and working definitions as empirical scaffolds that may be revised rather than as ontology fixed by definition.
Verbatim source quotation: "In this regard, it prioritises experimental results over a priori reasoning about criteria for cognition, evaluating hypotheses based on empirical findings." source

72. Finnur Dellsén, "Explanatory Rivals and the Ultimate Argument," 2016

73. **Fischer, B., & Barrett, M. (2025). "On the Utility of the Pathological Complexity Thesis." Philosophia.** [Peer-reviewed]
Shows that evidence for evaluative or valence systems does not automatically establish consciously felt valence.
Verbatim source quotation: "But nonconscious mentality threatens its usefulness on both counts." source

74. Fletcher 2017

75. **Fred C. Adams and Gregory Laughlin.** A dying universe: the long-term fate and evolution of astrophysical objects. 1997. Peer-reviewed theoretical review; publisher abstract and scope inspected.

76. Friston, K. (2010). “The free-energy principle: a unified brain theory?” *Nature Reviews Neuroscience*, 11, 127–138. doi:10.1038/nrn2787

77. Friston, K. J., et al. (2023). “The free energy principle made simpler but not too simple.” *Physics Reports*, 1024, 1–29. doi:10.1016/j.physrep.2023.07.001

78. From Bodily Causal Feeling to Whitehead’s Universal Prehension: Testing the Scope Inference (PDF). Cited resource

79. **Geiger, A., Ibeling, D., Zur, A., et al. (2025). “Causal Abstraction: A Theoretical Foundation for Mechanistic Interpretability.” Journal of Machine Learning Research, 26(83), 1–64.** [Peer-reviewed; full text verified]
Provides a formal example of cross-level bridge validation through state/intervention mappings and graded approximation.
Verbatim source quotation: “The other notions we study in this paper—namely, bijective translation and constructive abstraction—are special cases of exact transformation.” source

80. **Gröger, F., Wen, S. & Brbić, M. (2026). “Revisiting the Platonic Representation Hypothesis: An Aristotelian View.” ICML 2026, PMLR 306:36724–36766.** Null calibration removes much apparent global spectral convergence while local-neighborhood convergence remains more robust. PMLR.

81. **Gurnee, W. & Tegmark, M. (2024). “Language Models Represent Space and Time.” ICLR 2024.** Identifies multiscale spatial and temporal representations across prompts and entities. ML Anthology.

82. **Hanne De Jaegher and Ezequiel Di Paolo.** Participatory sense-making: An enactive approach to social cognition. 2007. Peer-reviewed theoretical paper, DOI 10.1007/s11097-007-9076-9; author-hosted full text inspected. Its autonomy and sense-making conditions require argument before transfer to an unfamiliar implementation.

83. **Harbecke, J. (2025). "Mechanistic Constitution as a Natural Law in Better Best Systems." Journal for General Philosophy of Science, 56, 325–344.** [Peer-reviewed]
Counterpressure against treating constitutive dependence as merely verbal or model-internal; some mechanistic constitutive claims may be empirically constrained and lawlike.
Verbatim source quotation: "This section aims to determine whether certain constitutive claims satisfy the conditions of lawlikeness." source

84. Hoel and colleagues' effective-information result. Cited resource

85. Hoel, E. P. (2013). "Quantifying causal emergence shows that macro can beat micro." *PNAS*, 110(49), 19790–19795.

86. Hoel, E. P. (2025). "Causal Emergence 2.0: Quantifying emergent complexity." Preprint, arXiv:2503.13395. arXiv:2503.13395

87. **Howard Robinson.** Dualism. Current entry inspected October 15, 2026. Authoritative reference; property/substance and interaction distinctions inspected.

88. IEP consciousness. Cited resource

89. IEP metaphysics of science. Cited resource

90. Information Value Theory, IEEE Transactions on Systems Science and Cybernetics 2(1), 22-26 (1966), DOI 10.1109/TSSC.1966.300074. Cited resource

91. Internet Encyclopedia of Philosophy — Karl Popper: Critical Rationalism

> **Verbatim source excerpts:** “Critical Rationalism” is the name Karl Popper (1902-1994) gave to a modest and self-critical rationalism. “Rationality does not need defense; it needs improvement, Agassi says.”

Framework implication: criticism must reach rationality and its methods themselves; RCF adds an explicit architecture for localizing and improving that correction.

92. Internet Encyclopedia of Philosophy — Scientific Change

> **Verbatim source quotation:** “A research program is successful if it leads to progressive problem-shifts and unsuccessful if it leads to degenerating problem-shifts.”

Framework implication: survival under criticism is not progress. RCF separately tracks conceptual clarification, added risky content, prospective corroboration, and degenerating accommodation.

93. Internet Encyclopedia of Philosophy — Scientific Realism and Antirealism

> **Verbatim source excerpts:** “Science aims to give a literally true account of the world.” “To accept a theory is to believe it is (approximately) true.”

Framework implication: these are stronger commitments than mere predictive success, which is why selective and graded realist commitment must be earned piecemeal.

▶ **Source notes**

94. **James Ladyman; M. H. A. Newman and subsequent critics.** Structural Realism. Current entry inspected October 15, 2026. Authoritative reference; Newman objection and responses inspected.

95. **James Woodward and Lauren Ross.** Causal Approaches to Scientific Explanation, SEP. Sections on mechanisms, intervention, and non-causal/mathematical explanation distinguish explanatory relations rather than stipulate that all legitimate explanation has one form.

96. **Jan Henrik Wasserziehr.** Are conscious machines valuers?. 2026, AI & SOCIETY 41, 4957-4969. Peer-reviewed philosophical argument; primary abstract and argument sections inspected.

97. Joel Dietz. Cited resource

98. John W. Carroll's survey of laws of nature. Cited resource

99. Justin Tiehen, "Physicalism," Analysis 78(3), 2018

100. Kawharu et al. 2023/2024

101. **Kim, J. & Lee, S. (2026). "Same Benchmark, Same Subspace: Task-Selective Convergence in LLM Representations." UAI 2026, PMLR 337:3101–3116.** Task-conditioned readout subspaces align across heterogeneous models, supporting local convergence claims. PMLR.

102. knowledge vocabulary. Cited resource

103. Kolchinsky, A., & Wolpert, D. H. (2018). “Semantic information, autonomous agency and non-equilibrium statistical physics.” *Interface Focus*, 8(6), 20180041. doi:10.1098/rsfs.2018.0041

104. Lakatos, I. (1970). “Falsification and the Methodology of Scientific Research Programmes.” In *Criticism and the Growth of Knowledge*.

105. Landauer (1996). Cited resource

106. Landauer, R. (1961). “Irreversibility and Heat Generation in the Computing Process.” *IBM Journal of Research and Development*, 5(3), 183–191. doi:10.1147/rd.53.0183

107. **Lawlor, D. A., Tilling, K., & Davey Smith, G. (2016/2017). “Triangulation in aetiological epidemiology.” International Journal of Epidemiology, 45(6), 1866–1886.** [Peer-reviewed]
Grounds triangulation in methods addressing the same question with importantly different and preferably differently directed biases.
Verbatim source quotation: “Triangulation aims to integrate data from different methodological approaches with different biases and to exploit these differences to draw qualitative conclusions.” source

108. **Lawrence M. Krauss.** The End of the Universe as We Know It. March 23, 2017.
Primary public essay by its author; relevant story and ending inspected.

109. **Lehtinen, A. (2026). “Inferential rules for confirmatory robustness.” European Journal for Philosophy of Science, 16, 34.** [Peer-reviewed]
Provides the constructive counterpoint: derivational robustness can contribute indirectly when linked to empirical confirmation, but does not generate empirical confirmation by itself.
Verbatim source quotation: “While similar rules apply to derivational and

experimental robustness, they are insufficient on their own to generate empirical confirmation from derivational robustness." source

110. **Leopold Stubenberg.** Neutral Monism. Current entry inspected October 15, 2026. Authoritative reference; neutral-base proposals inspected.

111. Leuridan, B. (2012). "Three Problems for the Mutual Manipulability Account of Constitutive Relevance in Mechanisms." *British Journal for the Philosophy of Science*, 63(2), 399–427. doi:10.1093/bjps/axr036

112. **Levin Brinkmann and colleagues.** Propagation and preservation of AI-discovered problem-solving strategies in human culture. 2026, published 18 August. Peer-reviewed experiment and simulations; primary methods, results, and limitations inspected and publisher page rechecked through its cookie-error URL.

113. **Lu, K., Gallagher, K., Michala, S., Fish, S. & Lindsey, J. (2026), "The Assistant Axis."** Developer-lab evidence that apparently convergent assistant personas can partly inherit shared cultural and pretraining genealogy. Primary research page.

114. Lyon, P., & Kuchling, F. (2021). "Valuing what happens: a biogenic approach to valence and (potentially) affect." *Philosophical Transactions of the Royal Society B*, 376, 20190752. doi:10.1098/rstb.2019.0752

115. Maria Caamaño-Alegre, "Empirical Underdetermination," 2025

116. Massimo Pigliucci — "Five allegedly bad 'isms' — Part II"

117. **Matthew David Segall (2026).** A Dialogue with Michael Levin, 30 June 2026. Primary account linking the programmes while recording substantive disagreements. Segall's interpretation of the biology is distinguished from the biology's experimental conclusions.

118. Matthew David Segall — Footnotes²Plato

119. **Mayne, Z. J. (2026). “Operationalizing Scientific Realism.” Philosophy of Science, First View.** [Peer-reviewed]
Supports localizing realism and ontological admission to specific questions and domains rather than applying one global principle indiscriminately.
Verbatim source quotation: “The local method tackles questions of realism in a radically piecemeal fashion, prioritizing the needs of particular domains of empirical inquiry.” source

120. Mayne, Z. J. (2026). “Operationalizing Scientific Realism.” *Philosophy of Science*, First View. doi:10.1017/psa.2026.10225

121. Mayo, D. G. (2018). *Statistical Inference as Severe Testing*. Cambridge University Press.

122. **McLoone, B., Hecht Orzack, S., & Sober, E. (2025). “The epistemic status of derivational robustness.” European Journal for Philosophy of Science, 15, 46.** [Peer-reviewed]
Challenges the assumption that agreement among models carries the same confirmatory status as agreement among empirical measurements.
Verbatim source quotation: “This is not true for derivational robustness.” source

123. **(2026).** Ingressing Minds: Causal, Non-Physical Patterns In-Form Natural, Synthetic, and Hybrid Embodiments. *Philosophies* 11(5), 161; DOI 10.3390/philosophies11050161. Published 9 September 2026. The supplied 53-page PDF, including its qualifications and notes, was read. Relevant locations: pp. 3–4, 12–13, 16–17, 22–26, 29–32, 37–40. Its own data statement says no new data were created or analyzed in this study (p. 38). The quoted and comparative conclusions below are an assessment of its argument, not a new experiment.

124. **Michael S. A. Graziano and Taylor W. Webb.** The attention schema theory: a mechanistic account of subjective awareness. 2015, Frontiers in Psychology 6, 500.

Peer-reviewed Hypothesis and Theory article; primary theory and evidence sections inspected.

125. **Milinkovic, B. & Aru, J. (2026). “On Biological and Artificial Consciousness: A Case for Biological Computationalism.” Neuroscience & Biobehavioral Reviews 181:106524.** A serious biological-computational rival emphasizing scale-inseparability and hybrid biological computation. DOI.

▶ **Source notes**

126. Mitchell, S. D. (2008). “Exporting Causal Knowledge in Evolutionary and Developmental Biology.” *Philosophy of Science*, 75(5), 697–706. doi:10.1086/594515

127. **Mohammadian, M. (2025 online / 2026 issue). “Theoretical Virtues, Truth, and the Epistemic Aim of Scientific Theorizing.” Philosophy of Science, 93(2), 402–422.** [Peer-reviewed]
Counterpressure to anti-IBE readings: defends theoretical virtues as a central epistemic aim of scientific theorizing.
Verbatim source quotation: “I argue that the epistemic aim of scientific theorizing (EAST) is producing theories with the highest possible number and degree of theoretical virtues.” source

128. Montévil, M., & Mossio, M. (2015). “Biological organisation as closure of constraints.” *Journal of Theoretical Biology*, 372, 179–191. doi:10.1016/j.jtbi.2015.02.029

129. Morgan 2014

130. **Morgan, J. P. (2025). “Design Selection for Model Discrimination.” Journal of Statistical Theory and Practice, 19, 48.** [Peer-reviewed; full text verified]

Supports separating mere model estimability from designs capable of discriminating among rivals.

Verbatim source quotation: “The selected design should be able to distinguish among the competing models.” source

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131. Moss, L. (2024). “Normativity, Autonomy, and Agency: A Critical Review…” *Biological Theory*, 19, 73–83. doi:10.1007/s13752-024-00462-8

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132. **Munafò, M. R., Nosek, B. A., Bishop, D. V. M., et al. (2017). “A manifesto for reproducible science.” Nature Human Behaviour, 1, 0021.** [Peer-reviewed] Supports making methodological reforms themselves testable, revisable, and subject to continuing empirical evaluation.

Verbatim source quotation: “Improving the reliability and efficiency of scientific research will increase the credibility of the published scientific literature and accelerate discovery.” source

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133. **Nathan Sweet (2026).** Religion With A Pipette: Is Michael Levin’s ‘Ingressing Minds’ Pseudoscience? A Scholarly Analysis. Author’s comparative criticism, read alongside the supplied PDF and pasted text. It supplies the D-versus-D+P analysis, rival comparisons, ethical distinctions, and provenance questions; it is not independent replication of the biological studies it discusses.

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134. **Nathan Sweet.** Synthbiosis: Deductive Proof That The Machine Did Not Need to Feel to Matter. 2026; live text checked 16 October. Primary formal argument. T0-G derives a specified moral constraint from protected capacities and constitutive impairment; T1 states conditional routes to direct standing with independently defended grounds. Actual application depends on the empirical and normative premises.

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135. **Nathan Sweet.** What Warrants the Scope Jump?. 2026; live text checked 16 October. Primary argument: compares the proposed universalization with positive causal and continuity-preserving alternatives. Not a peer-reviewed empirical result.

136. **Nosek, B. A., Ebersole, C. R., DeHaven, A. C., & Mellor, D. T. (2018). "The preregistration revolution." PNAS, 115(11), 2600–2606.** [Peer-reviewed] Supports separating free/exploratory hypothesis generation from prospective testing and not confusing postdiction with prediction.
Verbatim source quotation: "Progress in science relies in part on generating hypotheses with existing observations and testing hypotheses with new observations." source

137. **O. J. E. Maroney.** Generalizing Landauer's principle. 2009. Peer-reviewed primary theory reference carried from the current core thermodynamics discussion; publisher access was incomplete in this pass.

138. Okasha 2024. Cited resource

139. On computational irreducibility and the predictability of complex physical systems. Cited resource

140. **Park, K., Choe, Y. J. & Veitch, V. (2024). "The Linear Representation Hypothesis and the Geometry of Large Language Models." ICML 2024.** Connects representation claims to counterfactual intervention and steering rather than probe readability alone. PMLR.

141. **Paul Guyer and Rolf-Peter Horstmann.** Idealism. May 5, 2026 entry. Authoritative reference; epistemological and metaphysical arguments distinguished.

142. Pearl, J. (2009). *Causality: Models, Reasoning, and Inference*, 2nd ed. Cambridge University Press.

143. peer-reviewed analysis of duality and theoretical equivalence. Cited resource

144. Peter Visscher, William Hill, and Naomi Wray. Cited resource

145. Popper, K. R. (1959). *The Logic of Scientific Discovery*.

146. Potter & Mitchell 2022. Cited resource

147. Potter & Mitchell 2026. Cited resource

148. Potter, Ellis & Mitchell 2026. Cited resource

149. Quan Zhou, Margherita Malanchini, and colleagues. Cited resource

150. Real Patterns, The Journal of Philosophy 88(1), 27-51 (1991). Cited resource

151. Recommendation 1, Using Population Descriptors in Genetics and Genomics Research. Cited resource

152. The retained Popperian commitment is risky exposure to error. RCF extends that exposure to constructs, bridges, measurements, rival comparisons, dependency graphs, and the methodology itself rather than treating survival of a single falsification attempt as confirmation.

153. rigorous quantum-group construction. Cited resource

154. **Robert Van Gulick.** Consciousness. SEP, live entry consulted 16 October 2026. Authoritative reference distinguishing types, explanatory targets, and rival accounts. No single metric is treated as the definition of consciousness.

155. **Robinson, K. A., Saldanha, I. J., & Mckoy, N. A. (2011). "Development of a framework to identify research gaps from systematic reviews." Journal of Clinical Epidemiology, 64(12), 1325–1330.** [Peer-reviewed; full text verified] Supports explicit gap typing rather than treating every evidential shortfall as one generic absence.
Verbatim source quotation: "Our objective was to develop a framework to identify research gaps from systematic reviews." source

156. **Roche, W., & Sober, E. (2026). "Inference to the Best Explanation, Bayesianism, and the Screening-Off Challenge: A Critical Review." Journal for General Philosophy of Science.** [Peer-reviewed]
Shows how explanatory virtues can be screened off by the evidence/background and warns against double-counting explanatory attractiveness as extra confirmation. **Verbatim source quotation:** "We think that R&S's approach of using screening-off as a litmus test is useful in revealing how IBE often goes astray." source

157. Roher et al. 2024

158. Rule 30 technical account. Cited resource

159. Safi 2023

160. **Schaeffer, R., Miranda, B. & Koyejo, S. (2023). "Are Emergent Abilities of Large Language Models a Mirage?"** Shows that discontinuous metrics can manufacture apparent abrupt emergence. arXiv.

161. Seehawer 2018

162. Seifert, U. (2019). "From Stochastic Thermodynamics to Thermodynamic Inference." *Annual Review of Condensed Matter Physics*, 10, 171–192. doi:10.1146/annurev-conmatphys-031218-013554

163. SEP distinguishes methodological from ontological naturalism. Cited resource

164. **SEP reference checks.** Logical Consequence; Imre Lakatos; Prediction versus Accommodation. These anchor the distinctions among derivation, countermodel, programme progress, and predictive or accommodated success. Retrospective evidence is not disqualified merely by being retrospective; the question is which independent constraints and comparative consequences support which claim.

165. SEP, Pragmatism, sections on Peirce and inquiry. Cited resource

166. SEP, Rudolf Carnap, D. Methodology, section 1. Cited resource

167. SEP: Being and Becoming in Modern Physics. Cited resource

168. SEP: Church–Turing Thesis. Cited resource

169. SEP: Nothingness. Cited resource

170. SEP: Quantum Field Theory. Cited resource

171. **Şerban, M. (2026). "Rethinking interlevel experiments: no remainder from evidence for causal relations." European Journal for Philosophy of Science, 16, 12.** [Peer-reviewed]
Counterpressure in the opposite direction: matched interlevel evidence may support causal mediation without a distinct constitutive remainder.
Verbatim source quotation: "Drawing on paradigmatic action potential experiments, I demonstrate that practices satisfying MIE's formal requirements consistently establish causal mediation relationships without requiring constitutive interpretation." source

172. **Seth, A. K. (2025). "Conscious Artificial Intelligence and Biological Naturalism." Behavioral and Brain Sciences.** A serious biological-naturalist rival arguing that life-like organization may matter to consciousness. DOI.

173. Seth, A. K., & Bayne, T. (2022). "Theories of consciousness." *Nature Reviews Neuroscience*, 23, 439–452. doi:10.1038/s41583-022-00587-4

174. **Shen, G., Zhao, D., Dong, Y., Zhang, Q. & Zeng, Y. (2026). "Alignment between Brains and AI: Evidence for Convergent Evolution across Modalities, Scales and Training Trajectories." ICML 2026, PMLR 306:111522–111541.** Performance-related model–brain representational correspondence is suggestive but does not establish consciousness. PMLR.

175. Shevchenko 2024

176. **Sidney Coleman.** Fate of the false vacuum: Semiclassical theory. 1977. Peer-reviewed theoretical paper; publisher abstract, scope and erratum record inspected.

177. **Simon Storz and colleagues.** Loophole-free Bell inequality violation with superconducting circuits. 2023. Peer-reviewed primary experiment; publisher abstract and stated scope inspected.

178. Sleep and the Price of Plasticity. Cited resource

179. **Sofroniew, N., Kauvar, I., Saunders, W., Chen, R., et al. (2026). "Emotion Concepts and their Function in a Large Language Model." arXiv:2604.07729.** Reports causally consequential functional-emotion representations while explicitly separating them from subjective feeling. arXiv.

180. **Sommer, I., Sunder-Plassmann, V., Ratajczak, P., et al. (2023). "Full publication of preprint articles in prevention research: an analysis of publication proportions and results consistency." Scientific Reports, 13, 17034.** [Peer-reviewed; full text verified]
Supports retaining extra uncertainty around unreviewed results while checking whether the relevant claim survives later publication.
Verbatim source quotation: "We therefore warrant caution in using preprints of prevention research in decision-making." source

181. Stanford Encyclopedia of Philosophy — Karl Popper

> **Verbatim source quotation:** "Popper draws a clear distinction between the logic of falsifiability and its applied methodology."

Framework implication: a logical counter-instance and a responsible methodological revision are different questions; error localization, measurement, auxiliaries, and dependency structure matter.

182. Stanford Encyclopedia of Philosophy — Physicalism

> **Verbatim source excerpts:** “Physicalism is, in slogan form, the thesis that everything is physical.” “What does it mean to say that everything is physical?”

Framework implication: physical realization, causal completeness, supervenience, substrate necessity, absence of fundamental mentality, and exhaustive ontology must be factorized rather than bundled.

183. Stanford Encyclopedia of Philosophy — Underdetermination of Scientific Theory

> **Verbatim source quotation:** “Moreover, such differences in the character and strength of various claims of underdetermination turn out to be crucial for resolving the significance of the issue.”

Framework implication: underdetermination is typed and local. It does not entail equal priors, global skepticism, or automatic support for a rival.

▶ **Source notes**

184. **Stanford Encyclopedia of Philosophy, “Other Minds.”** Canonical orientation to

the epistemic problem of attributing mentality to others from indirect evidence. SEP.

185. **Stanford Encyclopedia of Philosophy. “Abduction.” Current entry, entry dated 2025.** [Reference taxonomy]
Reference taxonomy confirming that the form and normative status of abduction remain disputed; used for orientation, not as primary evidence.
Verbatim source quotation: “However, the exact form as well as the normative status of abduction are still matters of controversy.” source

186. Stanford, P. K. (2022). “Refusing the Devil’s Bargain: What Kind of Underdetermination Should We Take Seriously?” *Philosophy of Science*.

187. **Stegenga, J., & Menon, T. (2017). “Robustness and Independent Evidence.” Philosophy of Science, 84(3), 414–435.** [Peer-reviewed]
Warns that apparent ontic independence of evidence streams does not by itself guarantee confirmatory independence.
Verbatim source quotation: “However, we argue that, as typically construed, OI is not a sufficient independence condition for warranting robustness arguments.” source

188. Stengers, I. (2008). “A Constructivist Reading of Process and Reality.” *Theory, Culture & Society*, 25(4), 91–110. doi:10.1177/0263276408091985

189. Stenner, P. (2008). “A.N. Whitehead and Subjectivity.” *Subjectivity*, 22, 90–109. doi:10.1057/sub.2008.4

190. Strawson 1994. Cited resource

191. **Sungho Choi and Michael Fara.** Dispositions. SEP, live entry consulted 16 October 2026. Authoritative account of dispositional and categorical approaches and causal efficacy. These supply substantive rivals rather than an automatically established ontology.

192. **Sutter, D., Minder, J., Hofmann, T., & Pimentel, T. (2025). “The Non-Linear Representation Dilemma: Is Causal Abstraction Enough for Mechanistic Interpretability?” Advances in Neural Information Processing Systems 38.** [Peer-reviewed archival conference; full text verified]
Constrains bridge realism by showing that expressive mappings can overfit and fail to generalize.
Verbatim source quotation: “In this work, we critically examine the concept of causal abstraction by considering arbitrarily powerful alignment maps.” source

193. Sweet Rationalism. Cited resource

194. “Synthbiosis: The Machine Did Not Need to Feel to Matter Morally.”. Cited resource

195. **Thomas O. Nelson and Louis Narens.** Metamemory: A theoretical framework and new findings. 1990. Primary scholarly chapter; monitoring, control, and report-method sections inspected.

196. **Thompson, P., Andersson, B. J., Sundqvist, N., & Cedersund, G. (2025). “A new method for a priori practical identifiability.” PLOS ONE, 20(7), e0327593.** [Peer-reviewed; full text verified]
Shows why ontology should not infer a uniquely recoverable mechanism/property merely from model fit under idealized measurement assumptions.
Verbatim source quotation: “For all studied examples, traditional structural identifiability methods have widely overestimated the number of identifiable parameters.” source

197. **Tobias Bothwell and colleagues.** Resolving the gravitational redshift across a millimetre-scale atomic sample. 2022. Peer-reviewed primary experiment; publisher record and abstract inspected; updates an existing manuscript citation.

198. Toole 2023. Cited resource

199. **Veit, W. (2026). “Valence, Mirror Self-Recognition, and the Gradualist Evolution of Consciousness.” Philosophia, 54, 43–63.** [Peer-reviewed]
Even a defender of a stronger consciousness role distinguishes cognitive-scientific valence/value assignment from conscious realization.
Verbatim source quotation: “Since valence, in a cognitive science framework, refers simply to value assignment, it is silent on how such valence is neurologically implemented.” source

200. Waller 2023. Cited resource

201. Walsh, D. M., & Rupik, G. (2023). “The agential perspective: Countermapping the modern synthesis.” *Evolution & Development*, 25, 335–352. doi:10.1111/ede.12448

202. Wen, Yousif, and Sherman’s 2026 experiments. Cited resource

203. Woodward, J. (2003). *Making Things Happen*. Oxford University Press.

204. **Woodward, J. (2025). “The place of explanation in scientific inquiry: Inference to the best explanation vs inference to the only explanation.” European Journal for Philosophy of Science, 15, 21.** [Peer-reviewed]
Strong contemporary challenge to unrestricted inference-to-the-best-explanation; retained as a live rival rather than treated as consensus.
Verbatim source quotation: “It argues that IBE is not a defensible form of inference.” source

205. Zhang, J., Tao, R., Leong, K. H., et al. (2025). “Dynamical reversibility and a new theory of causal emergence based on SVD.” *npj Complexity*.

206. Zhang, Z., & Xu, F. (2024). “An Overview of the Free Energy Principle and Related Research.” *Neural Computation*, 36(5), 963–1021. doi:10.1162/neco_a_01642